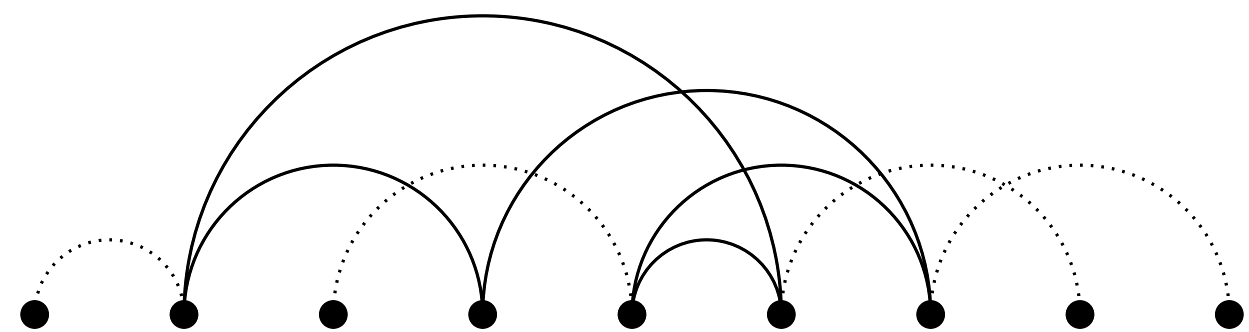
Hanani-Tutte for Level Planarity

Intuitive algorithms for finding planar drawings

Maximilian Süss

5.12.2024

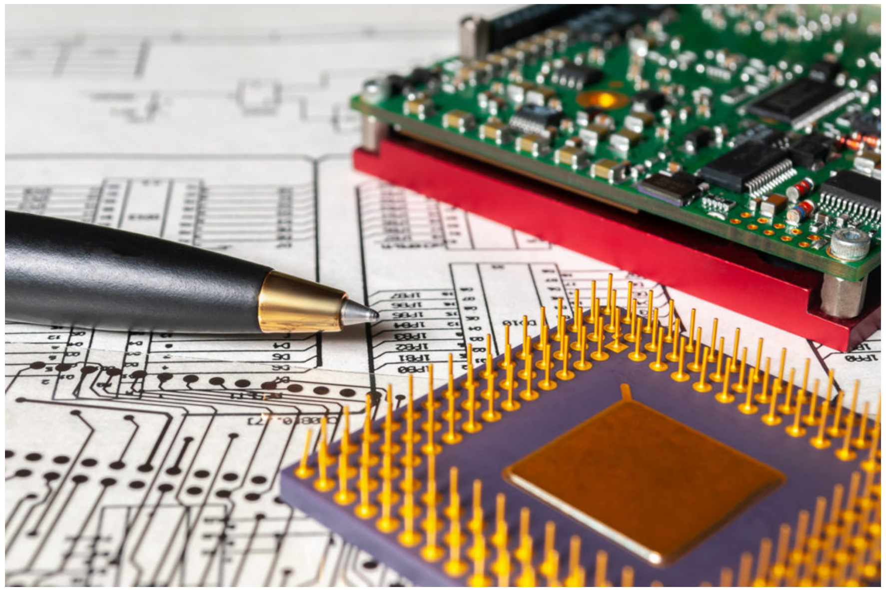


ALGORITHMS AND COMPLEXITY GROUP

Motivation

Divers Applications:

Motivation



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- Circuit Design

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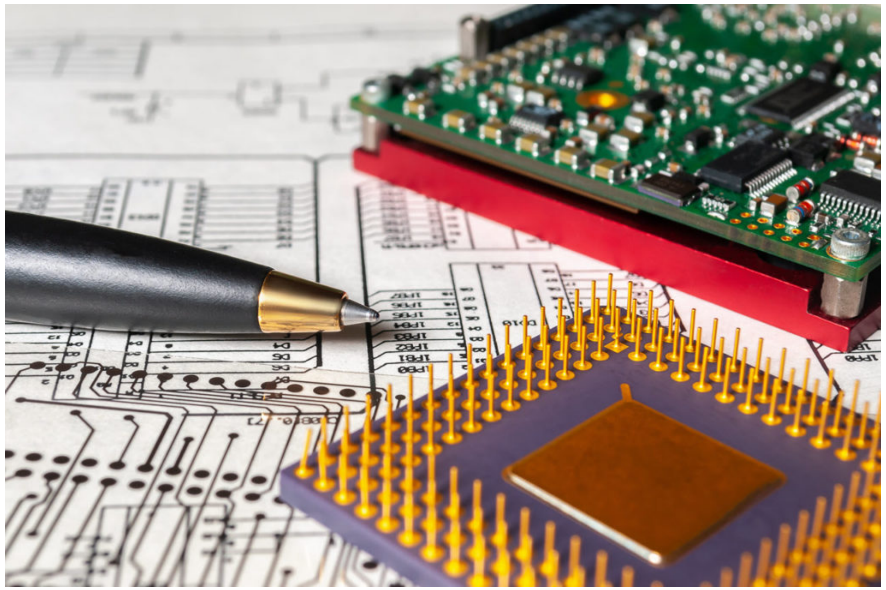

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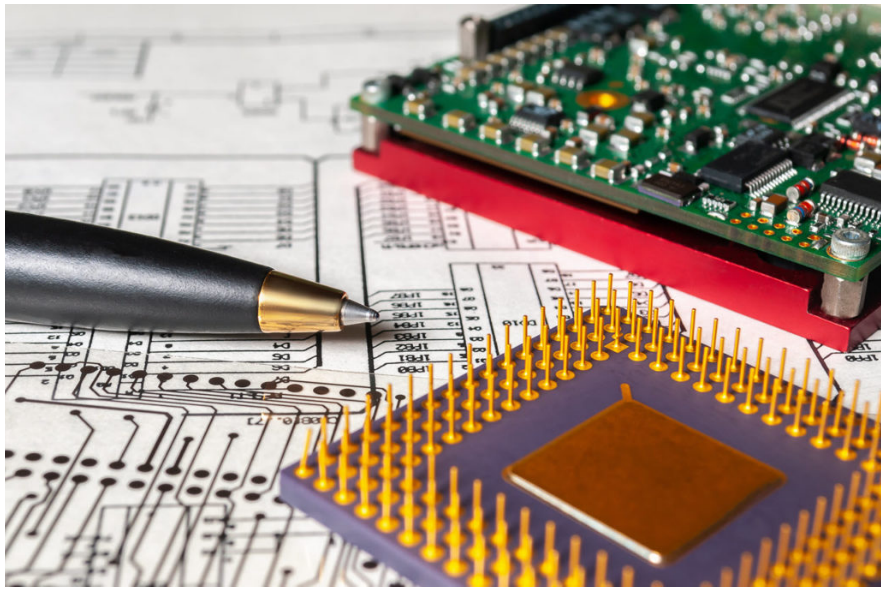

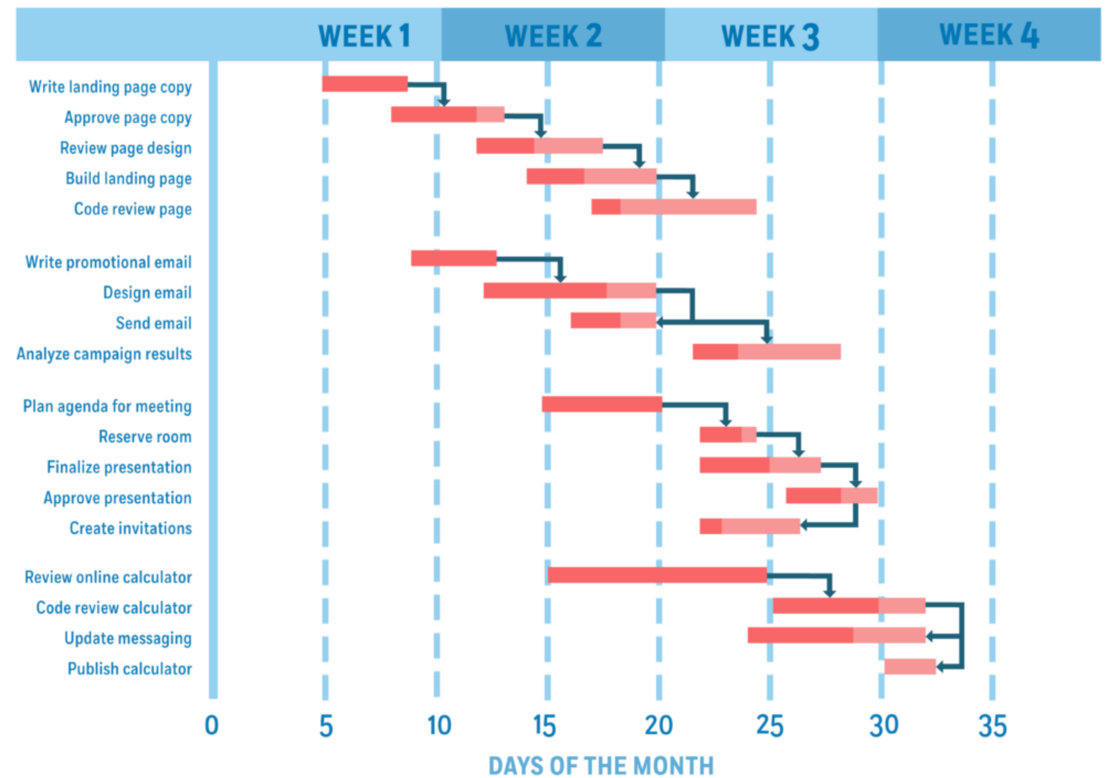

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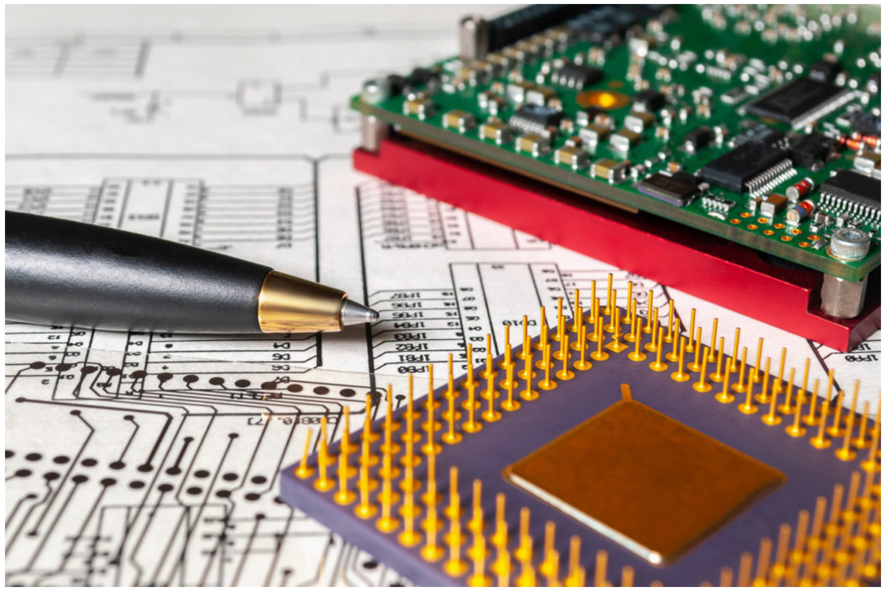

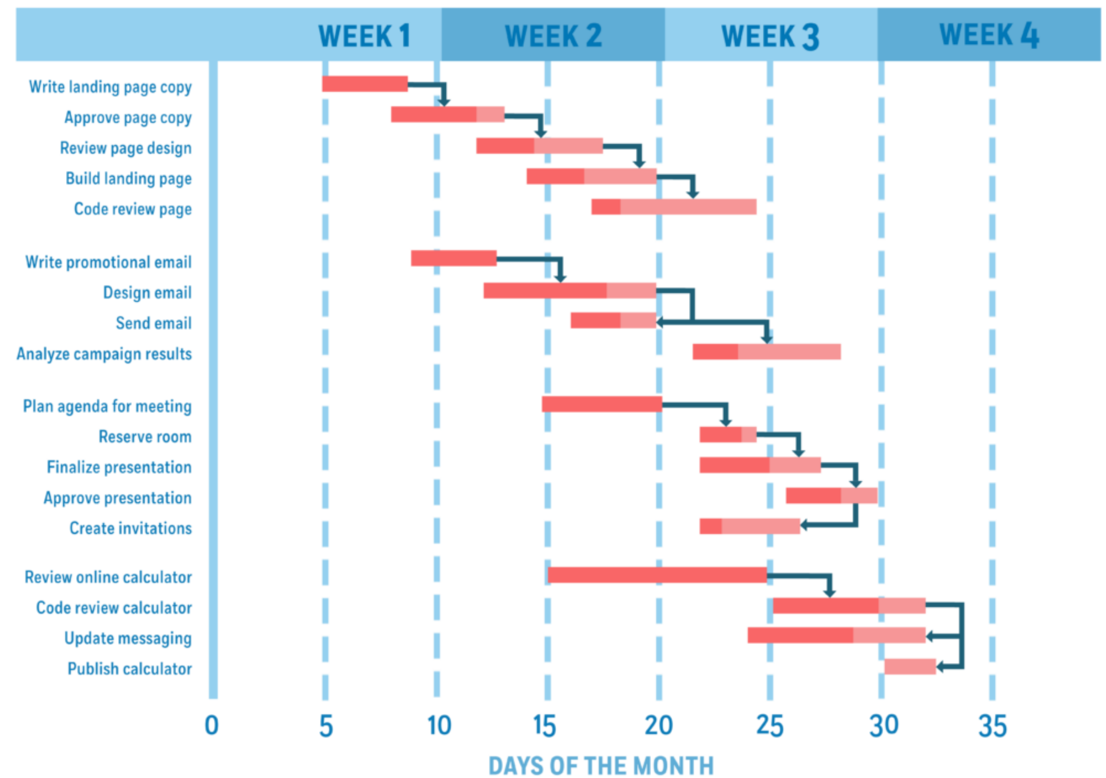

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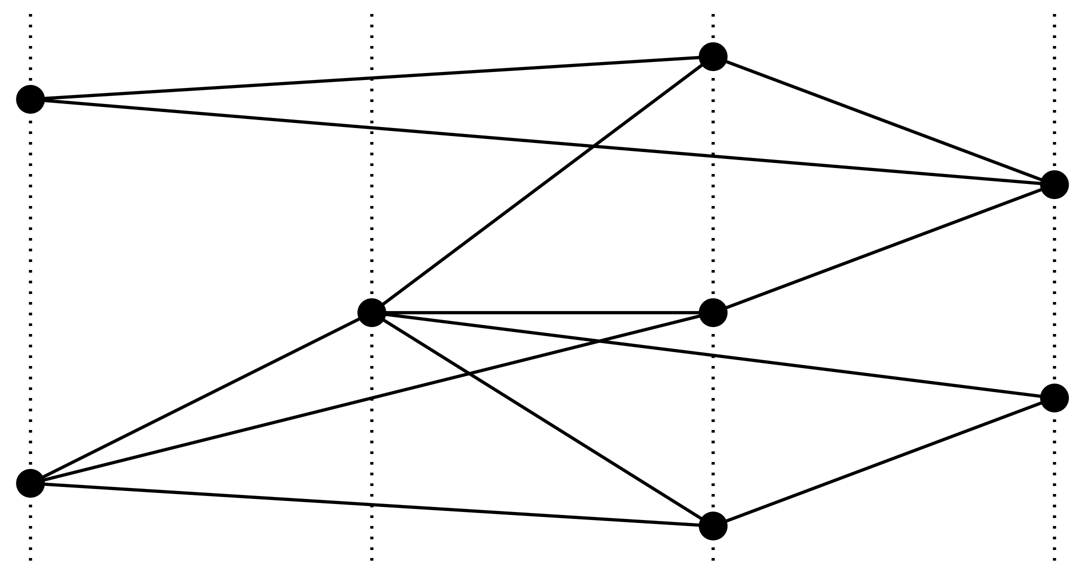

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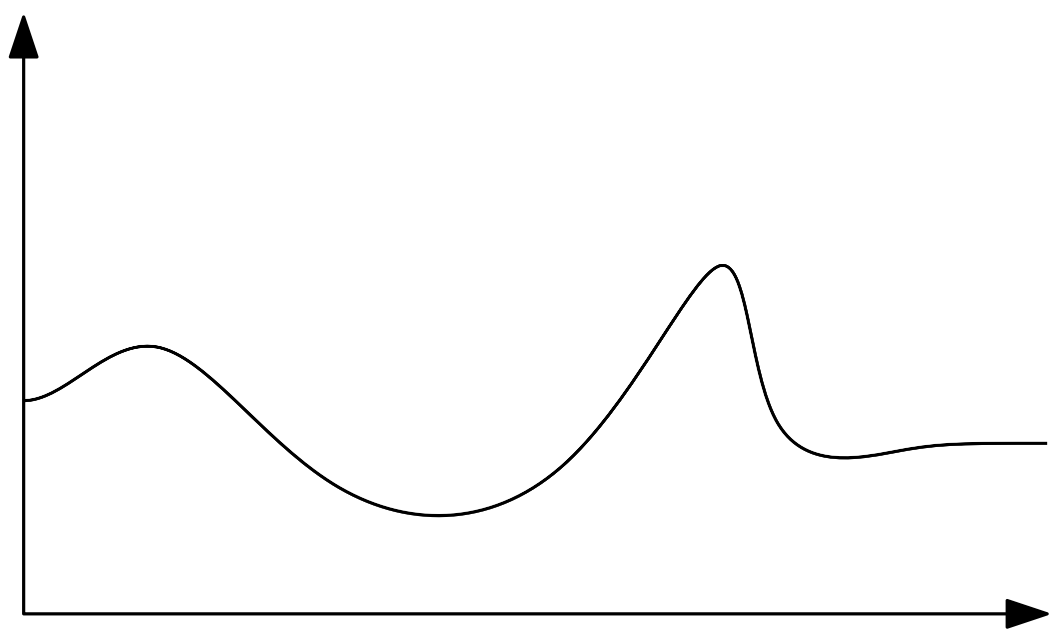

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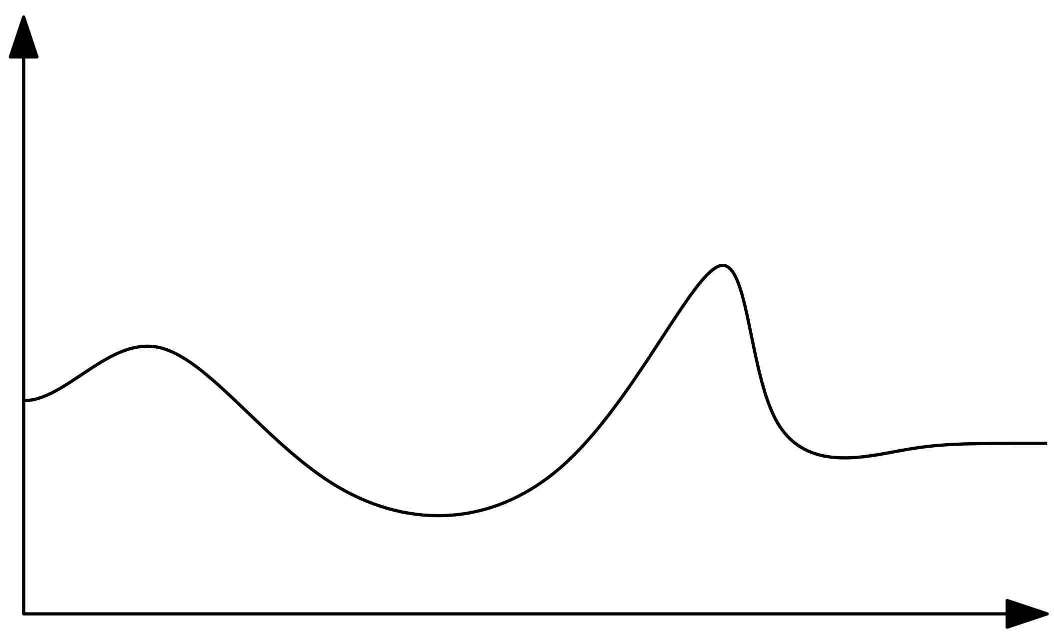

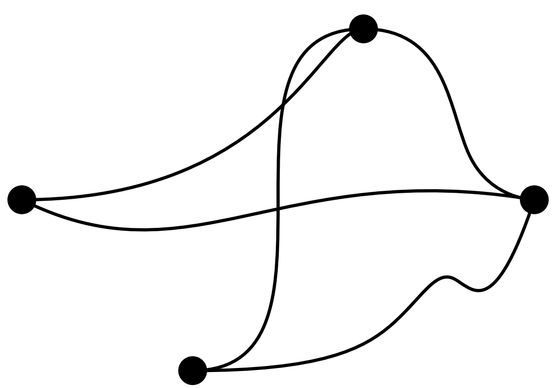

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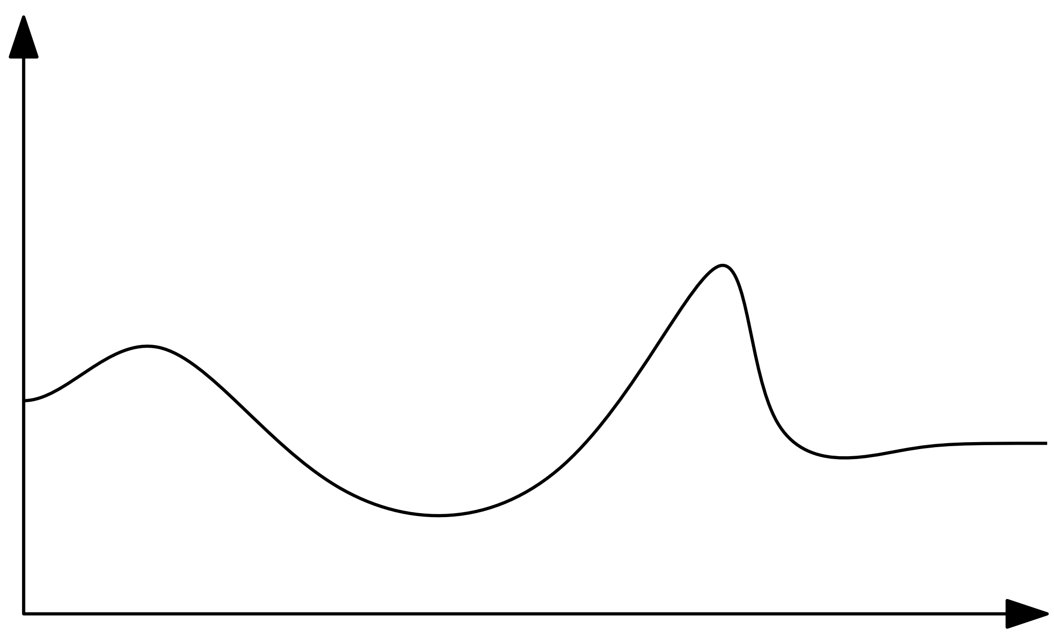

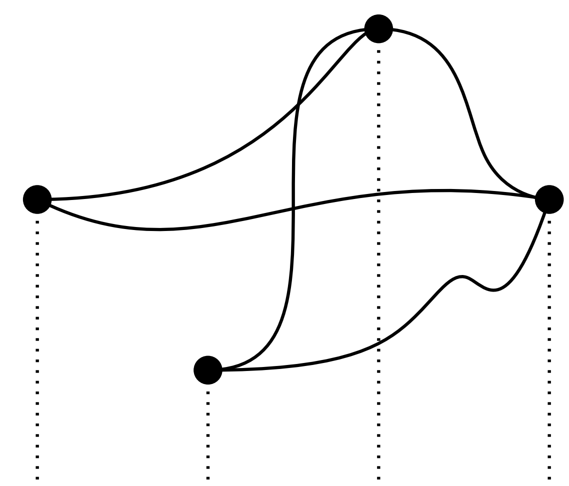

$$V_1 < V_2 < V_3 < V_4$$

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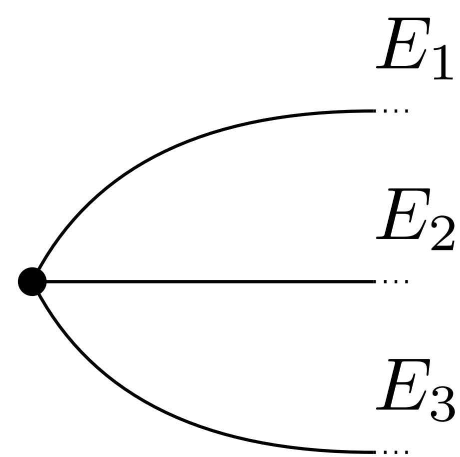

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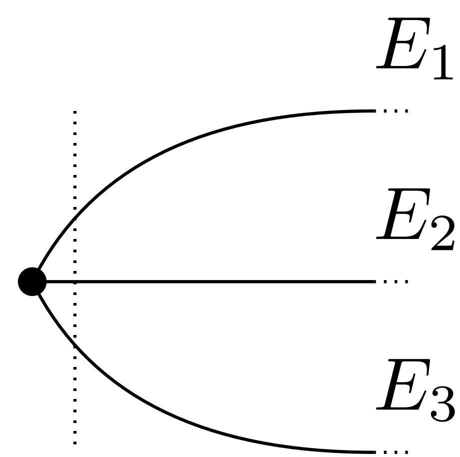

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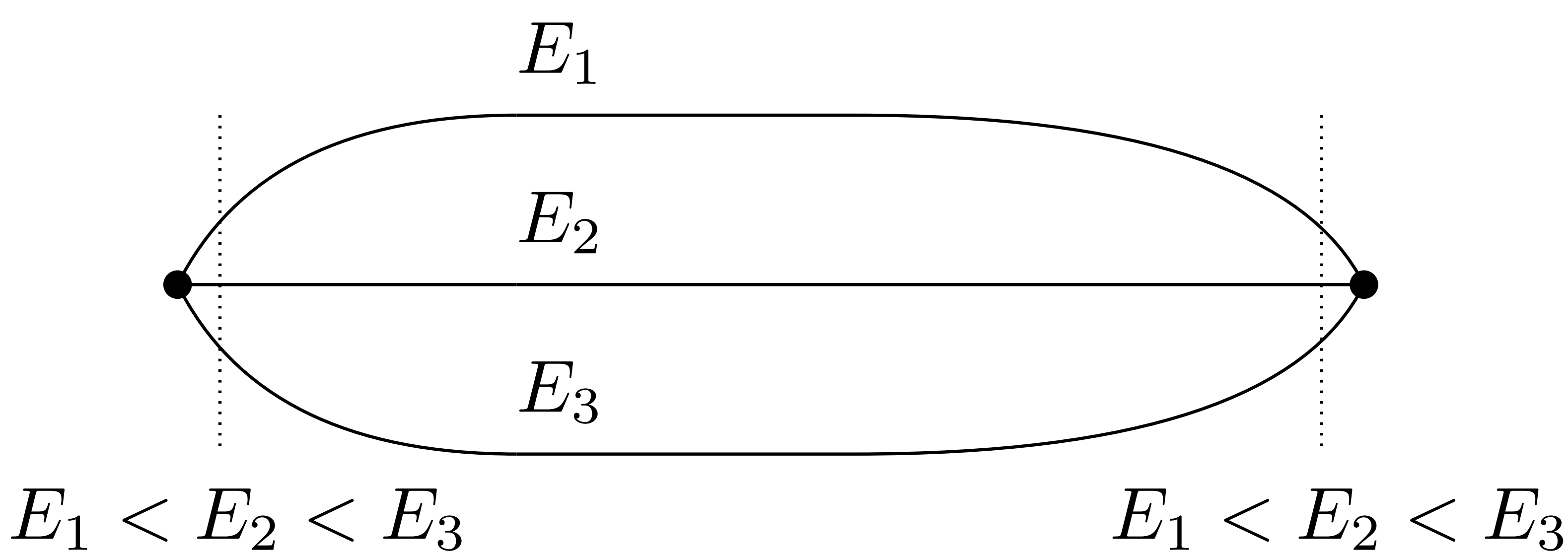

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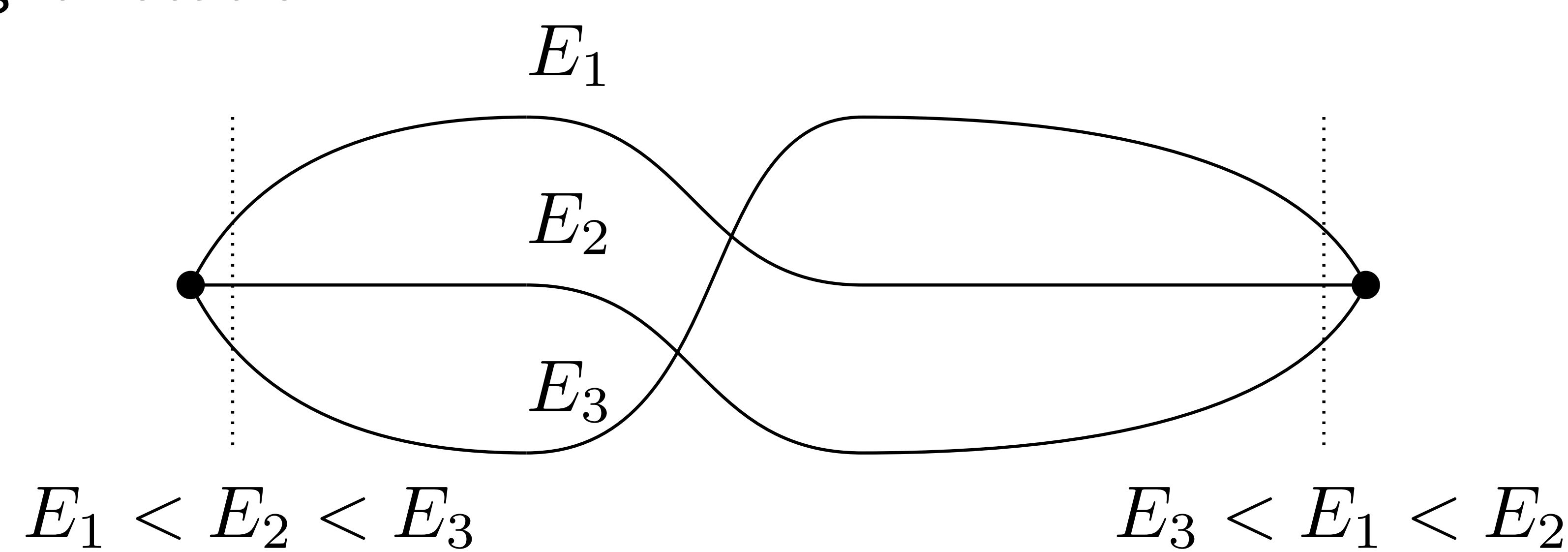

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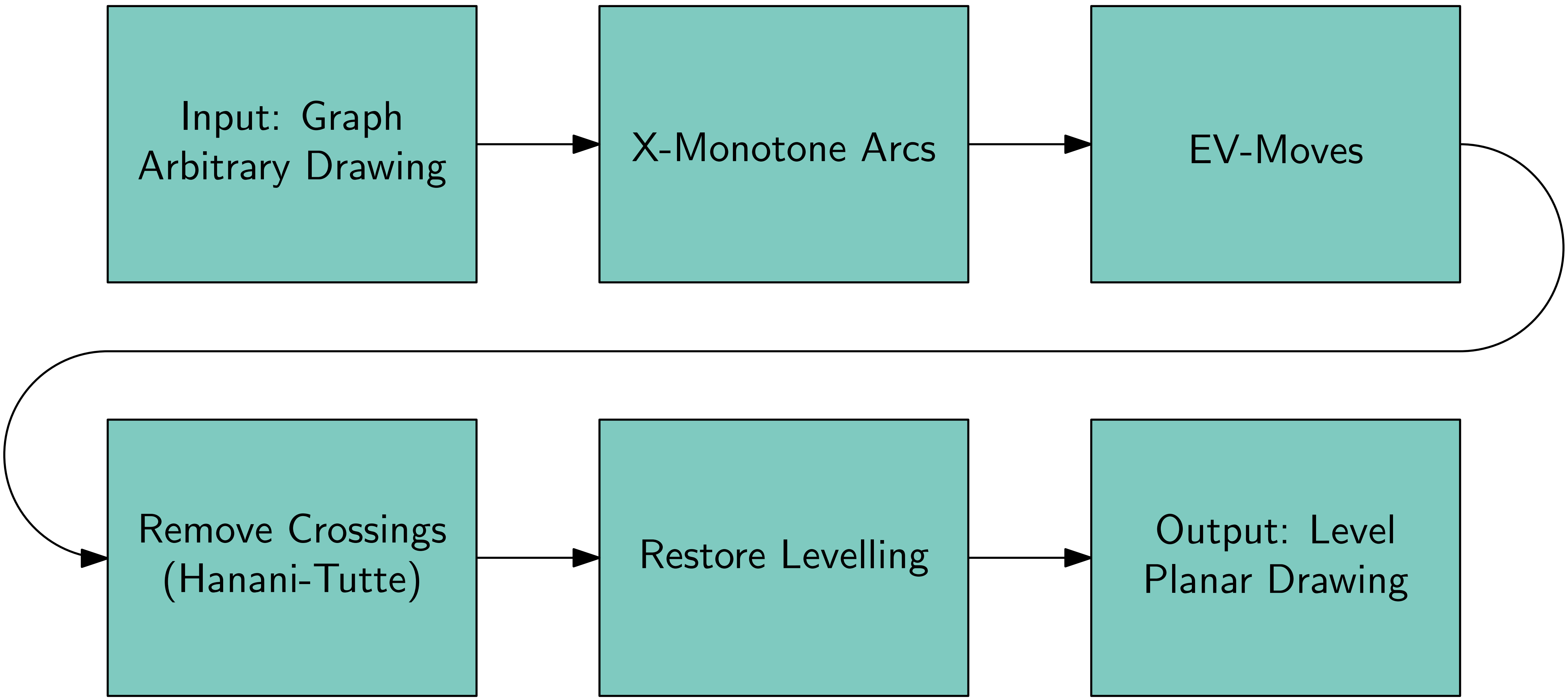
- Embedding:
  - Class of topologically equivalent drawings.
  - Same rotation system.
  - Planar.

Leveled Embedding Construction

Algorithm



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Graph $G = (V, E)$, Levelling l

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$$l = \{(A_1, l_A), (B_1, l_B)$$
$$(B_2, l_B), (C_1, l_C), (C_2, l_C)\}$$

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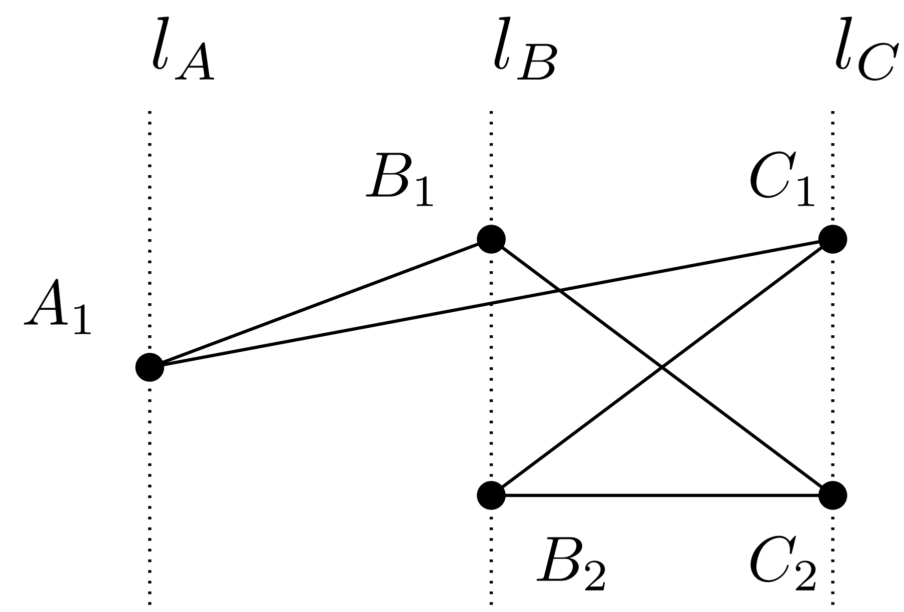

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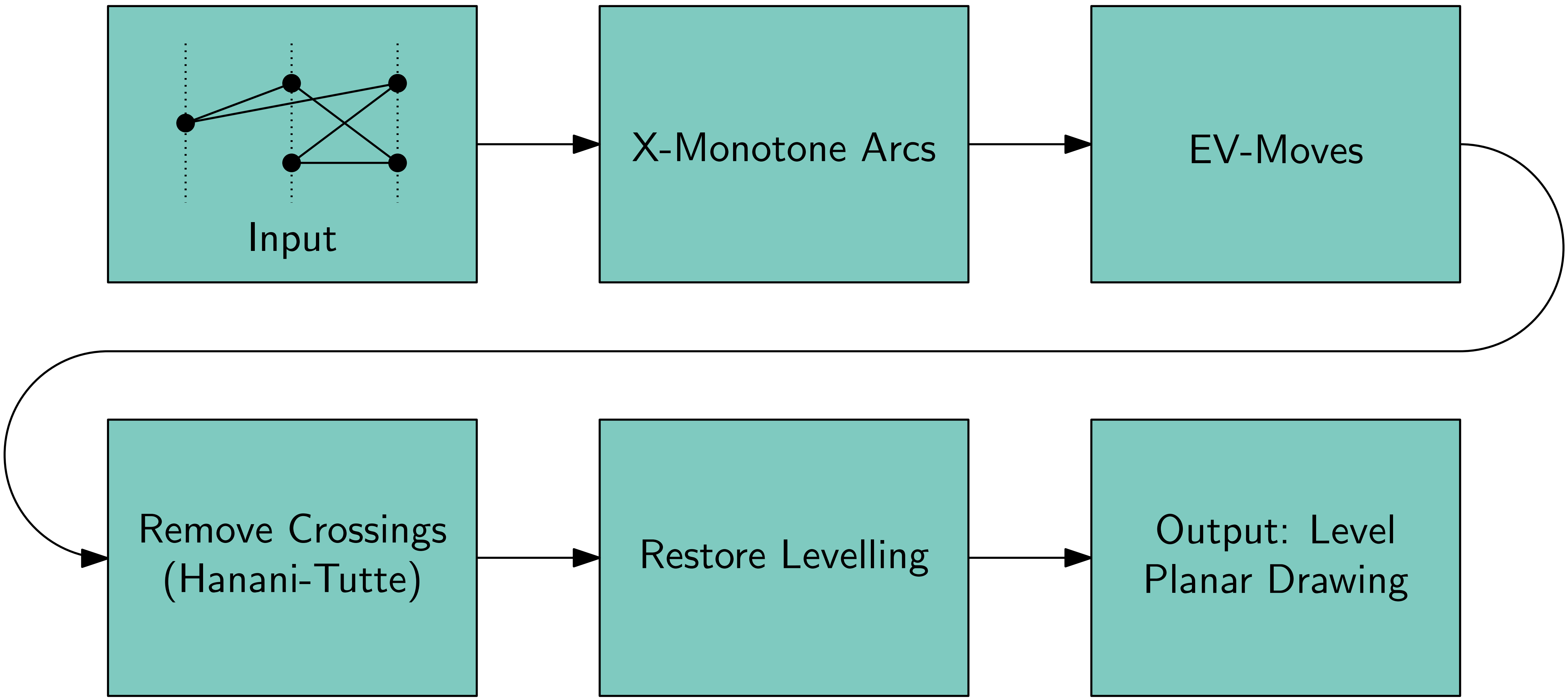
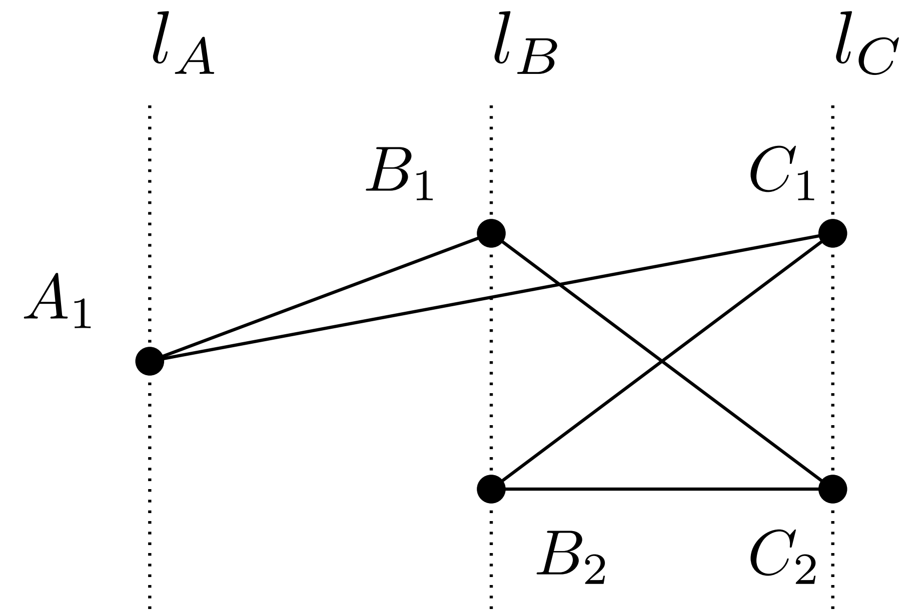


X-Monotone Arcs



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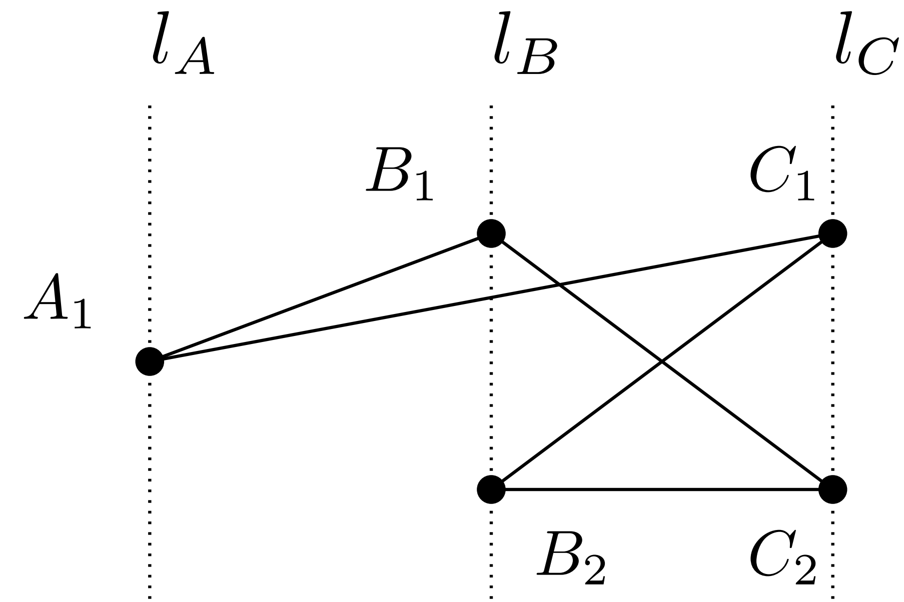

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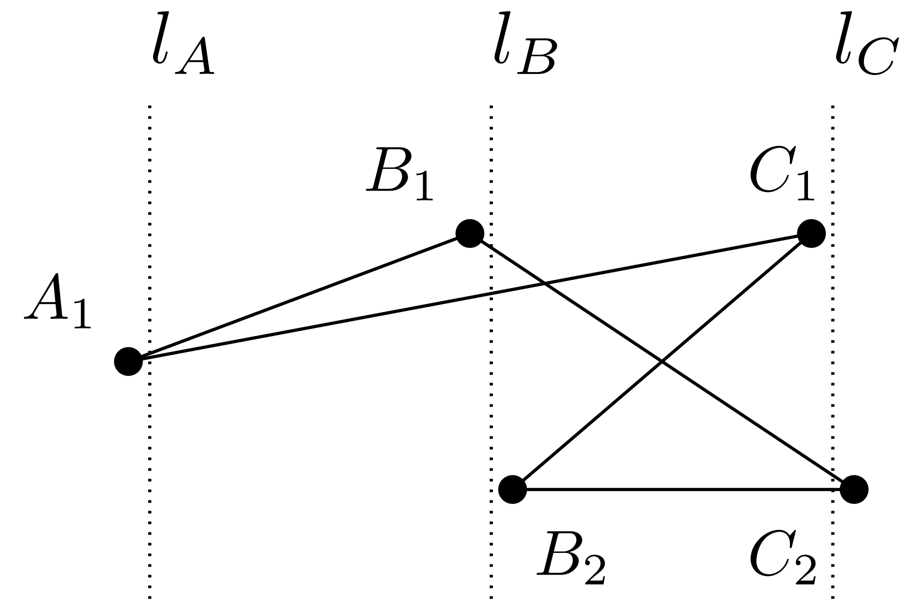

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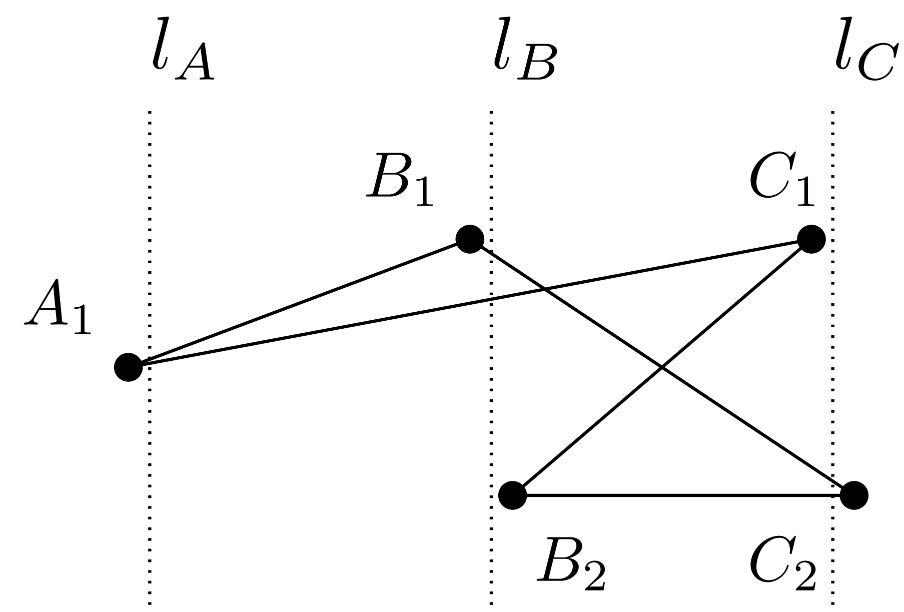
- add edges



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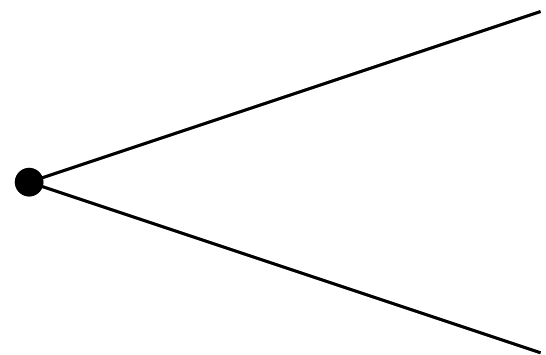

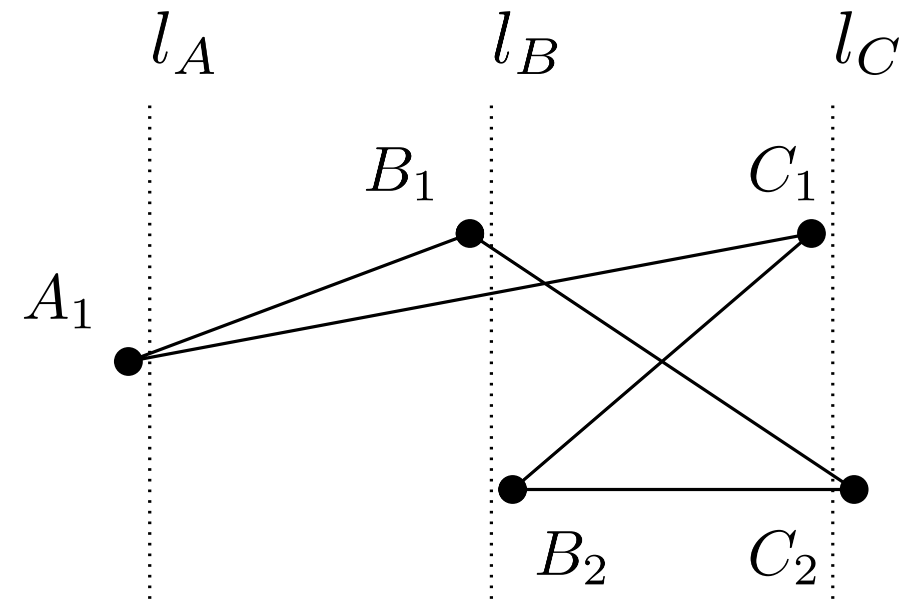

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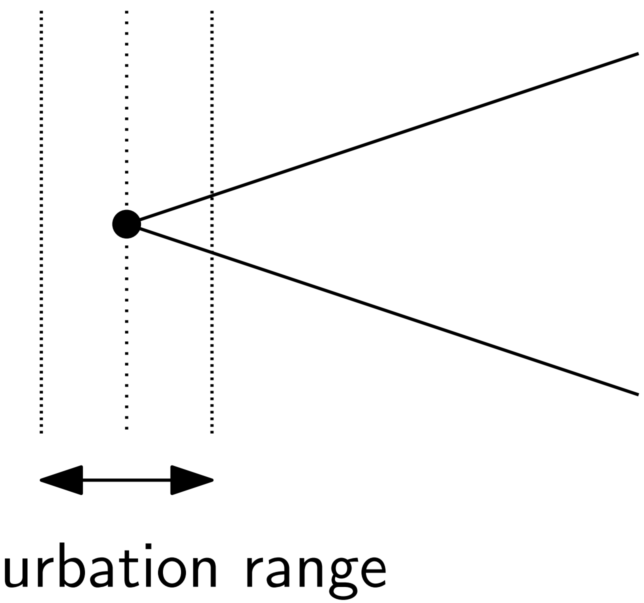

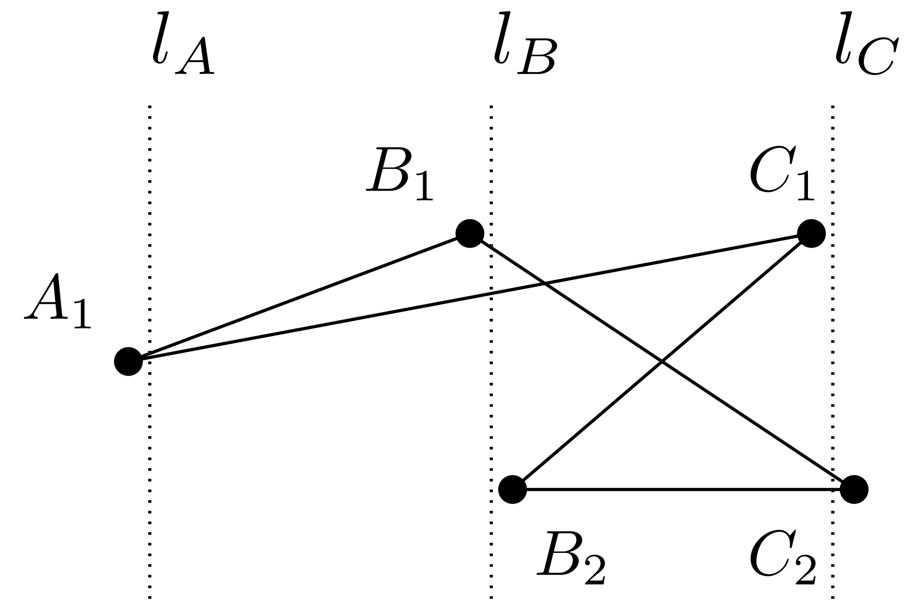

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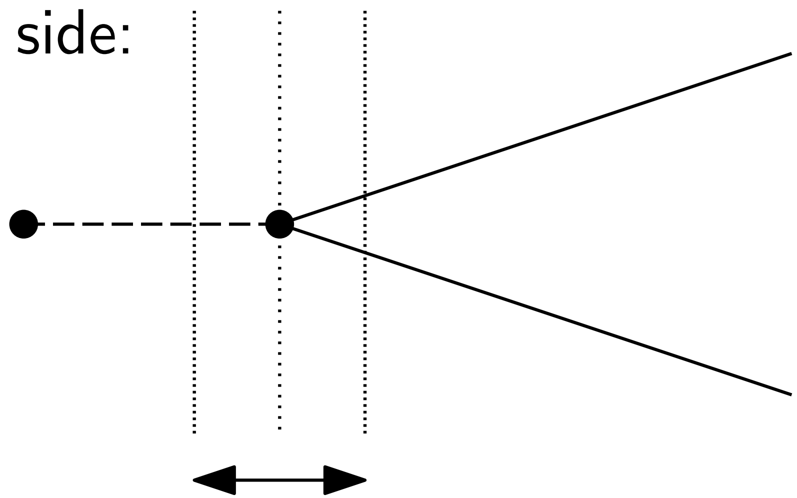
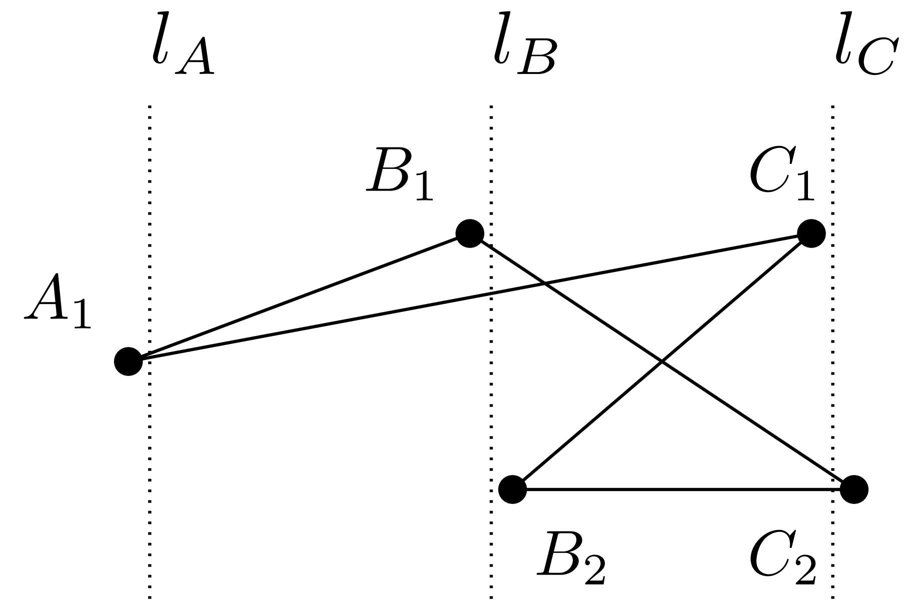


perturbation range



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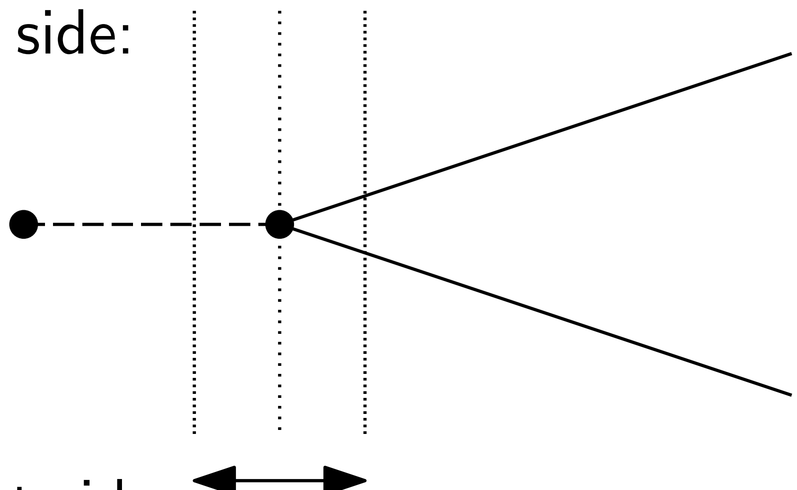

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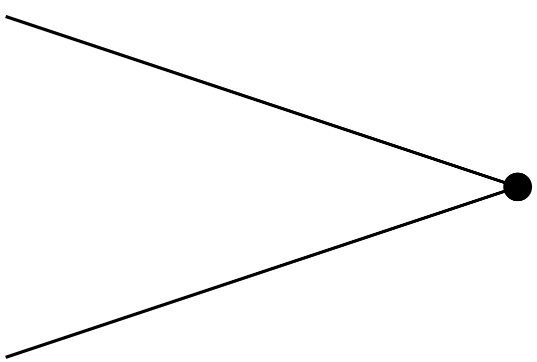

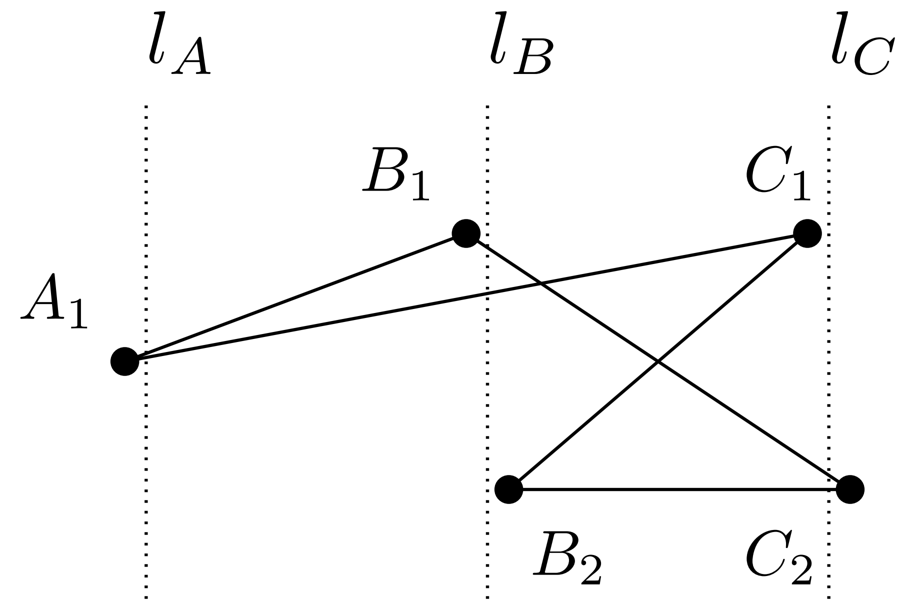

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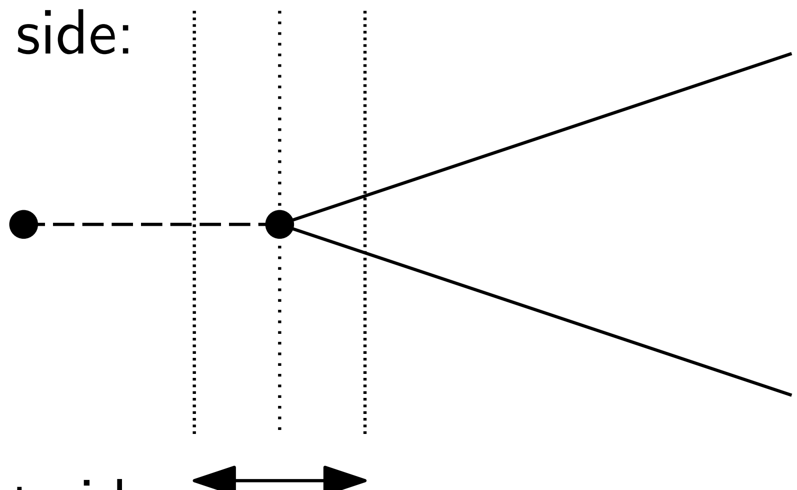

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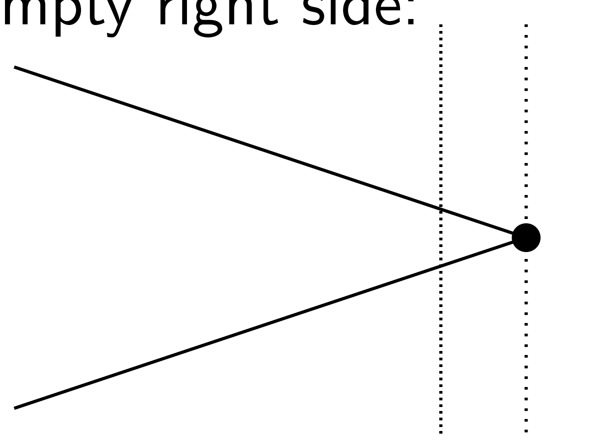

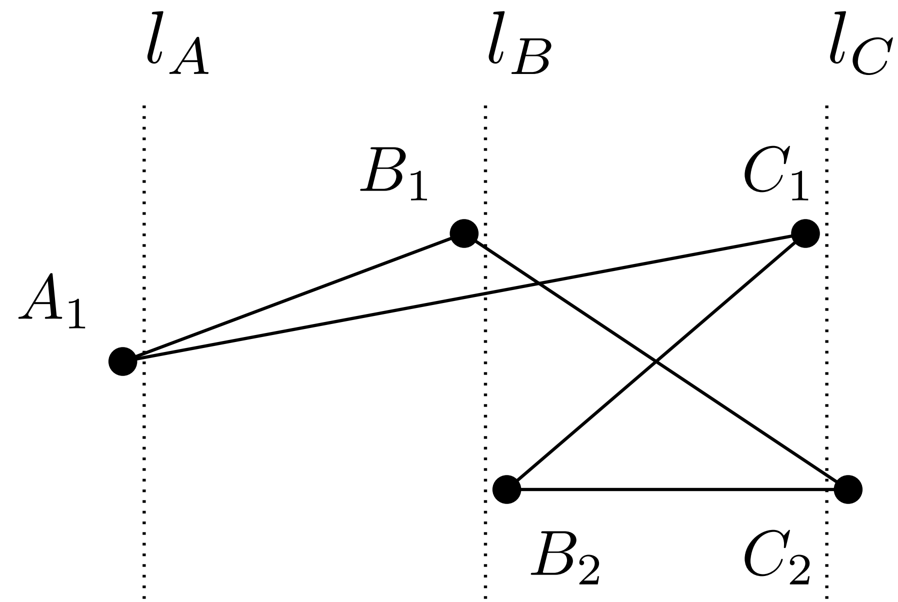

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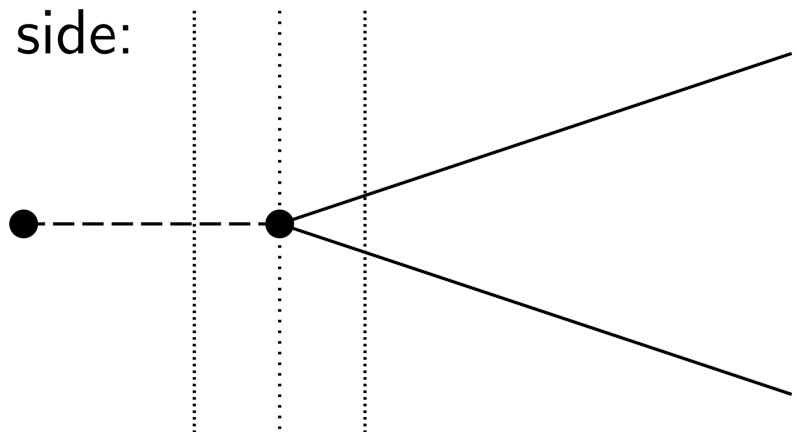

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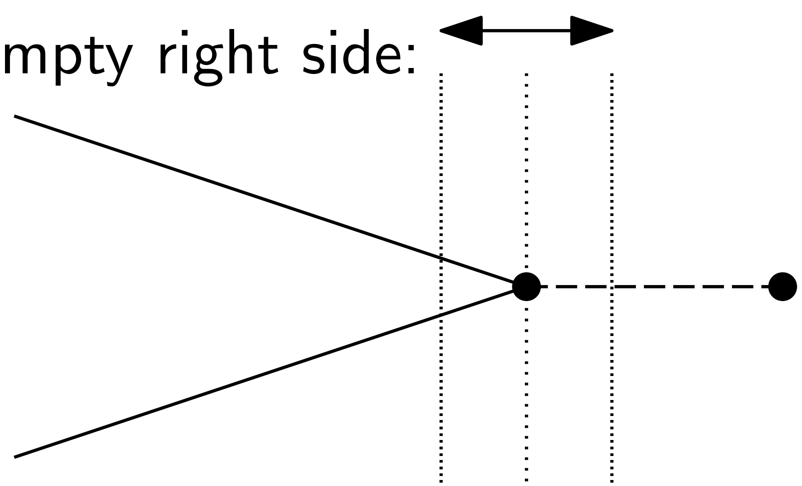

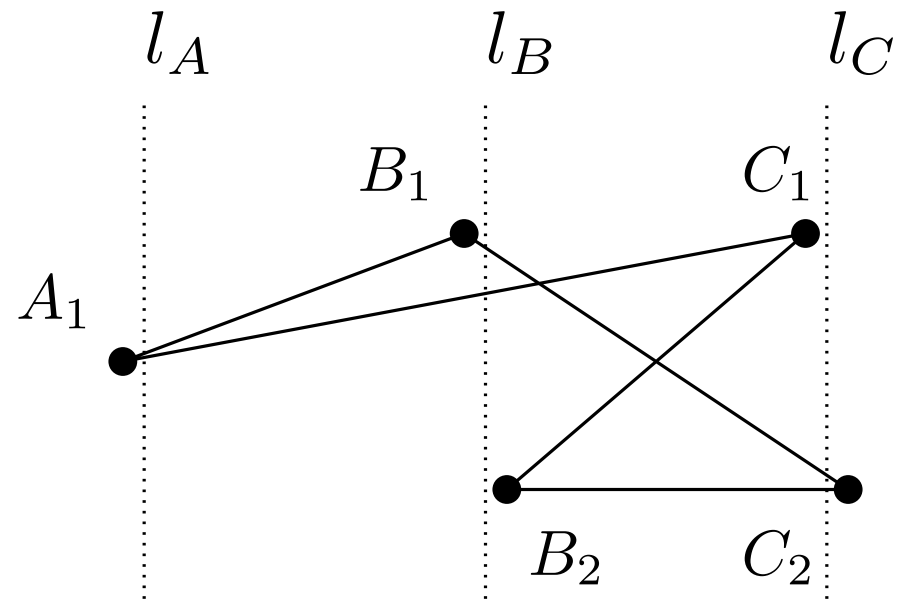

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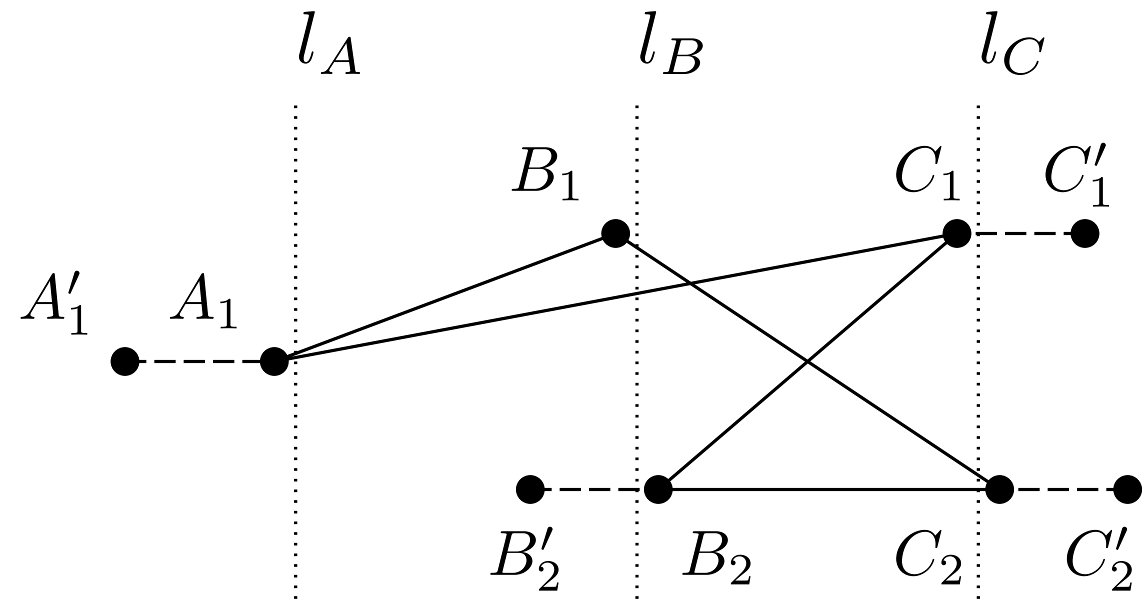

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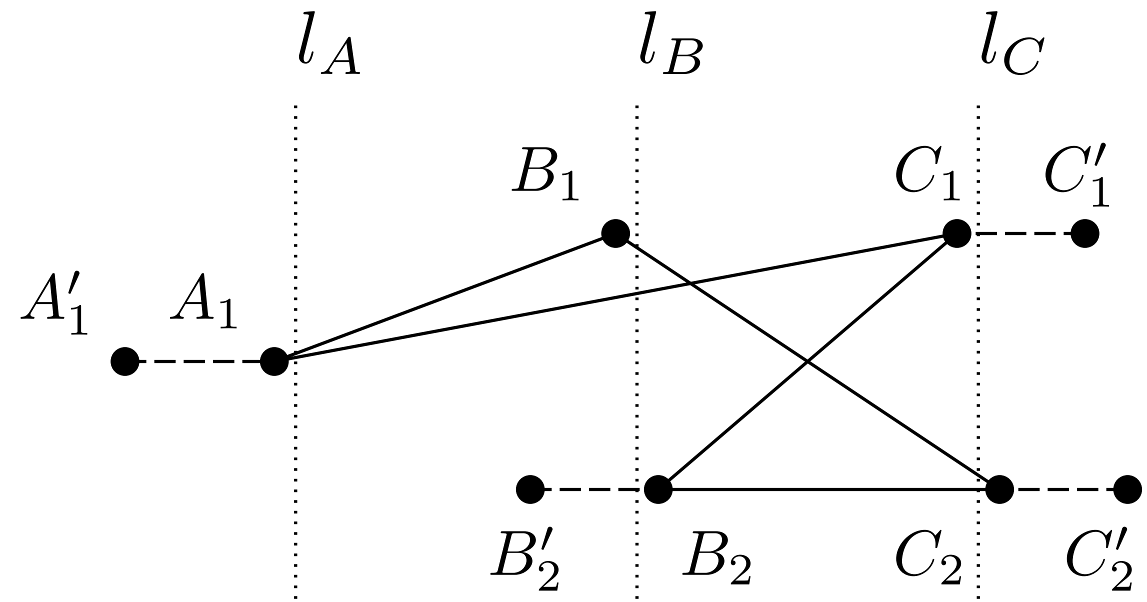

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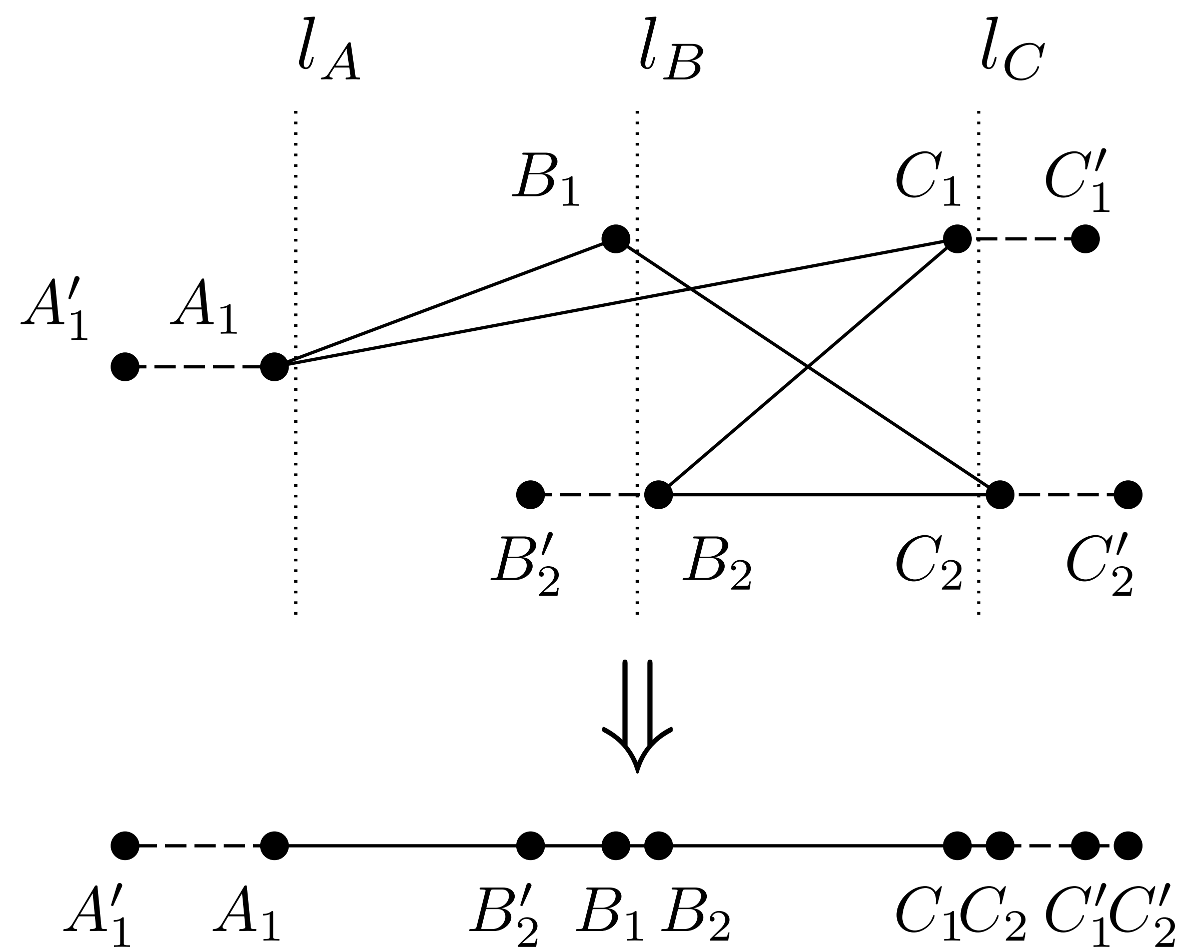

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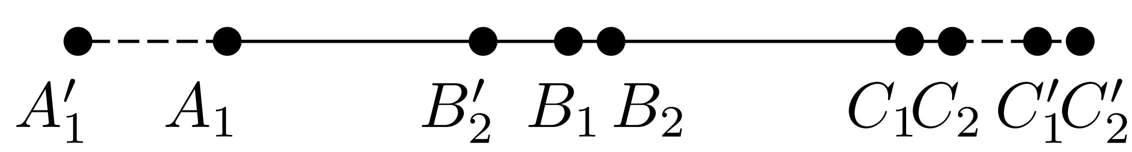

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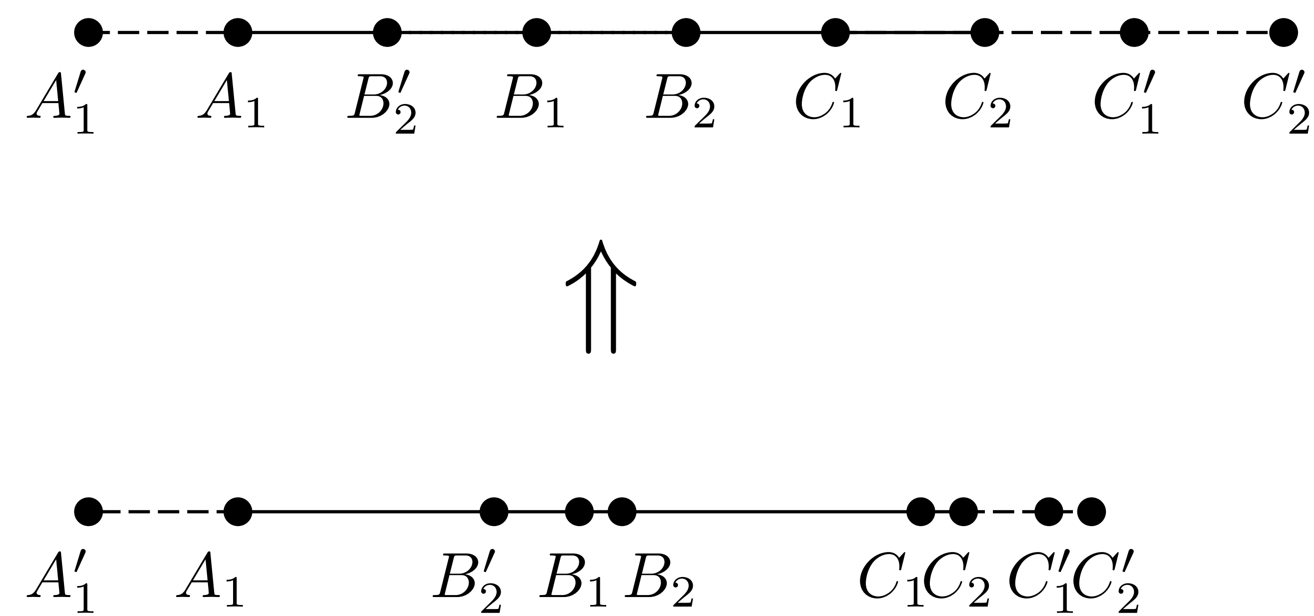

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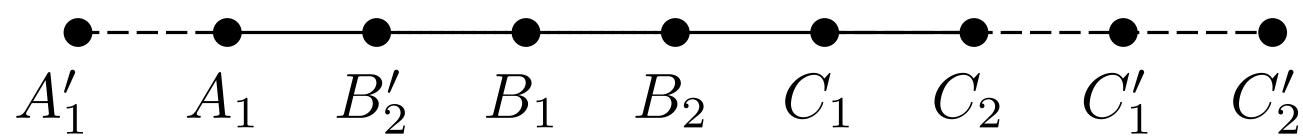

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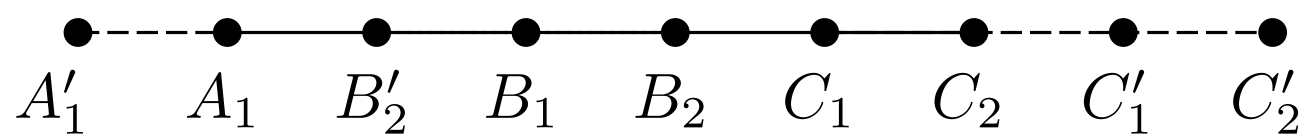

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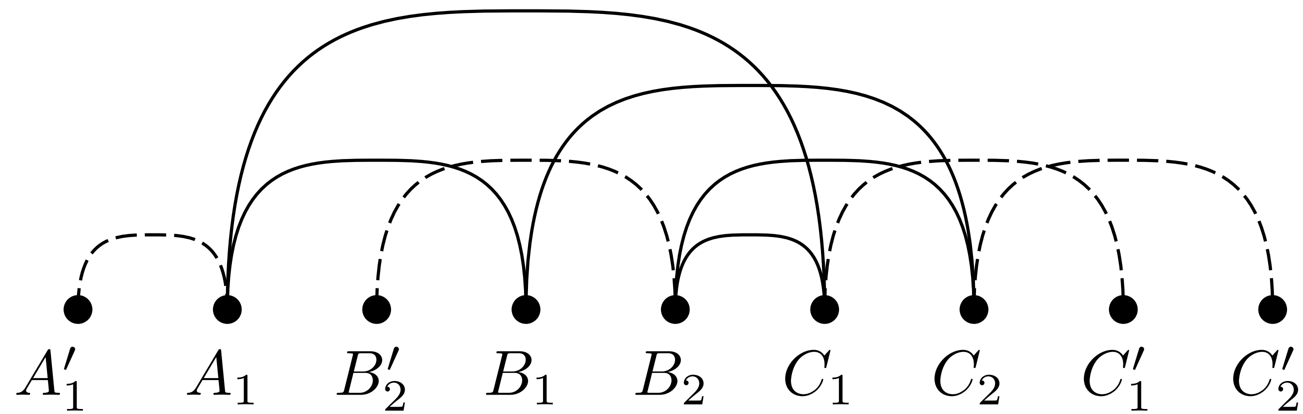

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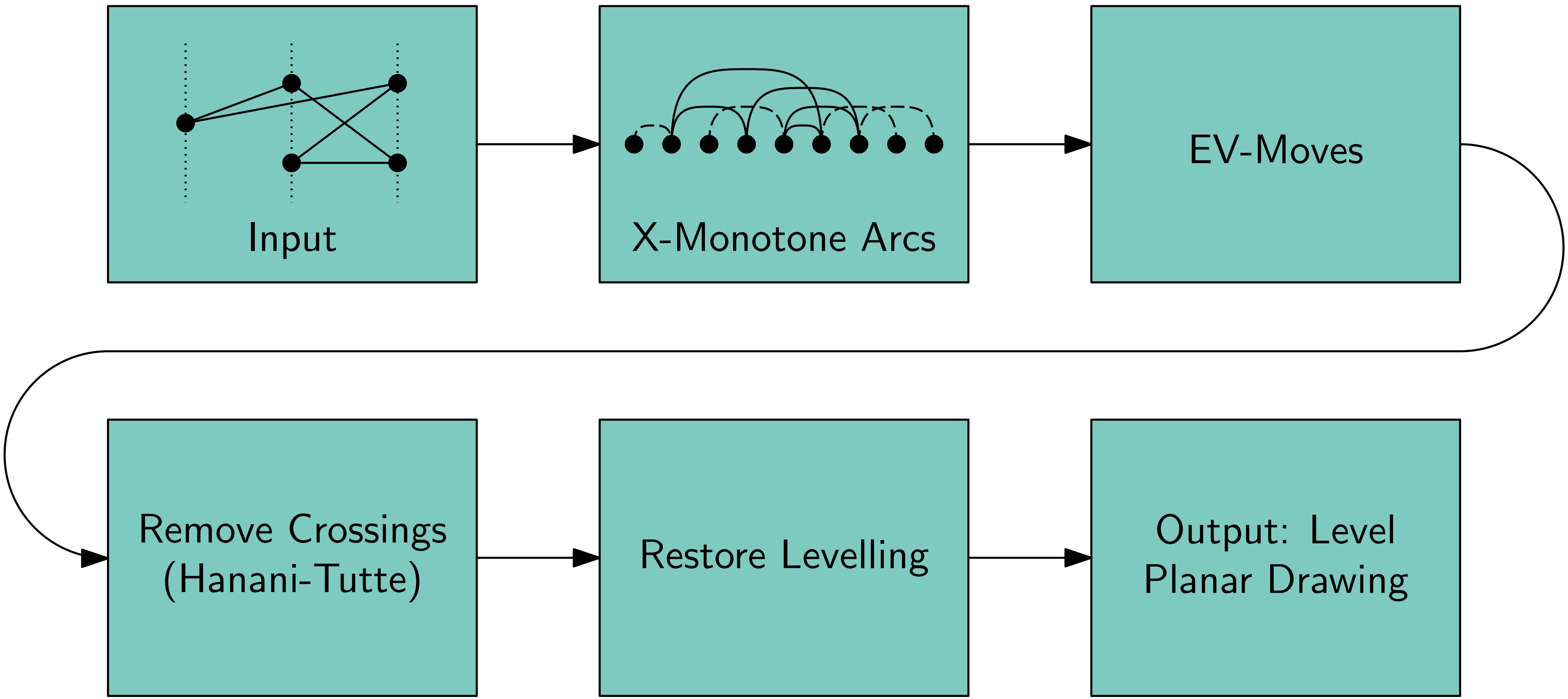

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- pairs of indepenent edges

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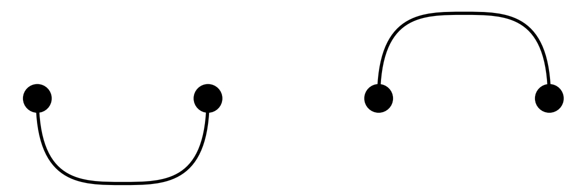

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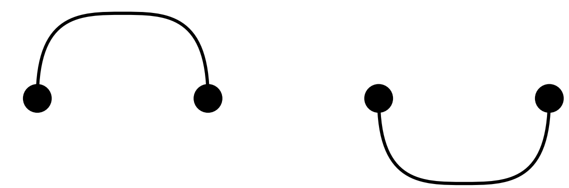

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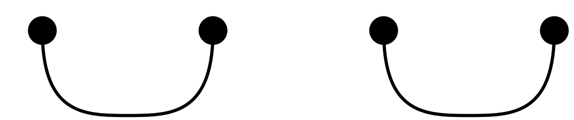

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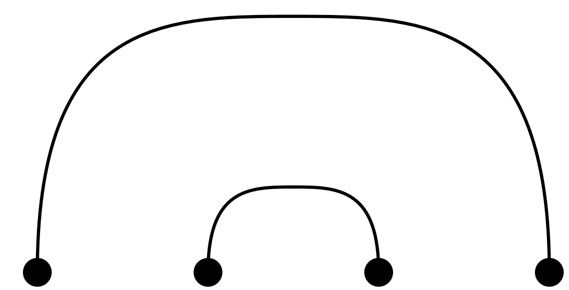

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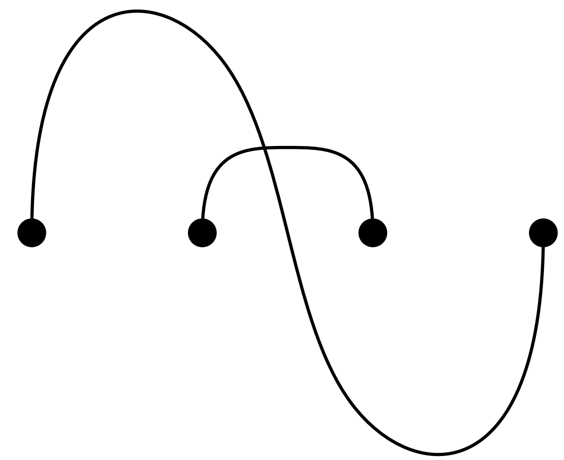

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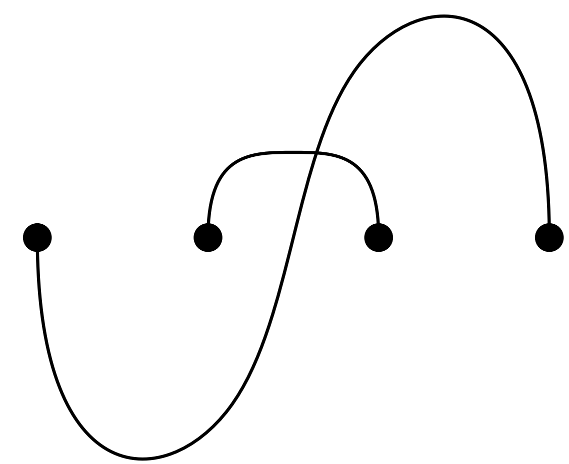

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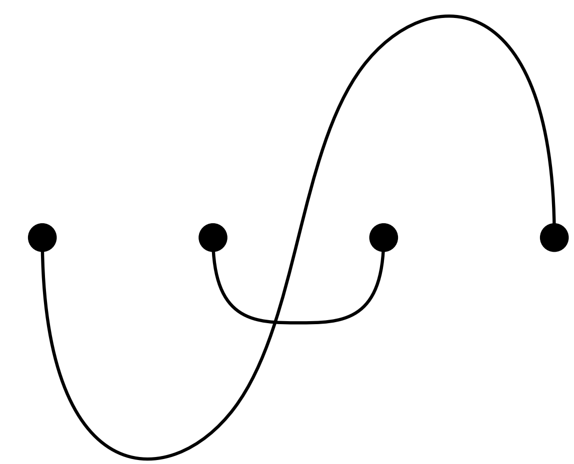
Case 0: Separated (irrelevant)

Case 1: Included

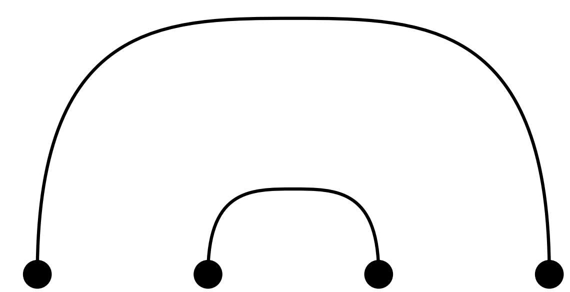


EV-Moves

- pairs of indepenent edges

Case 0: Separated (irrelevant)

Case 1: Included

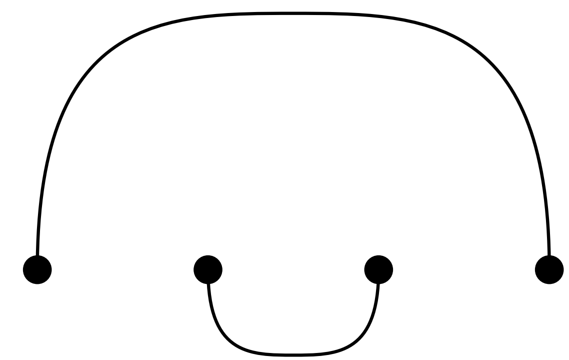


EV-Moves

- pairs of indepenent edges

Case 0: Separated (irrelevant)

Case 1: Included

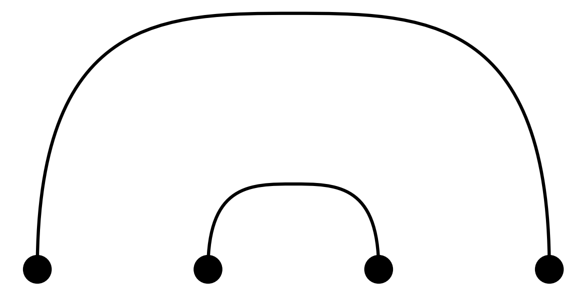


EV-Moves

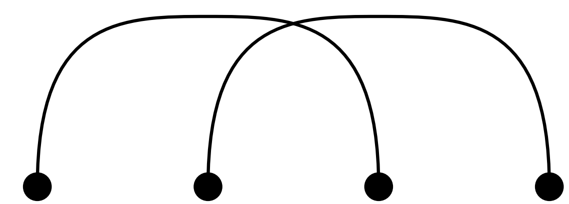
- pairs of indepenent edges

Case 0: Separated (irrelevant)

Case 1: Included



Case 2: Interleaved

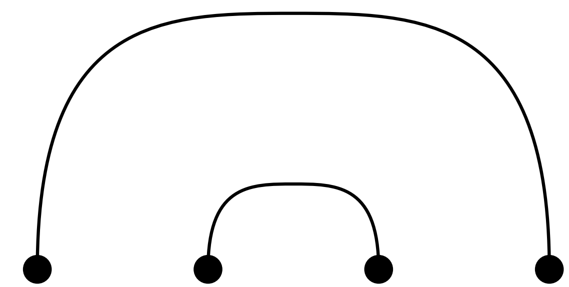


EV-Moves

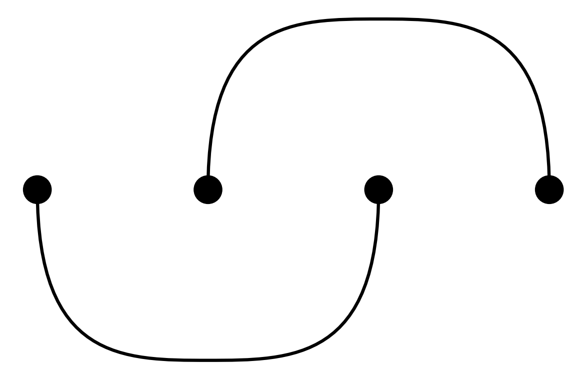
- pairs of indepenent edges

Case 0: Separated (irrelevant)

Case 1: Included



Case 2: Interleaved

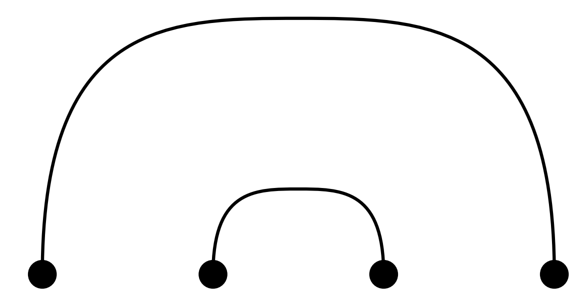


EV-Moves

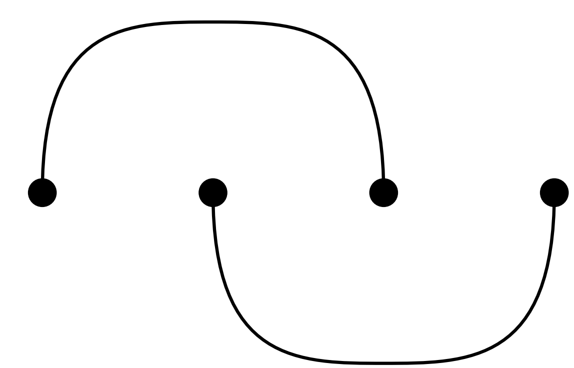
- pairs of indepenent edges

Case 0: Separated (irrelevant)

Case 1: Included



Case 2: Interleaved

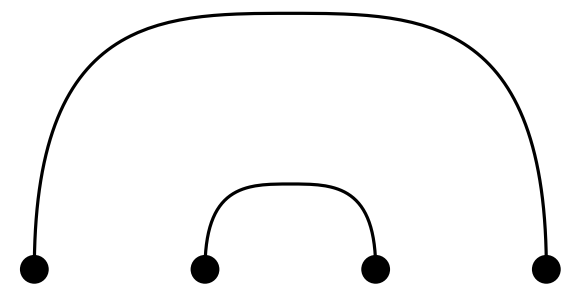


EV-Moves

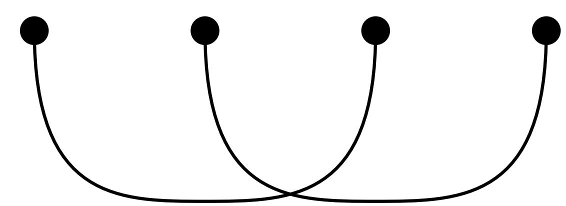
- pairs of indepenent edges

Case 0: Separated (irrelevant)

Case 1: Included

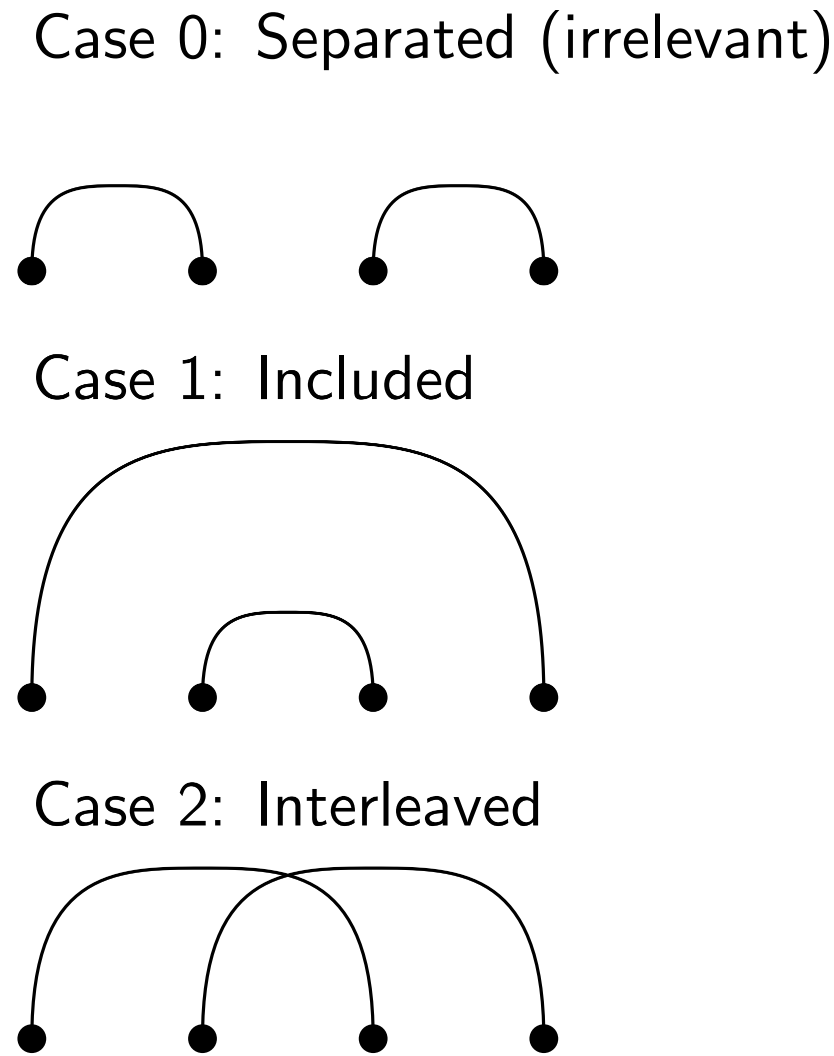


Case 2: Interleaved



EV-Moves

- pairs of indepenent edges
- case 1 - two possible ev-moves:
  - edge $\overline{TW}$ over vertex U
  - edge $\overline{TW}$ over vertex V

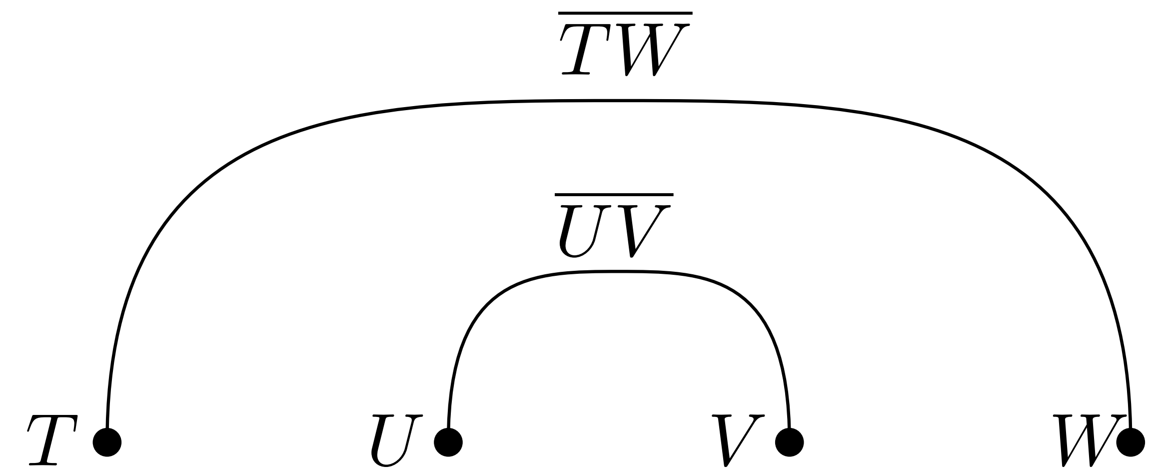


EV-Moves

- pairs of indepenent edges
- case 1 - two possible ev-moves:
  - edge $\overline{TW}$ over vertex U
  - edge $\overline{TW}$ over vertex V

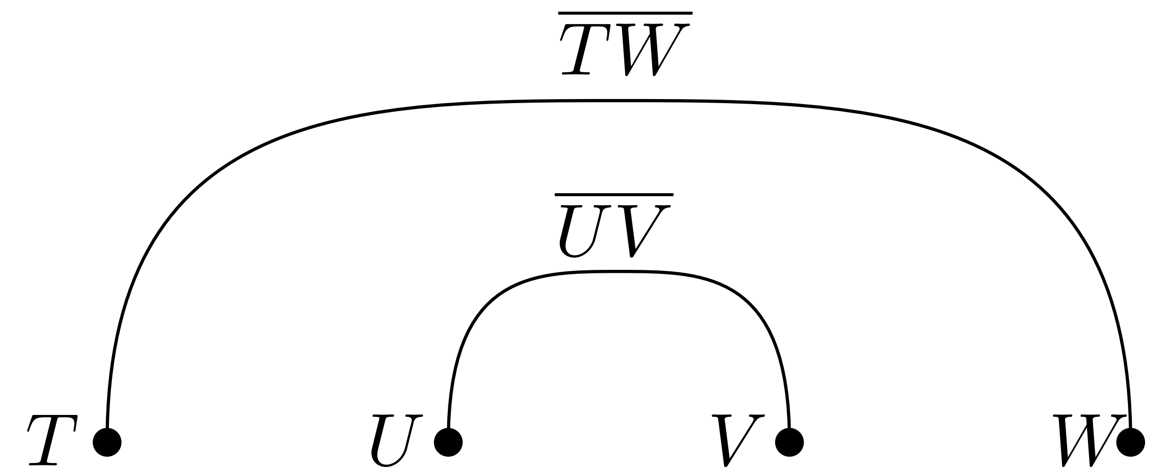
EV-Moves

- pairs of indepenent edges
- case 1 - two possible ev-moves:
  - edge $\overline{TW}$ over vertex U
  - edge $\overline{TW}$ over vertex V



EV-Moves

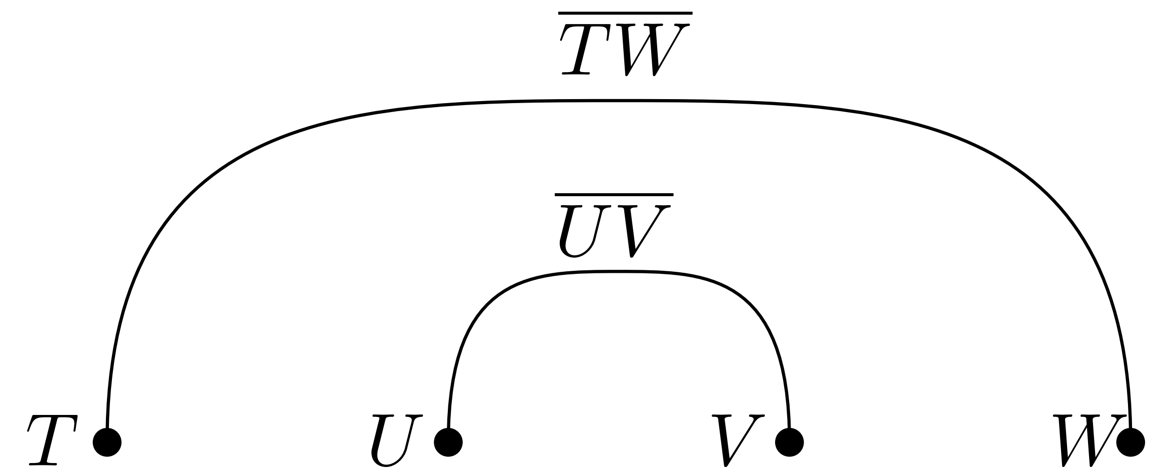
- pairs of indepenent edges
- case 1 - two possible ev-moves:
  - edge $\overline{TW}$ over vertex U
  - edge $\overline{TW}$ over vertex V
- x: defines ev-move



$$x_{\overline{TW},U} + x_{\overline{TW},V}$$

EV-Moves

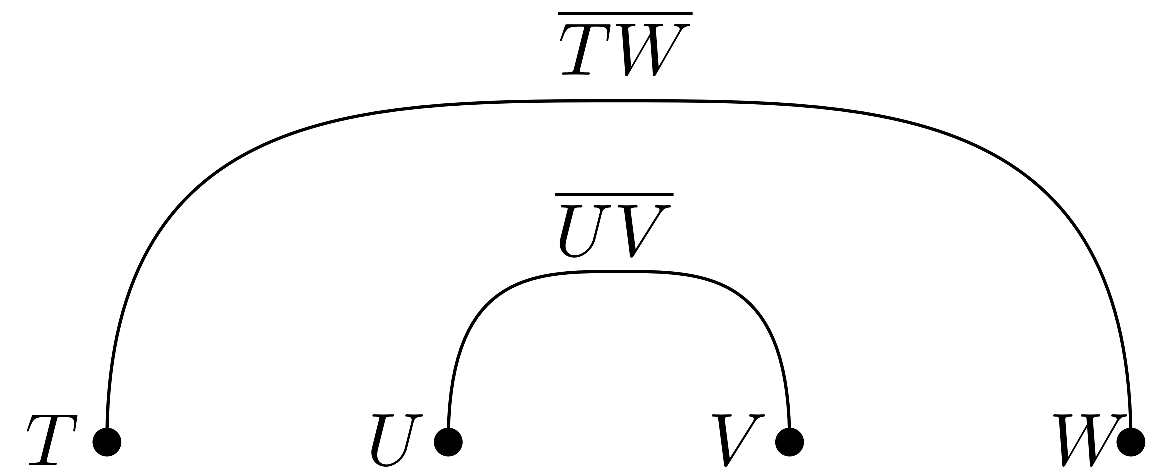
- pairs of indepenent edges
- case 1 - two possible ev-moves:
  - edge $\overline{TW}$ over vertex U
  - edge $\overline{TW}$ over vertex V
- x: defines ev-move
- i: number of crossings per pair



$$i_{\overline{TW},\overline{UV}} + x_{\overline{TW},U} + x_{\overline{TW},V}$$

EV-Moves

- pairs of indepenent edges
- case 1 - two possible ev-moves:
  - edge $\overline{TW}$ over vertex U
  - edge $\overline{TW}$ over vertex V
- x: defines ev-move
- i: number of crossings per pair
- force even parity



$$(i_{\overline{TW},\overline{UV}} + x_{\overline{TW},U} + x_{\overline{TW},V}) \mod 2 \stackrel{!}{=} 0$$

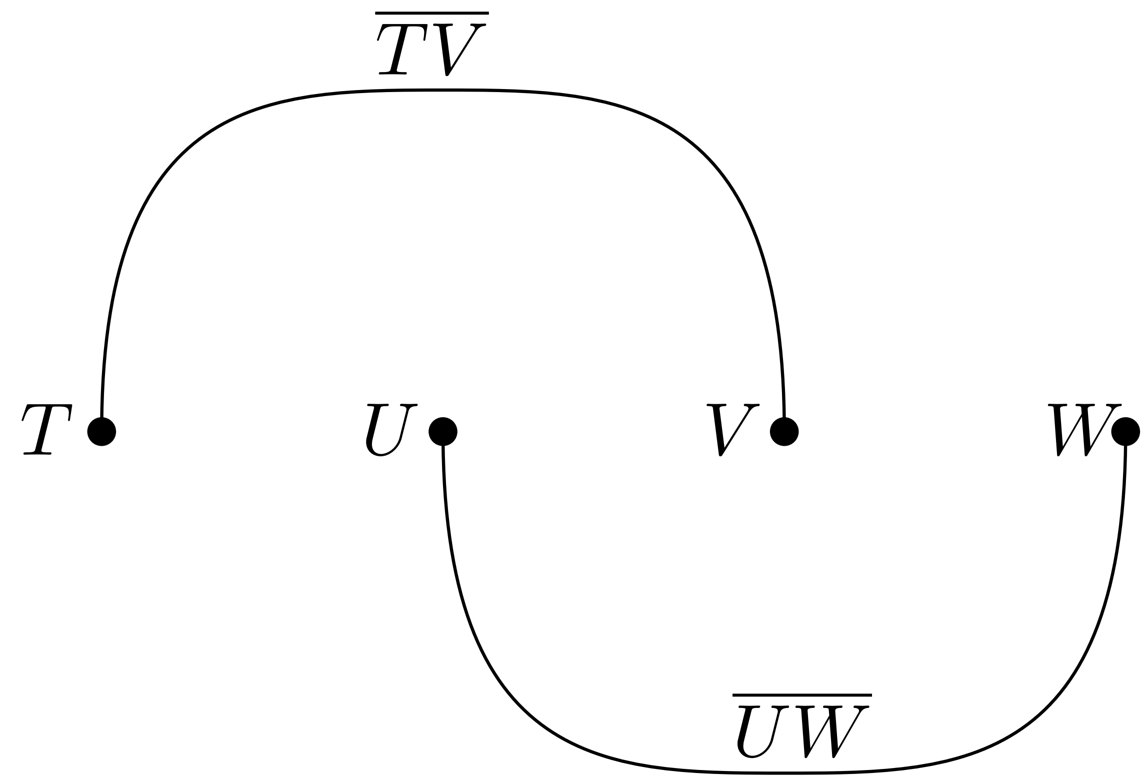
EV-Moves

- pairs of indepenent edges
- case 1 - two possible ev-moves:
  - edge $\overline{TW}$ over vertex U
  - edge $\overline{TW}$ over vertex V
- x: defines ev-move
- i: number of crossings per pair
- force even parity

$$(i_{\overline{TW},\overline{UV}} + x_{\overline{TW},U} + x_{\overline{TW},V}) \mod 2 \stackrel{!}{=} 0$$

EV-Moves

- pairs of indepenent edges
- case 1 - two possible ev-moves:
  - edge $\overline{TW}$ over vertex U
  - edge $\overline{TW}$ over vertex V
- x: defines ev-move
- i: number of crossings per pair
- force even parity
- case 2 analog:
  - edge $\overline{TV}$ over vertex U
  - edge $\overline{UW}$ over vertex V



$$\left(i_{\overline{TV},\overline{UW}} + x_{\overline{TV},U} + x_{\overline{UW},V}\right) \mod 2 \stackrel{!}{=} 0$$

EV-Moves

- pairs of indepenent edges
- case 1 - two possible ev-moves:
  - edge $\overline{TW}$ over vertex U
  - edge $\overline{TW}$ over vertex V
- x: defines ev-move
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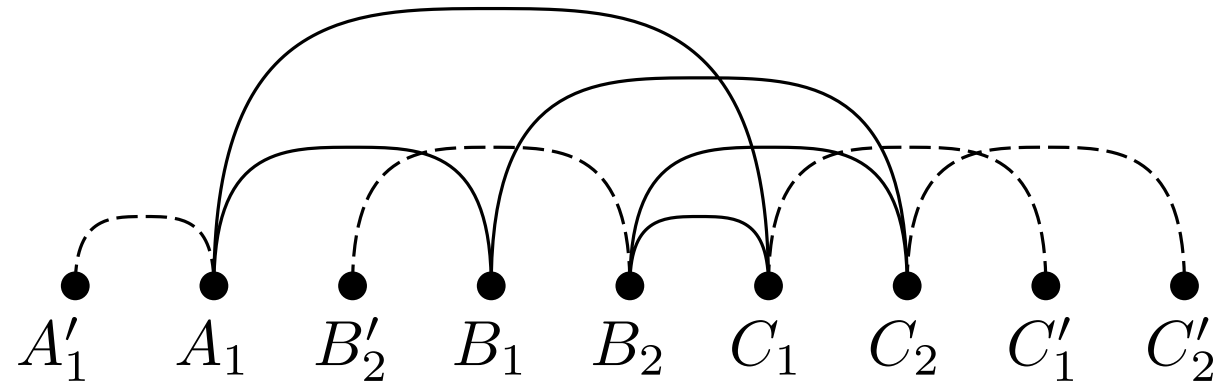
- case 2 analog:
  - edge $\overline{TV}$ over vertex U
  - edge $\overline{UW}$ over vertex V

$$(i_{\overline{TV},\overline{UW}} + x_{\overline{TV},U} + x_{\overline{UW},V}) \mod 2 \stackrel{!}{=} 0$$

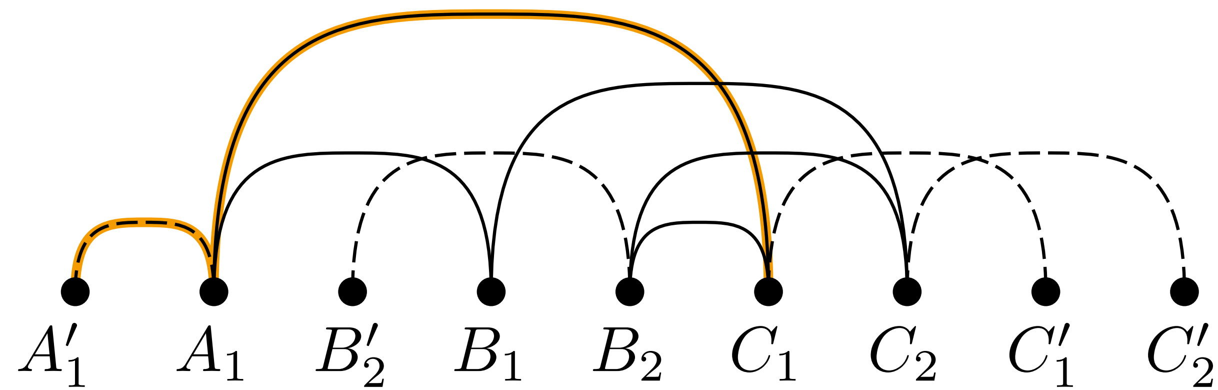
EV-Moves



- pairs of indepenent edges
- case 1 - two possible ev-moves:
  - edge $\overline{TW}$ over vertex U
  - edge $\overline{TW}$ over vertex V
- x: defines ev-move
- i: number of crossings per pair
- force even parity
- case 2 analog:
  - edge $\overline{TV}$ over vertex U
  - edge $\overline{UW}$ over vertex V

$$(i_{\overline{TV},\overline{UW}} + x_{\overline{TV},U} + x_{\overline{UW},V}) \mod 2 \stackrel{!}{=} 0$$

EV-Moves



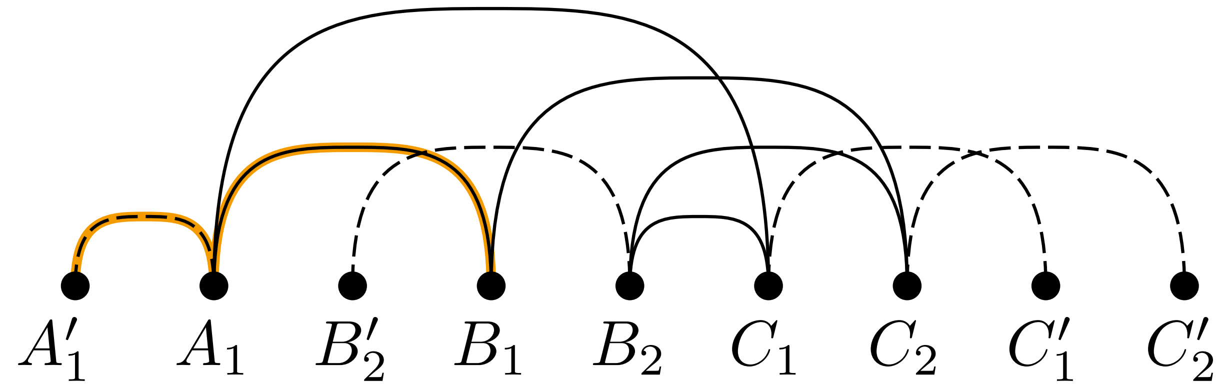
$\rightarrow$ not independent

- pairs of indepenent edges
- case 1 - two possible ev-moves:
  - edge $\overline{TW}$ over vertex U
  - edge $\overline{TW}$ over vertex V
- x: defines ev-move
- i: number of crossings per pair
- force even parity
- case 2 analog:
  - edge $\overline{TV}$ over vertex U
  - edge $\overline{UW}$ over vertex V

$$(i_{\overline{TV},\overline{UW}} + x_{\overline{TV},U} + x_{\overline{UW},V}) \mod 2 \overset{!}{=} 0$$

EV-Moves

- pairs of indepenent edges
- case 1 - two possible ev-moves:
  - edge $\overline{TW}$ over vertex U
  - edge $\overline{TW}$ over vertex V
- x: defines ev-move
- i: number of crossings per pair
- force even parity
- case 2 analog:
  - edge $\overline{TV}$ over vertex U
  - edge $\overline{UW}$ over vertex V

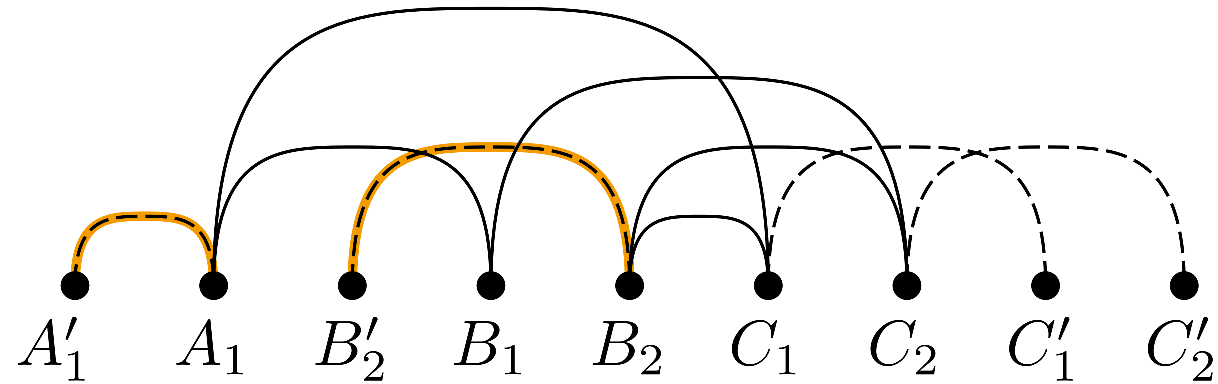


$\rightarrow$ not independent

$$(i_{\overline{TV},\overline{UW}} + x_{\overline{TV},U} + x_{\overline{UW},V}) \mod 2 \stackrel{!}{=} 0$$

EV-Moves

- pairs of indepenent edges
- case 1 - two possible ev-moves:
  - edge $\overline{TW}$ over vertex U
  - edge $\overline{TW}$ over vertex V
- x: defines ev-move
- i: number of crossings per pair
- force even parity
- case 2 analog:
  - edge $\overline{TV}$ over vertex U
  - edge $\overline{UW}$ over vertex V



$\rightarrow$ not interleaved

$$(i_{\overline{TV},\overline{UW}} + x_{\overline{TV},U} + x_{\overline{UW},V}) \mod 2 \overset{!}{=} 0$$

EV-Moves

- pairs of indepenent edges
- case 1 - two possible ev-moves:
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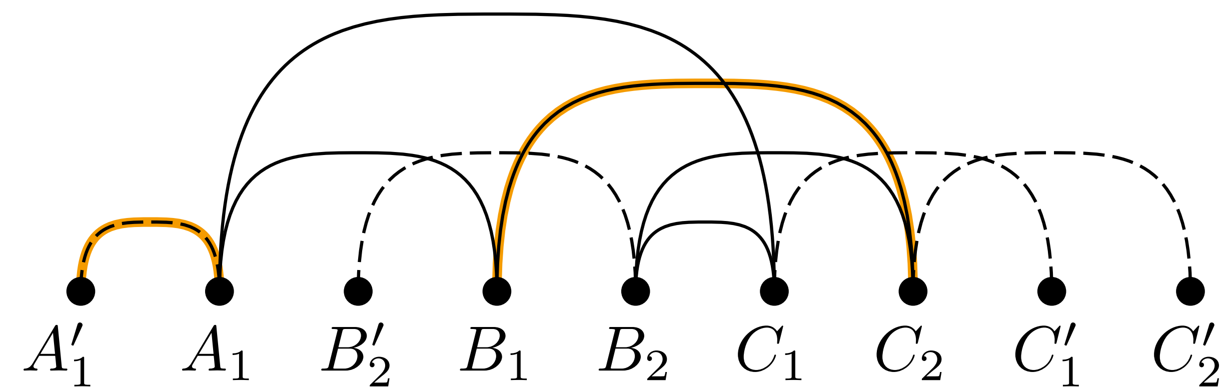
  - edge $\overline{TW}$ over vertex V
- x: defines ev-move
- i: number of crossings per pair
- force even parity
- case 2 analog:
  - edge $\overline{TV}$ over vertex U
  - edge $\overline{UW}$ over vertex V



$\rightarrow$ not interleaved

$$(i_{\overline{TV},\overline{UW}} + x_{\overline{TV},U} + x_{\overline{UW},V}) \mod 2 \overset{!}{=} 0$$

EV-Moves

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  - edge $\overline{TW}$ over vertex V
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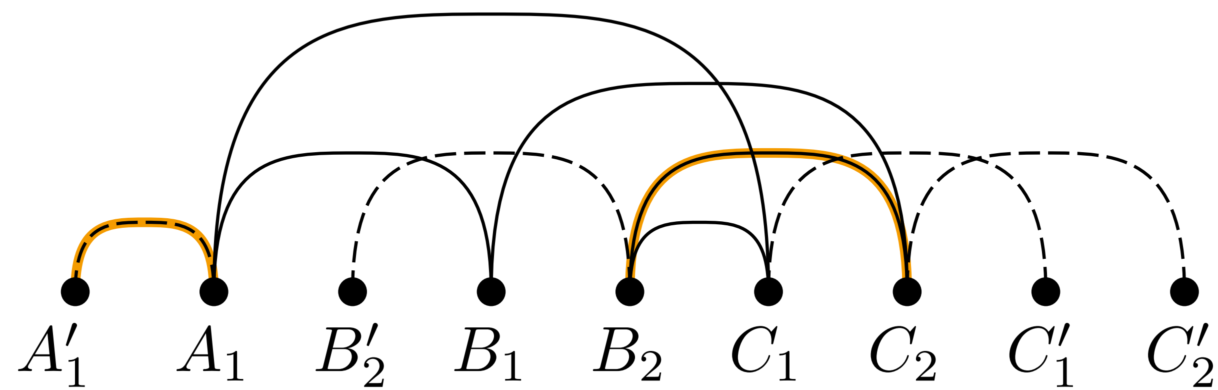
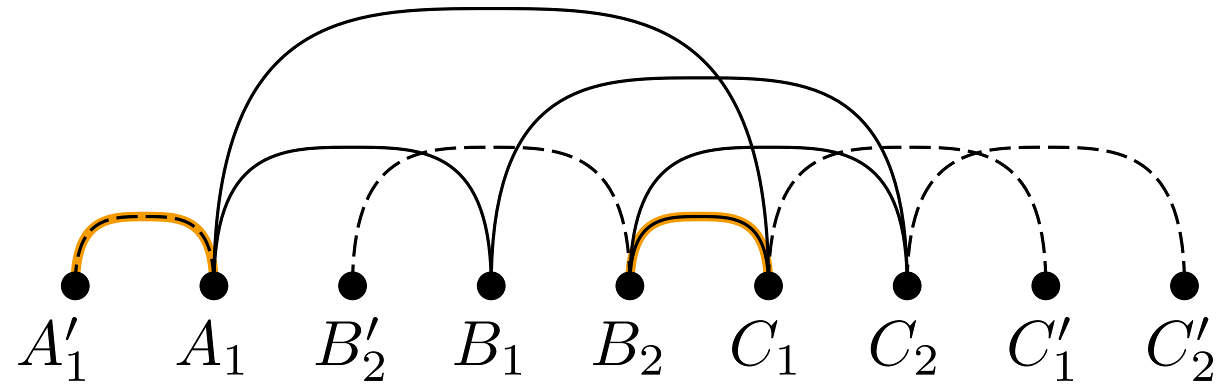
- i: number of crossings per pair
- force even parity
- case 2 analog:
  - edge $\overline{TV}$ over vertex U
  - edge $\overline{UW}$ over vertex V



$\rightarrow$ not interleaved

$$(i_{\overline{TV},\overline{UW}} + x_{\overline{TV},U} + x_{\overline{UW},V}) \mod 2 \overset{!}{=} 0$$

EV-Moves



$\rightarrow$ not interleaved

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  - edge $\overline{TW}$ over vertex U
  - edge $\overline{TW}$ over vertex V
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- case 2 analog:
  - edge $\overline{TV}$ over vertex U
  - edge $\overline{UW}$ over vertex V

$$(i_{\overline{TV},\overline{UW}} + x_{\overline{TV},U} + x_{\overline{UW},V}) \mod 2 \overset{!}{=} 0$$

EV-Moves

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  - edge $\overline{TV}$ over vertex U
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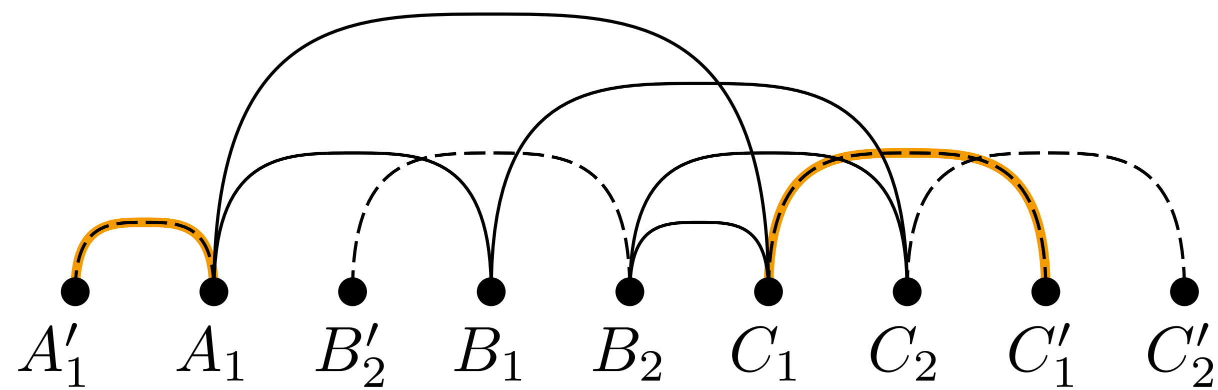

$\rightarrow$ not interleaved

$$(i_{\overline{TV},\overline{UW}} + x_{\overline{TV},U} + x_{\overline{UW},V}) \mod 2 \stackrel{!}{=} 0$$

EV-Moves

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- case 2 analog:
  - edge $\overline{TV}$ over vertex U
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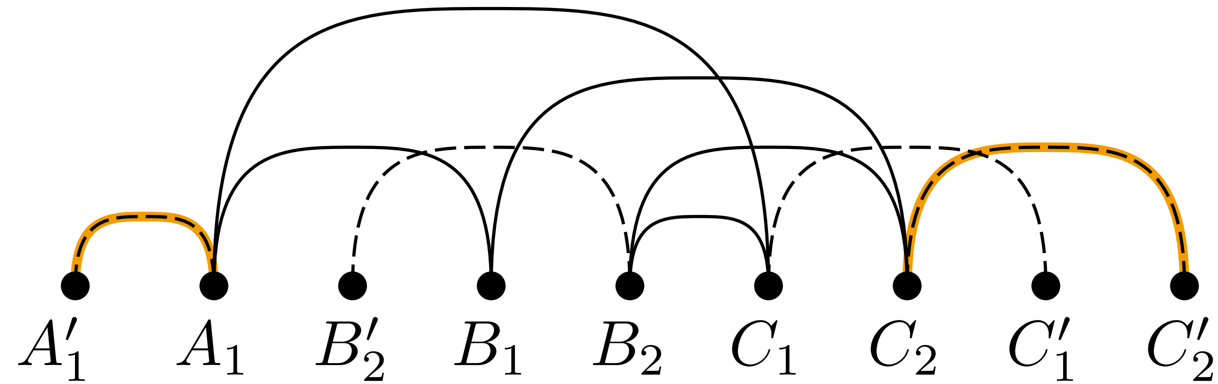
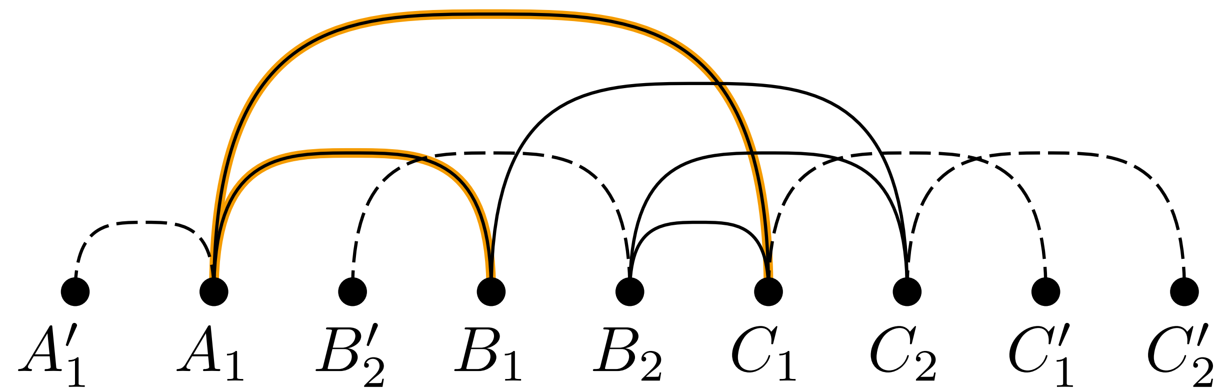


$\rightarrow$ not interleaved

$$(i_{\overline{TV},\overline{UW}} + x_{\overline{TV},U} + x_{\overline{UW},V}) \mod 2 \stackrel{!}{=} 0$$

EV-Moves

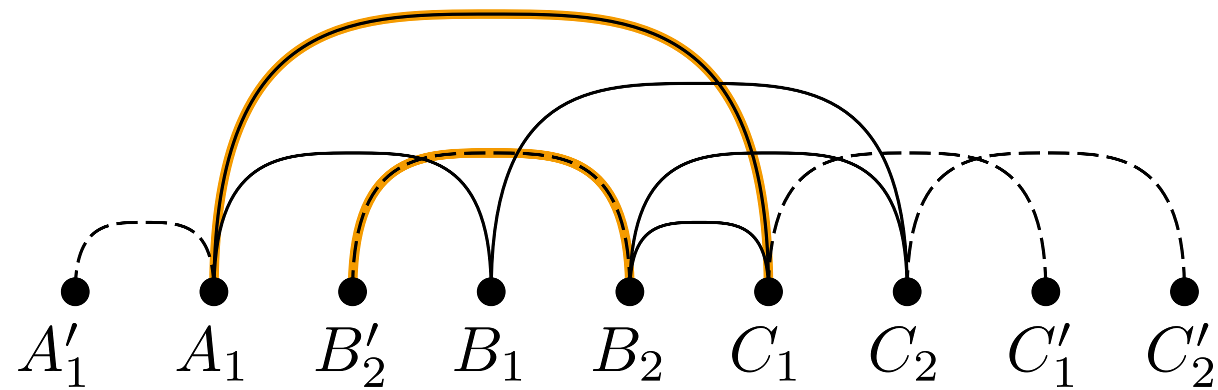


$\rightarrow$ not independent

- pairs of indepenent edges
- case 1 - two possible ev-moves:
  - edge $\overline{TW}$ over vertex U
  - edge $\overline{TW}$ over vertex V
- x: defines ev-move
- i: number of crossings per pair
- force even parity
- case 2 analog:
  - edge $\overline{TV}$ over vertex U
  - edge $\overline{UW}$ over vertex V

$$(i_{\overline{TV},\overline{UW}} + x_{\overline{TV},U} + x_{\overline{UW},V}) \mod 2 \stackrel{!}{=} 0$$

EV-Moves



$\rightarrow$ add equation

$$(0 + x_{\overline{A_1C_1},B'_2} + x_{\overline{A_1C_1},B_2})\bmod 2 = 0$$

- pairs of indepenent edges
- case 1 - two possible ev-moves:
  - edge $\overline{TW}$ over vertex U
  - edge $\overline{TW}$ over vertex V
- x: defines ev-move
- i: number of crossings per pair
- force even parity
- case 2 analog:
  - edge $\overline{TV}$ over vertex U
  - edge $\overline{UW}$ over vertex V

$$(i_{\overline{TW},\overline{UV}} + x_{\overline{TW},U} + x_{\overline{TW},V}) \mod 2 \stackrel{!}{=} 0$$

EV-Moves

- pairs of indepenent edges
- case 1 - two possible ev-moves:
  - edge $\overline{TW}$ over vertex U
  - edge $\overline{TW}$ over vertex V
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  - edge $\overline{TV}$ over vertex U
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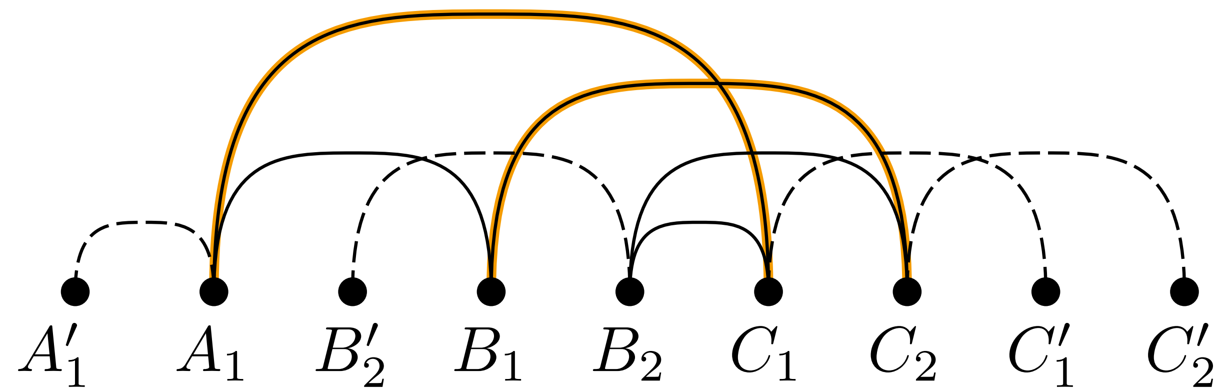

$\rightarrow$ add equation

$$(0 + x_{\overline{A_1C_1},B'_2} + x_{\overline{A_1C_1},B_2})\bmod 2 = 0$$

$$(1 + x_{\overline{A_1C_1},B_1} + x_{\overline{B_1C_2},C_1})\bmod 2 = 0$$

$$(i_{\overline{TV},\overline{UW}} + x_{\overline{TV},U} + x_{\overline{UW},V}) \mod 2 \overset{!}{=} 0$$

EV-Moves

- pairs of indepenent edges
- case 1 - two possible ev-moves:
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  - edge $\overline{TW}$ over vertex V
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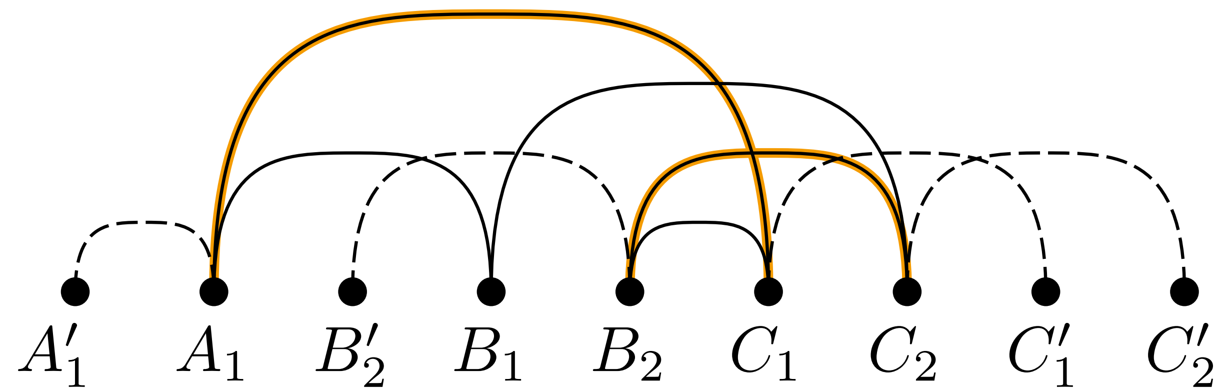

$\rightarrow$ add equation

$$(0 + x_{\overline{A_1C_1},B'_2} + x_{\overline{A_1C_1},B_2})\bmod 2 = 0$$
$$(1 + x_{\overline{A_1C_1},B_1} + x_{\overline{B_1C_2},C_1})\bmod 2 = 0$$
$$(1 + x_{\overline{A_1C_1},B_2} + x_{\overline{B_2C_2},C_1})\bmod 2 = 0$$

$$(i_{\overline{TV},\overline{UW}} + x_{\overline{TV},U} + x_{\overline{UW},V}) \mod 2 \stackrel{!}{=} 0$$

EV-Moves

- pairs of indepenent edges
- case 1 - two possible ev-moves:
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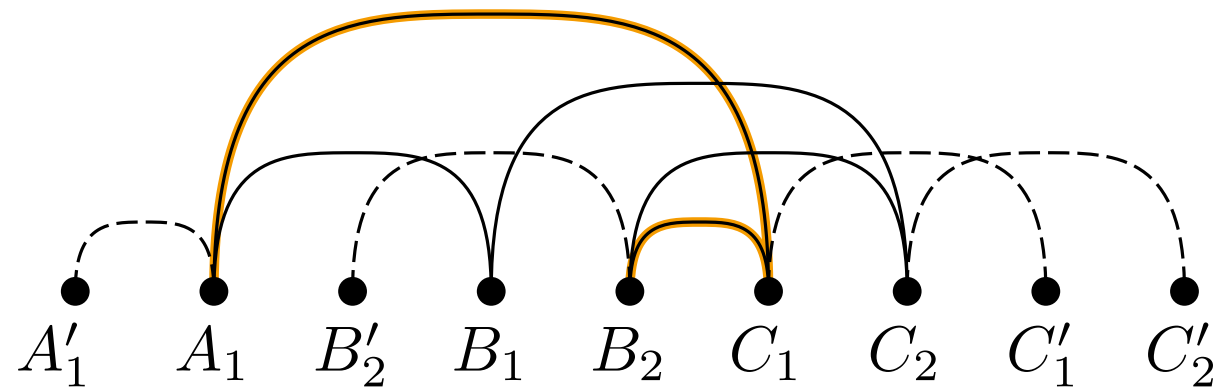

$\rightarrow$ not independent

$$(0+x_{\overline{A_1C_1},B'_2} + x_{\overline{A_1C_1},B_2})\bmod 2 = 0$$

$$(1+x_{\overline{A_1C_1},B_1} + x_{\overline{B_1C_2},C_1})\bmod 2 = 0$$

$$(1+x_{\overline{A_1C_1},B_2} + x_{\overline{B_2C_2},C_1})\bmod 2 = 0$$

$$(i_{\overline{TV},\overline{UW}} + x_{\overline{TV},U} + x_{\overline{UW},V}) \mod 2 \overset{!}{=} 0$$

EV-Moves

- pairs of indepenent edges
- case 1 - two possible ev-moves:
  - edge $\overline{TW}$ over vertex U
  - edge $\overline{TW}$ over vertex V
- x: defines ev-move
- i: number of crossings per pair
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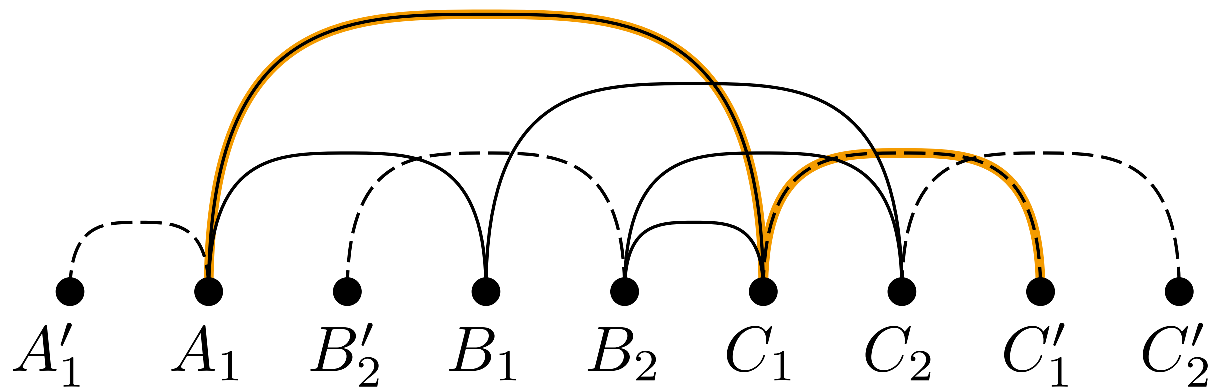
- case 2 analog:
  - edge $\overline{TV}$ over vertex U
  - edge $\overline{UW}$ over vertex V



$\rightarrow$ not independent

$$(0 + x_{\overline{A_1C_1},B'_2} + x_{\overline{A_1C_1},B_2})\bmod 2 = 0$$
$$(1 + x_{\overline{A_1C_1},B_1} + x_{\overline{B_1C_2},C_1})\bmod 2 = 0$$
$$(1 + x_{\overline{A_1C_1},B_2} + x_{\overline{B_2C_2},C_1})\bmod 2 = 0$$

$$(i_{\overline{TV},\overline{UW}} + x_{\overline{TV},U} + x_{\overline{UW},V}) \mod 2 \stackrel{!}{=} 0$$

EV-Moves

- pairs of indepenent edges
- case 1 - two possible ev-moves:
  - edge $\overline{TW}$ over vertex U
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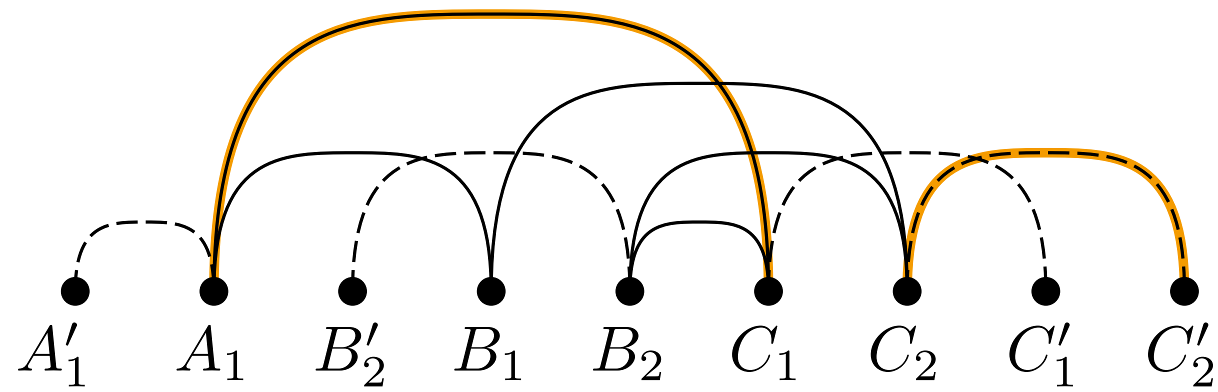
- x: defines ev-move
- i: number of crossings per pair
- force even parity
- case 2 analog:
  - edge $\overline{TV}$ over vertex U
  - edge $\overline{UW}$ over vertex V



$\rightarrow$ not interleaved

$$(0 + x_{\overline{A_1C_1},B_2'} + x_{\overline{A_1C_1},B_2})\bmod 2 = 0$$
$$(1 + x_{\overline{A_1C_1},B_1} + x_{\overline{B_1C_2},C_1})\bmod 2 = 0$$
$$(1 + x_{\overline{A_1C_1},B_2} + x_{\overline{B_2C_2},C_1})\bmod 2 = 0$$

$$(i_{\overline{TV},\overline{UW}} + x_{\overline{TV},U} + x_{\overline{UW},V}) \mod 2 \stackrel{!}{=} 0$$

EV-Moves

- pairs of indepenent edges
- case 1 - two possible ev-moves:
  - edge $\overline{TW}$ over vertex U
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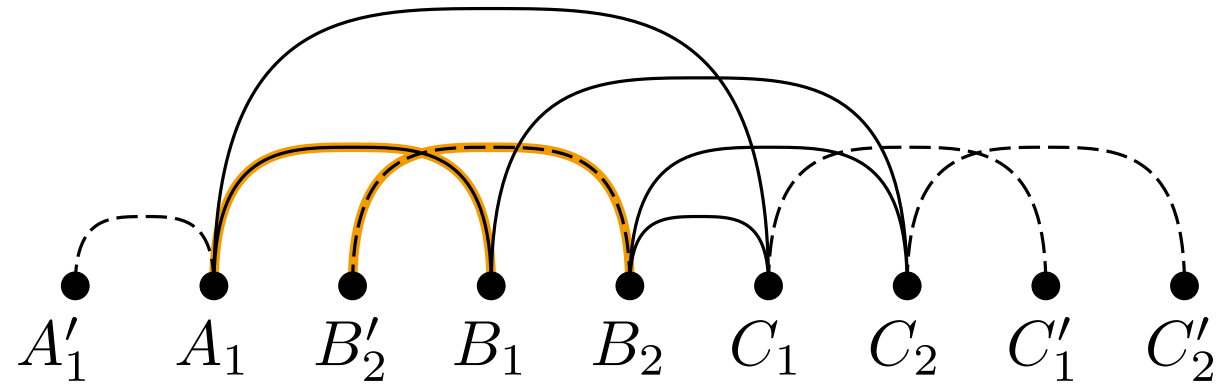

$\rightarrow$ add equation

$$(0 + x_{\overline{A_1C_1},B_2'} + x_{\overline{A_1C_1},B_2})\bmod 2 = 0$$
$$(1 + x_{\overline{A_1C_1},B_1} + x_{\overline{B_1C_2},C_1})\bmod 2 = 0$$
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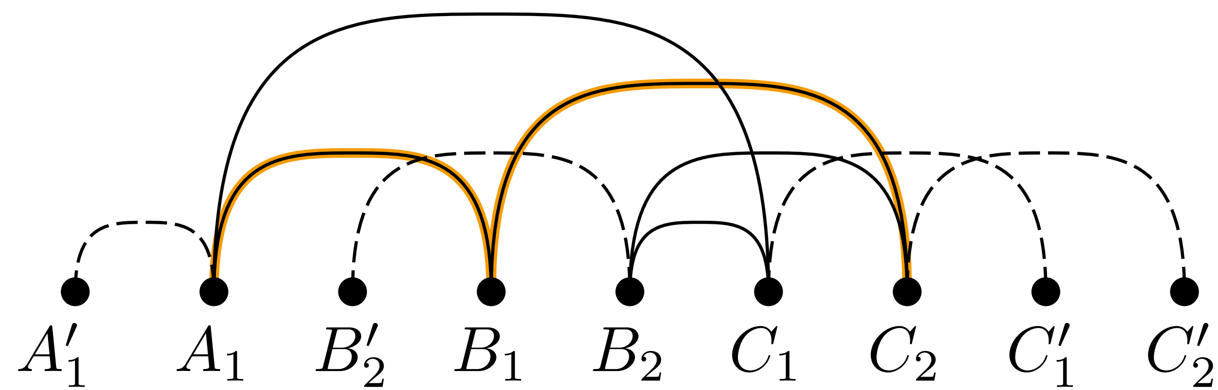

$\rightarrow$ not independent

$$(0 + x_{\overline{A_1C_1},B'_2} + x_{\overline{A_1C_1},B_2})\bmod 2 = 0$$
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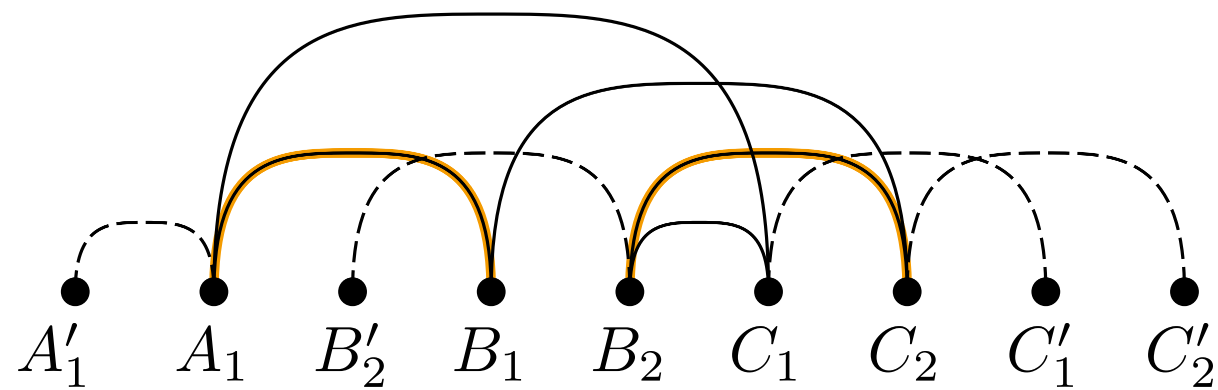

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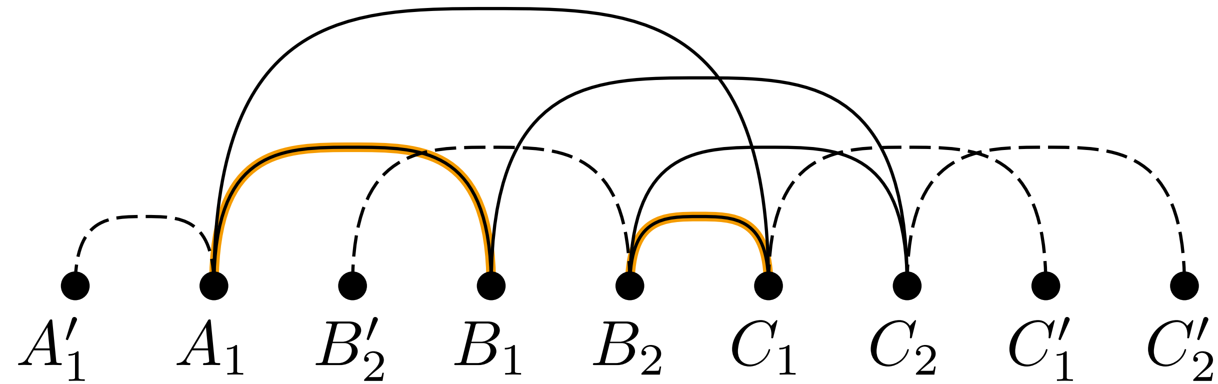

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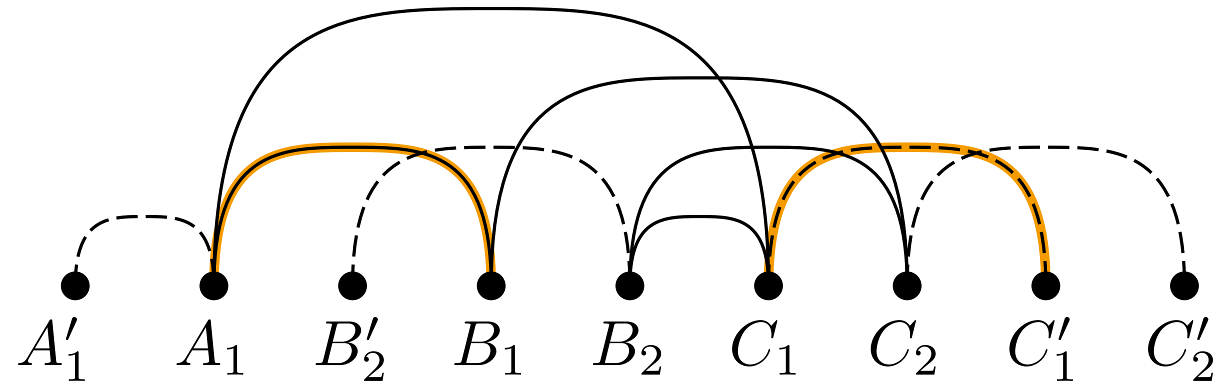

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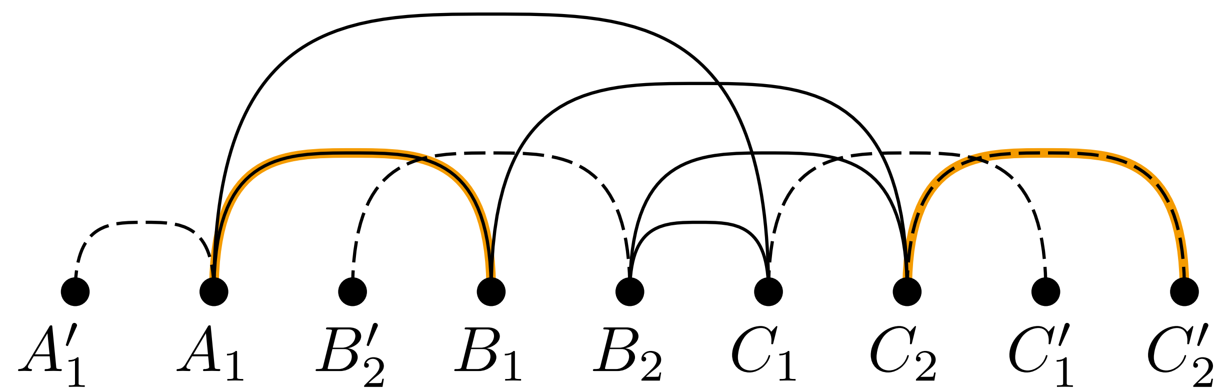

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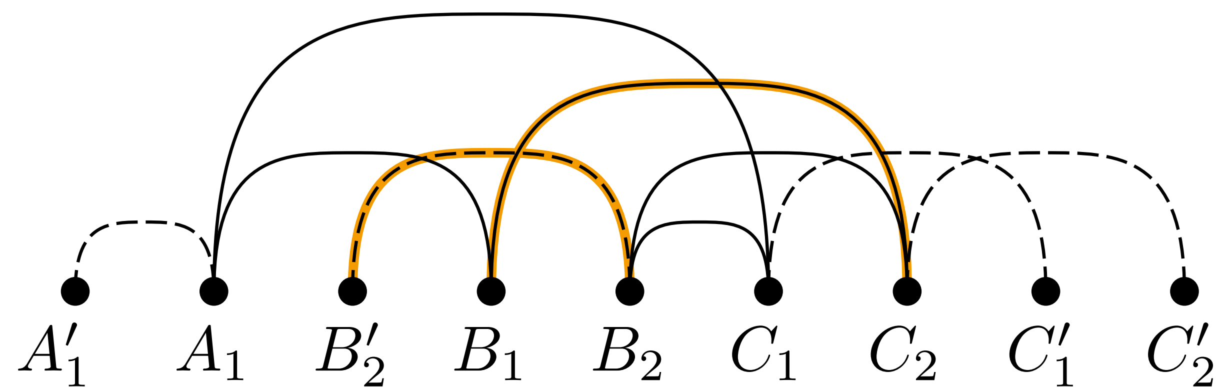

$\rightarrow$ add equation

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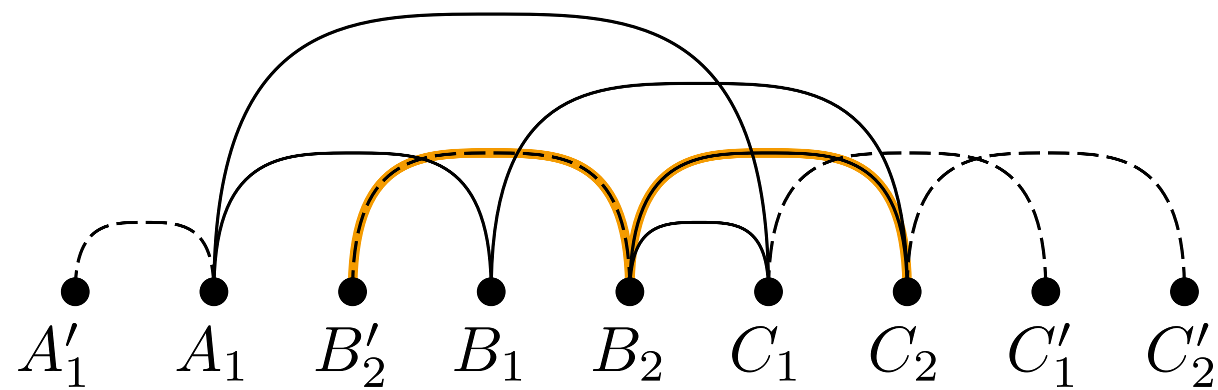

$\rightarrow$ not independent

$$(0 + x_{\overline{A_1C_1},B'_2} + x_{\overline{A_1C_1},B_2})\bmod 2 = 0$$
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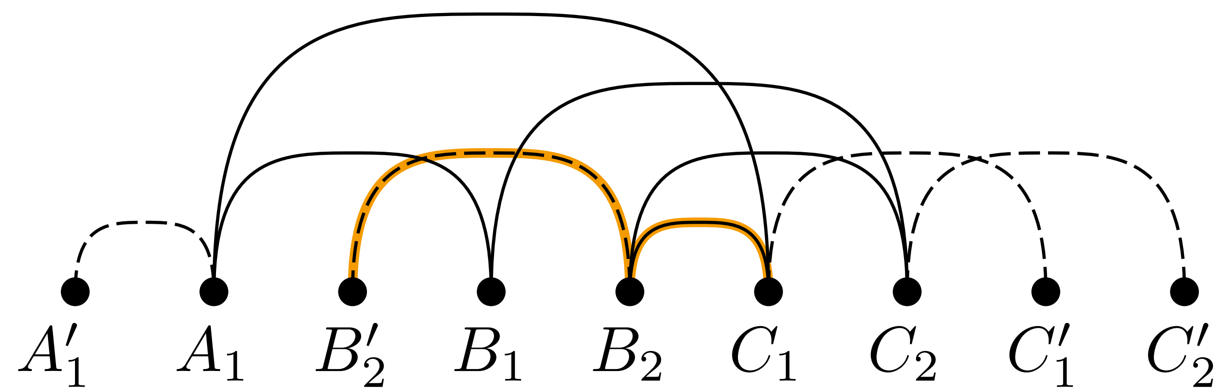

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$$(0+x_{\overline{A_1C_1},B'_2}+x_{\overline{A_1C_1},B_2})\bmod 2=0$$
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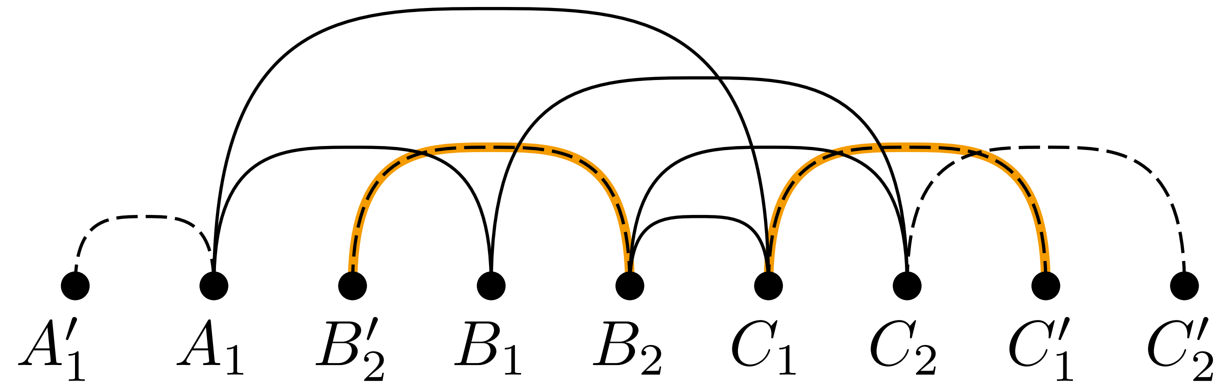

$\rightarrow$ not interleaved

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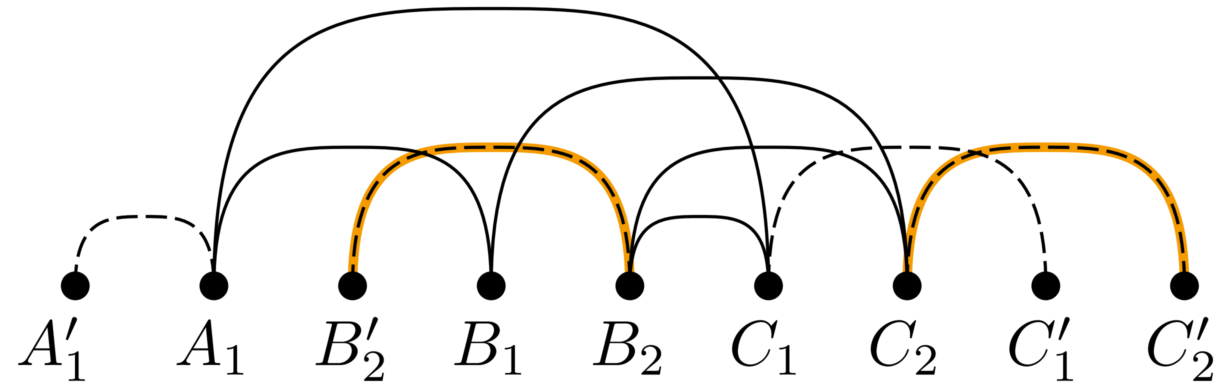

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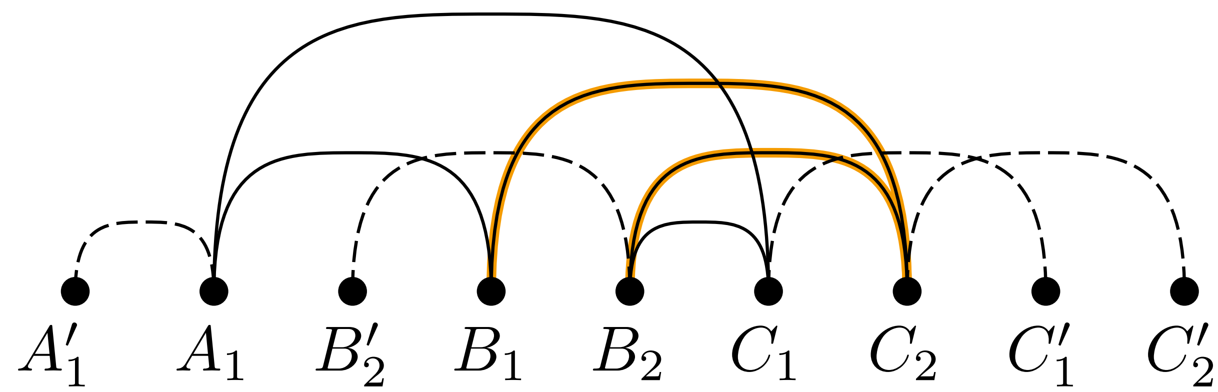

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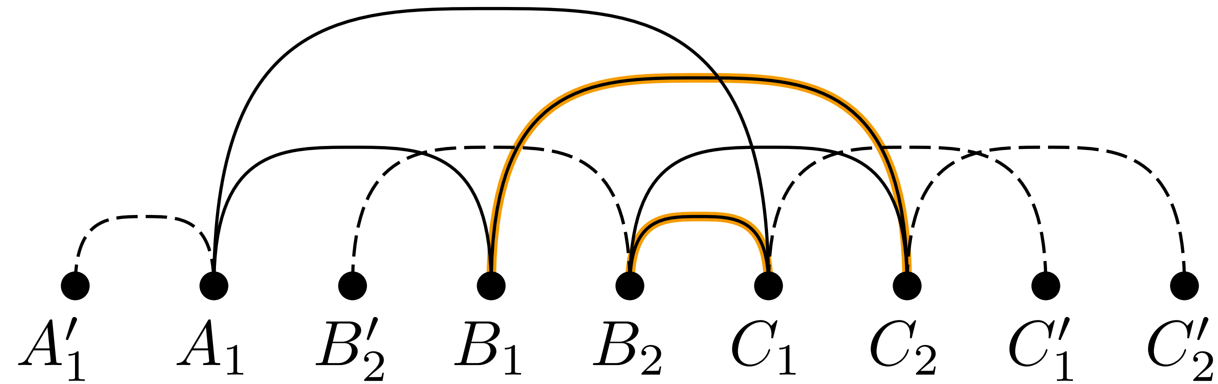

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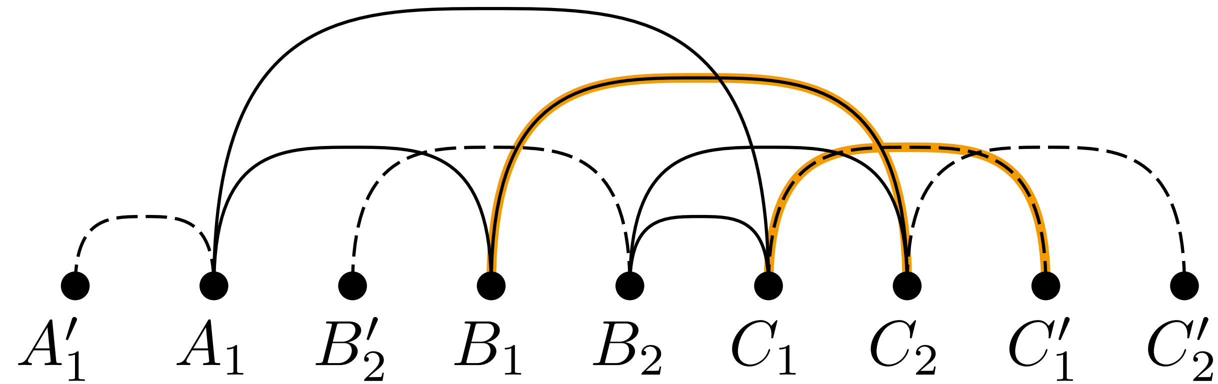

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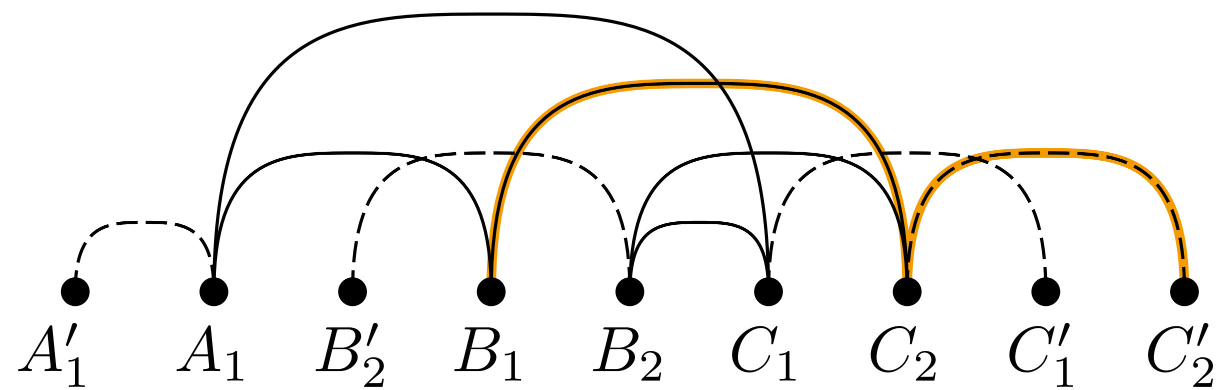
EV-Moves



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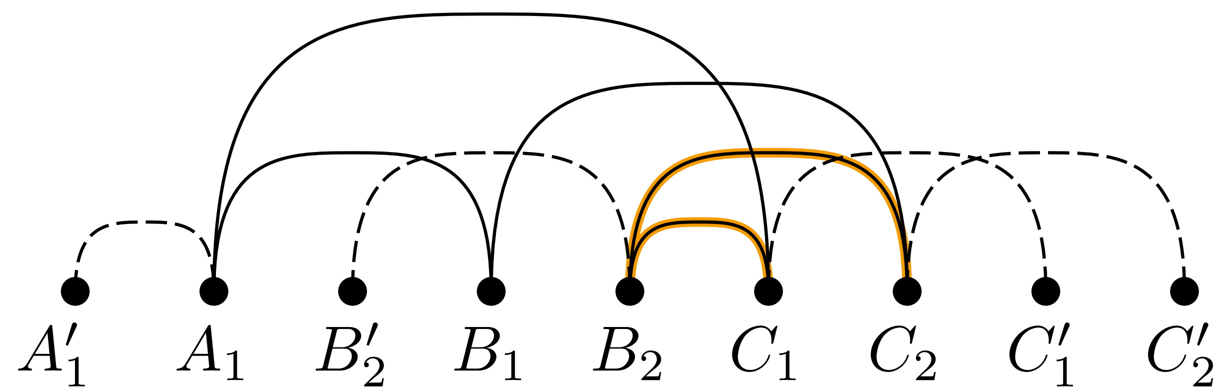

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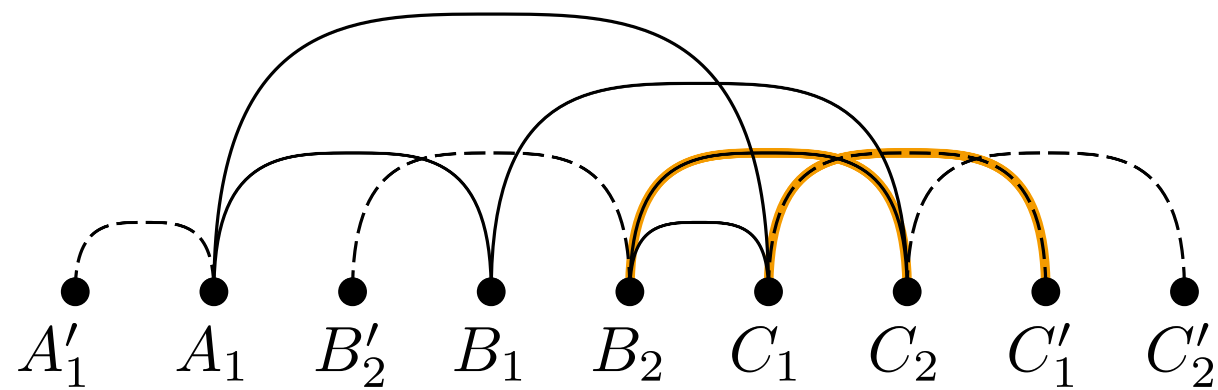

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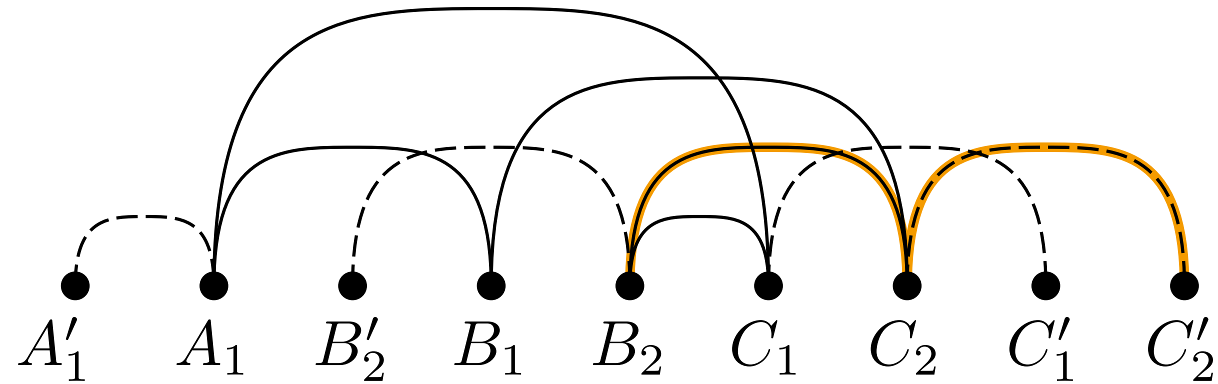
- force even parity
- case 2 analog:
  - edge $\overline{TV}$ over vertex U
  - edge $\overline{UW}$ over vertex V

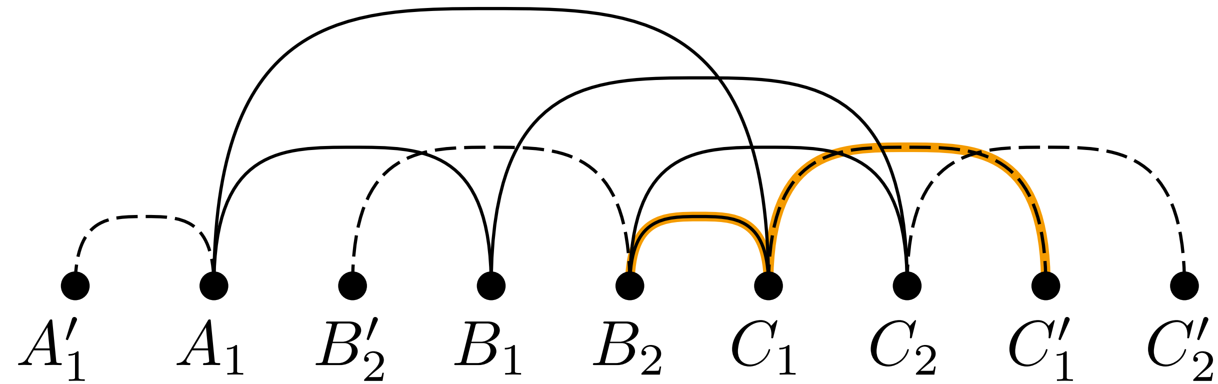


$\rightarrow$ not independent

$$(0 + x_{\overline{A_1C_1},B'_2} + x_{\overline{A_1C_1},B_2})\bmod 2 = 0$$
$$(1 + x_{\overline{A_1C_1},B_1} + x_{\overline{B_1C_2},C_1})\bmod 2 = 0$$
$$(1 + x_{\overline{A_1C_1},B_2} + x_{\overline{B_2C_2},C_1})\bmod 2 = 0$$
$$(1 + x_{\overline{A_1B_1},B'_2} + x_{\overline{B'_2B_2},B_1})\bmod 2 = 0$$
$$(1 + x_{\overline{B'_2B_2},B_1} + x_{\overline{B_1C_2},B_2})\bmod 2 = 0$$
$$(0 + x_{\overline{B_1C_2},B_2} + x_{\overline{B_1C_2},C_1})\bmod 2 = 0$$
$$(1 + x_{\overline{B_2C_2},C_1} + x_{\overline{C_1C'_1},C_2})\bmod 2 = 0$$
$$(1 + x_{\overline{B_2C_2},C_1} + x_{\overline{C_1C'_1},C_2})\bmod 2 = 0$$

EV-Moves

- pairs of indepenent edges
- case 1 - two possible ev-moves:
  - edge $\overline{TW}$ over vertex U
  - edge $\overline{TW}$ over vertex V
- x: defines ev-move
- i: number of crossings per pair
- force even parity
- case 2 analog:
  - edge $\overline{TV}$ over vertex U
  - edge $\overline{UW}$ over vertex V

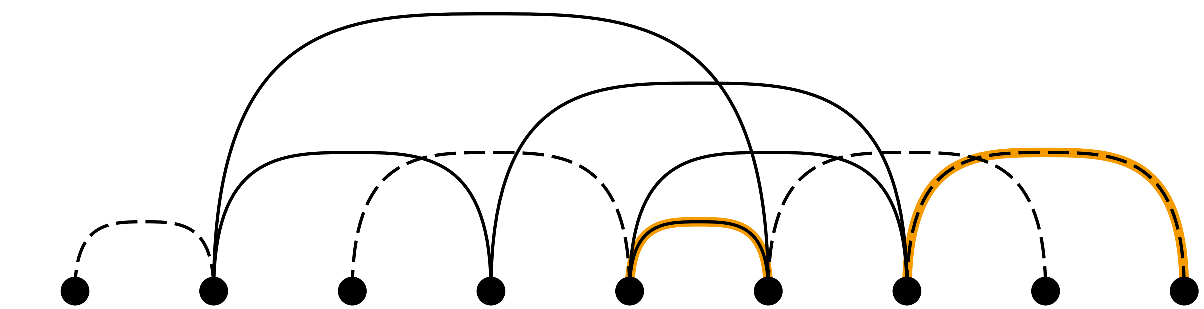


$\rightarrow$ not independent

$$(0 + x_{\overline{A_1C_1},B'_2} + x_{\overline{A_1C_1},B_2})\bmod 2 = 0$$
$$(1 + x_{\overline{A_1C_1},B_1} + x_{\overline{B_1C_2},C_1})\bmod 2 = 0$$
$$(1 + x_{\overline{A_1C_1},B_2} + x_{\overline{B_2C_2},C_1})\bmod 2 = 0$$
$$(1 + x_{\overline{A_1B_1},B'_2} + x_{\overline{B'_2B_2},B_1})\bmod 2 = 0$$
$$(1 + x_{\overline{B'_2B_2},B_1} + x_{\overline{B_1C_2},B_2})\bmod 2 = 0$$
$$(0 + x_{\overline{B_1C_2},B_2} + x_{\overline{B_1C_2},C_1})\bmod 2 = 0$$
$$(1 + x_{\overline{B_2C_2},C_1} + x_{\overline{C_1C'_1},C_2})\bmod 2 = 0$$
$$(1 + x_{\overline{B_2C_2},C_1} + x_{\overline{C_1C'_1},C_2})\bmod 2 = 0$$

EV-Moves

- pairs of indepenent edges
- case 1 - two possible ev-moves:
  - edge $\overline{TW}$ over vertex U
  - edge $\overline{TW}$ over vertex V
- x: defines ev-move
- i: number of crossings per pair
- force even parity
- case 2 analog:
  - edge $\overline{TV}$ over vertex U
  - edge $\overline{UW}$ over vertex V

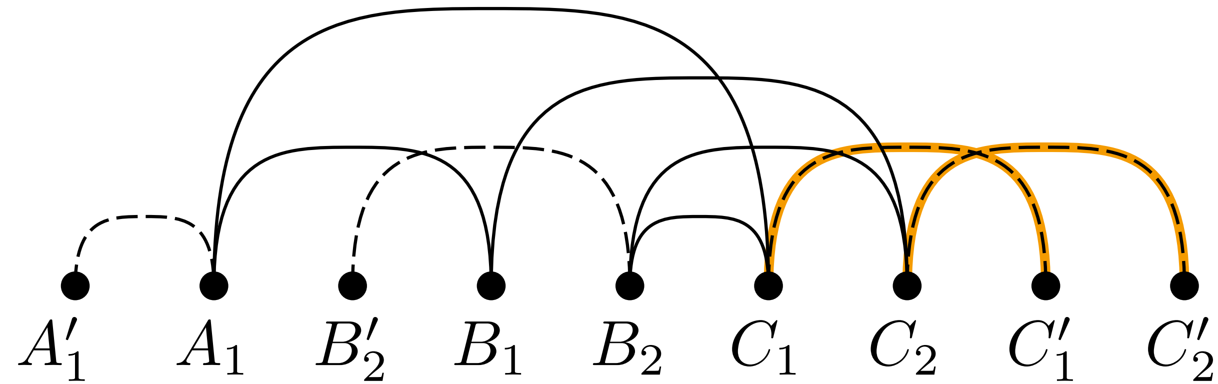


$\rightarrow$ not interleaved

$$(0 + x_{\overline{A_1C_1},B'_2} + x_{\overline{A_1C_1},B_2}) \bmod 2 = 0$$
$$(1 + x_{\overline{A_1C_1},B_1} + x_{\overline{B_1C_2},C_1}) \bmod 2 = 0$$
$$(1 + x_{\overline{A_1C_1},B_2} + x_{\overline{B_2C_2},C_1}) \bmod 2 = 0$$
$$(1 + x_{\overline{A_1B_1},B'_2} + x_{\overline{B'_2B_2},B_1}) \bmod 2 = 0$$
$$(1 + x_{\overline{B'_2B_2},B_1} + x_{\overline{B_1C_2},B_2}) \bmod 2 = 0$$
$$(0 + x_{\overline{B_1C_2},B_2} + x_{\overline{B_1C_2},C_1}) \bmod 2 = 0$$
$$(1 + x_{\overline{B_2C_2},C_1} + x_{\overline{C_1C'_1},C_2}) \bmod 2 = 0$$
$$(1 + x_{\overline{B_2C_2},C_1} + x_{\overline{C_1C'_1},C_2}) \bmod 2 = 0$$

EV-Moves

- pairs of indepenent edges
- case 1 - two possible ev-moves:
  - edge $\overline{TW}$ over vertex U
  - edge $\overline{TW}$ over vertex V
- x: defines ev-move
- i: number of crossings per pair
- force even parity
- case 2 analog:
  - edge $\overline{TV}$ over vertex U
  - edge $\overline{UW}$ over vertex V



$\rightarrow$ add equation

$$(0 + x_{\overline{A_1C_1},B_2'} + x_{\overline{A_1C_1},B_2})\bmod 2 = 0$$
$$(1 + x_{\overline{A_1C_1},B_1} + x_{\overline{B_1C_2},C_1})\bmod 2 = 0$$
$$(1 + x_{\overline{A_1C_1},B_2} + x_{\overline{B_2C_2},C_1})\bmod 2 = 0$$
$$(1 + x_{\overline{A_1B_1},B_2'} + x_{\overline{B_2'B_2},B_1})\bmod 2 = 0$$
$$(1 + x_{\overline{B_2'B_2},B_1} + x_{\overline{B_1C_2},B_2})\bmod 2 = 0$$
$$(0 + x_{\overline{B_1C_2},B_2} + x_{\overline{B_1C_2},C_1})\bmod 2 = 0$$
$$(1 + x_{\overline{B_2C_2},C_1} + x_{\overline{C_1C_1'},C_2})\bmod 2 = 0$$
$$(1 + x_{\overline{B_2C_2},C_1} + x_{\overline{C_1C_1'},C_2})\bmod 2 = 0$$
$$(1 + x_{\overline{C_1C_1'},C_2} + x_{\overline{C_2,C_2'},C_1'})\bmod 2 = 0$$

Solve Equation System

Reduction to 2-SAT

$$(0 + x_{\overline{A_1C_1},B_2'} + x_{\overline{A_1C_1},B_2})\bmod 2 = 0$$
$$(1 + x_{\overline{A_1C_1},B_1} + x_{\overline{B_1C_2},C_1})\bmod 2 = 0$$
$$(1 + x_{\overline{A_1C_1},B_2} + x_{\overline{B_2C_2},C_1})\bmod 2 = 0$$
$$(1 + x_{\overline{A_1B_1},B_2'} + x_{\overline{B_2'B_2},B_1})\bmod 2 = 0$$
$$(1 + x_{\overline{B_2'B_2},B_1} + x_{\overline{B_1C_2},B_2})\bmod 2 = 0$$
$$(0 + x_{\overline{B_1C_2},B_2} + x_{\overline{B_1C_2},C_1})\bmod 2 = 0$$
$$(1 + x_{\overline{B_2C_2},C_1} + x_{\overline{C_1C_1'},C_2})\bmod 2 = 0$$
$$(1 + x_{\overline{B_2C_2},C_1} + x_{\overline{C_1C_1'},C_2})\bmod 2 = 0$$
$$(1 + x_{\overline{C_1C_1'},C_2} + x_{\overline{C_2,C_2'},C_1'})\bmod 2 = 0$$

Solve Equation System

Reduction to 2-SAT

$$(\mathrm{x}_1 \vee \neg \mathrm{x}_2) \wedge (\neg \mathrm{x}_1 \vee \mathrm{x}_2) \iff (0 + x_1 + x_2) \quad \mathrm{mod}\ 2 = 0$$

$$
\begin{aligned}
&(0 + x_{\overline{A_1C_1},B_2'} + x_{\overline{A_1C_1},B_2})\bmod 2 = 0\\
&(1 + x_{\overline{A_1C_1},B_1} + x_{\overline{B_1C_2},C_1})\bmod 2 = 0\\
&(1 + x_{\overline{A_1C_1},B_2} + x_{\overline{B_2C_2},C_1})\bmod 2 = 0\\
&(1 + x_{\overline{A_1B_1},B_2'} + x_{\overline{B_2'B_2},B_1})\bmod 2 = 0\\
&(1 + x_{\overline{B_2'B_2},B_1} + x_{\overline{B_1C_2},B_2})\bmod 2 = 0\\
&(0 + x_{\overline{B_1C_2},B_2} + x_{\overline{B_1C_2},C_1})\bmod 2 = 0\\
&(1 + x_{\overline{B_2C_2},C_1} + x_{\overline{C_1C_1'},C_2})\bmod 2 = 0\\
&(1 + x_{\overline{B_2C_2},C_1} + x_{\overline{C_1C_1'},C_2})\bmod 2 = 0\\
&(1 + x_{\overline{C_1C_1'},C_2} + x_{\overline{C_2,C_2'},C_1'})\bmod 2 = 0
\end{aligned}
$$

Solve Equation System

Reduction to 2-SAT

$$(\mathrm{x}_1 \vee \neg \mathrm{x}_2) \wedge (\neg \mathrm{x}_1 \vee \mathrm{x}_2) \iff (0 + x_1 + x_2) \mod 2 = 0$$

x_1	x_2	$(0 + x_1 + x_2) \mod 2 = 0$
0	0	$\top$
0	1	$\perp$
1	0	$\perp$
1	1	$\top$

x_1	x_2	$(\mathrm{x}_1 \vee \neg \mathrm{x}_2) \wedge (\neg \mathrm{x}_1 \vee \mathrm{x}_2)$
$\perp$	$\perp$	$\top$
$\perp$	$\top$	$\perp$
$\top$	$\perp$	$\perp$
$\top$	$\top$	$\top$

$$(0 + x_{\overline{A_1C_1}, B_2'} + x_{\overline{A_1C_1}, B_2}) \mod 2 = 0$$
$$(1 + x_{\overline{A_1C_1}, B_1} + x_{\overline{B_1C_2}, C_1}) \mod 2 = 0$$
$$(1 + x_{\overline{A_1C_1}, B_2} + x_{\overline{B_2C_2}, C_1}) \mod 2 = 0$$
$$(1 + x_{\overline{A_1B_1}, B_2'} + x_{\overline{B_2'B_2}, B_1}) \mod 2 = 0$$
$$(1 + x_{\overline{B_2'B_2}, B_1} + x_{\overline{B_1C_2}, B_2}) \mod 2 = 0$$
$$(0 + x_{\overline{B_1C_2}, B_2} + x_{\overline{B_1C_2}, C_1}) \mod 2 = 0$$
$$(1 + x_{\overline{B_2C_2}, C_1} + x_{\overline{C_1C_1'}, C_2}) \mod 2 = 0$$
$$(1 + x_{\overline{B_2C_2}, C_1} + x_{\overline{C_1C_1'}, C_2}) \mod 2 = 0$$
$$(1 + x_{\overline{C_1C_1'}, C_2} + x_{\overline{C_2, C_2'}, C_1'}) \mod 2 = 0$$

Solve Equation System

Reduction to 2-SAT

$$(\mathrm{x}_1 \vee \neg \mathrm{x}_2) \wedge (\neg \mathrm{x}_1 \vee \mathrm{x}_2) \iff (0 + x_1 + x_2) \mod 2 = 0$$

$$\begin{aligned}
(0 + x_{\overline{A_1C_1},B_2'} + x_{\overline{A_1C_1},B_2})\bmod 2 &= 0\\
(1 + x_{\overline{A_1C_1},B_1} + x_{\overline{B_1C_2},C_1})\bmod 2 &= 0\\
(1 + x_{\overline{A_1C_1},B_2} + x_{\overline{B_2C_2},C_1})\bmod 2 &= 0\\
(1 + x_{\overline{A_1B_1},B_2'} + x_{\overline{B_2'B_2},B_1})\bmod 2 &= 0\\
(1 + x_{\overline{B_2'B_2},B_1} + x_{\overline{B_1C_2},B_2})\bmod 2 &= 0\\
(0 + x_{\overline{B_1C_2},B_2} + x_{\overline{B_1C_2},C_1})\bmod 2 &= 0\\
(1 + x_{\overline{B_2C_2},C_1} + x_{\overline{C_1C_1'},C_2})\bmod 2 &= 0\\
(1 + x_{\overline{B_2C_2},C_1} + x_{\overline{C_1C_1'},C_2})\bmod 2 &= 0\\
(1 + x_{\overline{C_1C_1'},C_2} + x_{\overline{C_2,C_2'},C_1'})\bmod 2 &= 0
\end{aligned}$$

Solve Equation System

Reduction to 2-SAT

$$(\mathrm{x}_1 \lor \neg \mathrm{x}_2) \land (\neg \mathrm{x}_1 \lor \mathrm{x}_2) \iff (0 + x_1 + x_2) \mod 2 = 0$$

$$(\mathrm{x}_1 \lor \mathrm{x}_2) \land (\neg \mathrm{x}_1 \lor \neg \mathrm{x}_2) \iff (1 + x_1 + x_2) \mod 2 = 0$$

$$(0 + x_{\overline{A_1C_1},B_2'} + x_{\overline{A_1C_1},B_2})\mod 2 = 0$$
$$(1 + x_{\overline{A_1C_1},B_1} + x_{\overline{B_1C_2},C_1})\mod 2 = 0$$
$$(1 + x_{\overline{A_1C_1},B_2} + x_{\overline{B_2C_2},C_1})\mod 2 = 0$$
$$(1 + x_{\overline{A_1B_1},B_2'} + x_{\overline{B_2'B_2},B_1})\mod 2 = 0$$
$$(1 + x_{\overline{B_2'B_2},B_1} + x_{\overline{B_1C_2},B_2})\mod 2 = 0$$
$$(0 + x_{\overline{B_1C_2},B_2} + x_{\overline{B_1C_2},C_1})\mod 2 = 0$$
$$(1 + x_{\overline{B_2C_2},C_1} + x_{\overline{C_1C_1'},C_2})\mod 2 = 0$$
$$(1 + x_{\overline{B_2C_2},C_1} + x_{\overline{C_1C_1'},C_2})\mod 2 = 0$$
$$(1 + x_{\overline{C_1C_1'},C_2} + x_{\overline{C_2,C_2'},C_1'})\mod 2 = 0$$

Solve Equation System

Reduction to 2-SAT

$$(\mathrm{x}_1 \vee \neg \mathrm{x}_2) \wedge (\neg \mathrm{x}_1 \vee \mathrm{x}_2) \iff (0 + x_1 + x_2) \mod 2 = 0$$

$$(\mathrm{x}_1 \vee \mathrm{x}_2) \wedge (\neg \mathrm{x}_1 \vee \neg \mathrm{x}_2) \iff (1 + x_1 + x_2) \mod 2 = 0$$

x_1	x_2	$(1 + x_1 + x_2) \mod 2 = 0$
0	0	$\bot$
0	1	$\top$
1	0	$\top$
1	1	$\bot$

x_1	x_2	$(\mathrm{x}_1 \vee \mathrm{x}_2) \wedge (\neg \mathrm{x}_1 \vee \neg \mathrm{x}_2)$
$\bot$	$\bot$	$\bot$
$\bot$	$\top$	$\top$
$\top$	$\bot$	$\top$
$\top$	$\top$	$\bot$

$$(0 + x_{\overline{A_1 C_1}, B_2'} + x_{\overline{A_1 C_1}, B_2}) \mod 2 = 0$$
$$(1 + x_{\overline{A_1 C_1}, B_1} + x_{\overline{B_1 C_2}, C_1}) \mod 2 = 0$$
$$(1 + x_{\overline{A_1 C_1}, B_2} + x_{\overline{B_2 C_2}, C_1}) \mod 2 = 0$$
$$(1 + x_{\overline{A_1 B_1}, B_2'} + x_{\overline{B_2' B_2}, B_1}) \mod 2 = 0$$
$$(1 + x_{\overline{B_2' B_2}, B_1} + x_{\overline{B_1 C_2}, B_2}) \mod 2 = 0$$
$$(0 + x_{\overline{B_1 C_2}, B_2} + x_{\overline{B_1 C_2}, C_1}) \mod 2 = 0$$
$$(1 + x_{\overline{B_2 C_2}, C_1} + x_{\overline{C_1 C_1'}, C_2}) \mod 2 = 0$$
$$(1 + x_{\overline{B_2 C_2}, C_1} + x_{\overline{C_1 C_1'}, C_2}) \mod 2 = 0$$
$$(1 + x_{\overline{C_1 C_1'}, C_2} + x_{\overline{C_2, C_2'}, C_1'}) \mod 2 = 0$$

Solve Equation System

Reduction to 2-SAT

$$(\mathrm{x}_1 \vee \neg \mathrm{x}_2) \wedge (\neg \mathrm{x}_1 \vee \mathrm{x}_2) \iff (0 + x_1 + x_2) \mod 2 = 0$$

$$(\mathrm{x}_1 \vee \mathrm{x}_2) \wedge (\neg \mathrm{x}_1 \vee \neg \mathrm{x}_2) \iff (1 + x_1 + x_2) \mod 2 = 0$$

$$\begin{aligned}
&(0 + x_{\overline{A_1C_1},B_2'} + x_{\overline{A_1C_1},B_2})\bmod 2 = 0\\
&(1 + x_{\overline{A_1C_1},B_1} + x_{\overline{B_1C_2},C_1})\bmod 2 = 0\\
&(1 + x_{\overline{A_1C_1},B_2} + x_{\overline{B_2C_2},C_1})\bmod 2 = 0\\
&(1 + x_{\overline{A_1B_1},B_2'} + x_{\overline{B_2'B_2},B_1})\bmod 2 = 0\\
&(1 + x_{\overline{B_2'B_2},B_1} + x_{\overline{B_1C_2},B_2})\bmod 2 = 0\\
&(0 + x_{\overline{B_1C_2},B_2} + x_{\overline{B_1C_2},C_1})\bmod 2 = 0\\
&(1 + x_{\overline{B_2C_2},C_1} + x_{\overline{C_1C_1'},C_2})\bmod 2 = 0\\
&(1 + x_{\overline{B_2C_2},C_1} + x_{\overline{C_1C_1'},C_2})\bmod 2 = 0\\
&(1 + x_{\overline{C_1C_1'},C_2} + x_{\overline{C_2,C_2'},C_1'})\bmod 2 = 0
\end{aligned}$$

Solve Equation System

Reduction to 2-SAT

$$(\mathrm{x}_1 \vee \neg \mathrm{x}_2) \wedge (\neg \mathrm{x}_1 \vee \mathrm{x}_2) \iff (0 + x_1 + x_2) \mod 2 = 0$$

$$(\mathrm{x}_1 \vee \mathrm{x}_2) \wedge (\neg \mathrm{x}_1 \vee \neg \mathrm{x}_2) \iff (1 + x_1 + x_2) \mod 2 = 0$$

$$\begin{aligned}
&(\mathrm{x}_{\overline{A_1C_1},B_1} \vee \mathrm{x}_{\overline{B_1C_2},C_1}) \wedge (\neg \mathrm{x}_{\overline{A_1C_1},B_1} \vee \neg \mathrm{x}_{\overline{B_1C_2},C_1})\\
&\wedge (\mathrm{x}_{\overline{A_1C_1},B_2'} \vee \neg \mathrm{x}_{\overline{A_1C_1},B_2}) \wedge (\neg \mathrm{x}_{\overline{A_1C_1},B_2'} \vee \mathrm{x}_{\overline{A_1C_1},B_2})\\
&\wedge (\mathrm{x}_{\overline{A_1C_1},B_2} \vee \mathrm{x}_{\overline{B_2C_2},C_1}) \wedge (\neg \mathrm{x}_{\overline{A_1C_1},B_2} \vee \neg \mathrm{x}_{\overline{B_2C_2},C_1})\\
&\wedge (\mathrm{x}_{\overline{A_1B_1},B_2'} \vee \mathrm{x}_{\overline{B_2'B_2},B_1}) \wedge (\neg \mathrm{x}_{\overline{A_1B_1},B_2'} \vee \neg \mathrm{x}_{\overline{B_2'B_2},B_1})\\
&\wedge (\mathrm{x}_{\overline{B_2'B_2},B_1} \vee \mathrm{x}_{\overline{B_1C_2},B_2}) \wedge (\neg \mathrm{x}_{\overline{B_2'B_2},B_1} \vee \neg \mathrm{x}_{\overline{B_1C_2},B_2})\\
&\wedge (\mathrm{x}_{\overline{B_1C_2},B_2} \vee \neg \mathrm{x}_{\overline{B_1C_2},C_1}) \wedge (\neg \mathrm{x}_{\overline{B_1C_2},B_2} \vee \mathrm{x}_{\overline{B_1C_2},C_1})\\
&\wedge (\mathrm{x}_{\overline{B_1C_2},C_1} \vee \mathrm{x}_{\overline{C_1C_1'},C_2}) \wedge (\neg \mathrm{x}_{\overline{B_1C_2},C_1} \vee \neg \mathrm{x}_{\overline{C_1C_1'},C_2})\\
&\wedge (\mathrm{x}_{\overline{B_2C_2},C_1} \vee \mathrm{x}_{\overline{C_1C_1'},C_2}) \wedge (\neg \mathrm{x}_{\overline{B_2C_2},C_1} \vee \neg \mathrm{x}_{\overline{C_1C_1'},C_2})\\
&\wedge (\mathrm{x}_{\overline{C_1C_1'},C_2} \vee \mathrm{x}_{\overline{C_2,C_2'},C_1'}) \wedge (\neg \mathrm{x}_{\overline{C_1C_1'},C_2} \vee \neg \mathrm{x}_{\overline{C_2,C_2'},C_1'})
\end{aligned}$$

$$\begin{aligned}
&(0 + x_{\overline{A_1C_1},B_2'} + x_{\overline{A_1C_1},B_2}) \bmod 2 = 0\\
&(1 + x_{\overline{A_1C_1},B_1} + x_{\overline{B_1C_2},C_1}) \bmod 2 = 0\\
&(1 + x_{\overline{A_1C_1},B_2} + x_{\overline{B_2C_2},C_1}) \bmod 2 = 0\\
&(1 + x_{\overline{A_1B_1},B_2'} + x_{\overline{B_2'B_2},B_1}) \bmod 2 = 0\\
&(1 + x_{\overline{B_2'B_2},B_1} + x_{\overline{B_1C_2},B_2}) \bmod 2 = 0\\
&(0 + x_{\overline{B_1C_2},B_2} + x_{\overline{B_1C_2},C_1}) \bmod 2 = 0\\
&(1 + x_{\overline{B_2C_2},C_1} + x_{\overline{C_1C_1'},C_2}) \bmod 2 = 0\\
&(1 + x_{\overline{B_2C_2},C_1} + x_{\overline{C_1C_1'},C_2}) \bmod 2 = 0\\
&(1 + x_{\overline{C_1C_1'},C_2} + x_{\overline{C_2,C_2'},C_1'}) \bmod 2 = 0
\end{aligned}$$

Solve Equation System

$$x_{\overline{A_1 B_1}, B_2'} \leftarrow \perp$$

$$x_{\overline{A_1 C_1}, B_1} \leftarrow \top$$

$$x_{\overline{A_1 C_1}, B_2'} \leftarrow \top$$

$$x_{\overline{A_1 C_1}, B_2} \leftarrow \top$$

$$x_{\overline{B_1 C_2}, C_1} \leftarrow \perp$$

$$x_{\overline{B_1 C_2}, B_2} \leftarrow \perp$$

$$x_{\overline{B_1 C_2}, C_1} \leftarrow \perp$$

$$x_{\overline{B_2 C_2}, C_1} \leftarrow \perp$$

$$x_{\overline{B_2' B_2}, B_1} \leftarrow \top$$

$$x_{\overline{C_1 C_1'}, C_2} \leftarrow \top$$

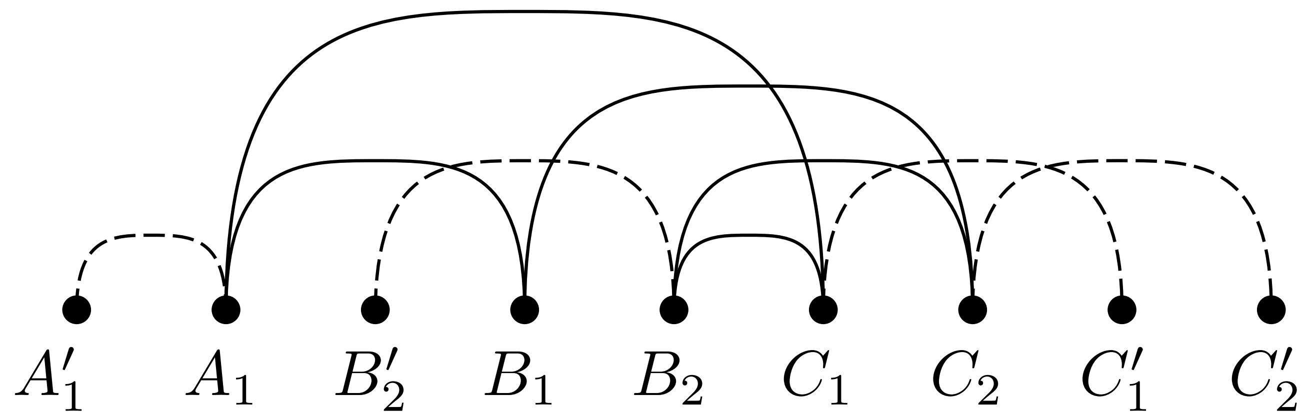
$$x_{\overline{C_2, C_2'}, C_1'} \leftarrow \perp$$

Solve Equation System

$$x_{\overline{A_1B_1},B_2'} = 0$$
$$x_{\overline{A_1C_1},B_1} = 1$$
$$x_{\overline{A_1C_1},B_2'} = 1$$
$$x_{\overline{A_1C_1},B_2} = 1$$
$$x_{\overline{B_1C_2},C_1} = 0$$
$$x_{\overline{B_1C_2},B_2} = 0$$
$$x_{\overline{B_1C_2},C_1} = 0$$
$$x_{\overline{B_2C_2},C_1} = 0$$
$$x_{\overline{B_2'B_2},B_1} = 1$$
$$x_{\overline{C_1C_1'},C_2} = 1$$
$$x_{\overline{C_2,C_2'},C_1'} = 0$$

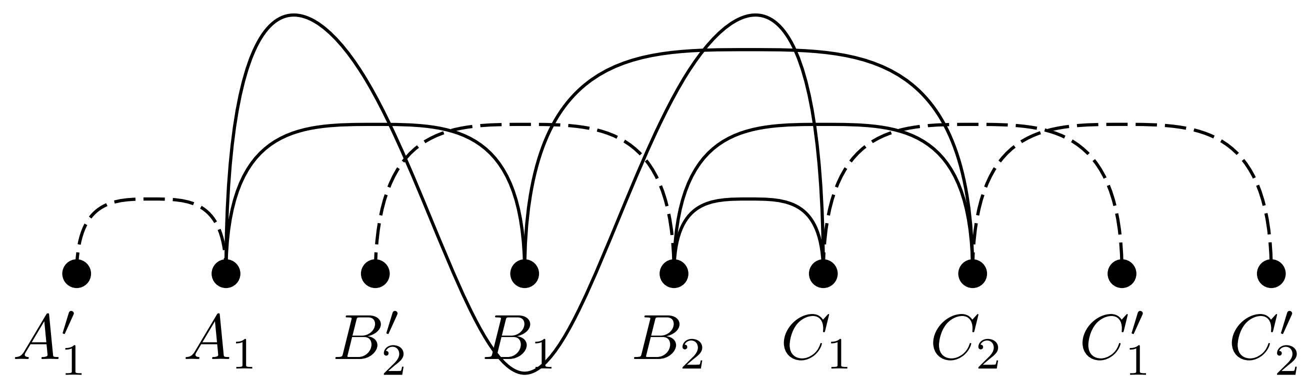
Solve Equation System

$$x_{\overline{A_1B_1},B_2'} = 0$$
$$x_{\overline{A_1C_1},B_1} = 1$$
$$x_{\overline{A_1C_1},B_2'} = 1$$
$$x_{\overline{A_1C_1},B_2} = 1$$
$$x_{\overline{B_1C_2},C_1} = 0$$
$$x_{\overline{B_1C_2},B_2} = 0$$
$$x_{\overline{B_1C_2},C_1} = 0$$
$$x_{\overline{B_2C_2},C_1} = 0$$
$$x_{\overline{B_2'B_2},B_1} = 1$$
$$x_{\overline{C_1C_1'},C_2} = 1$$
$$x_{\overline{C_2,C_2'},C_1'} = 0$$



Solve Equation System

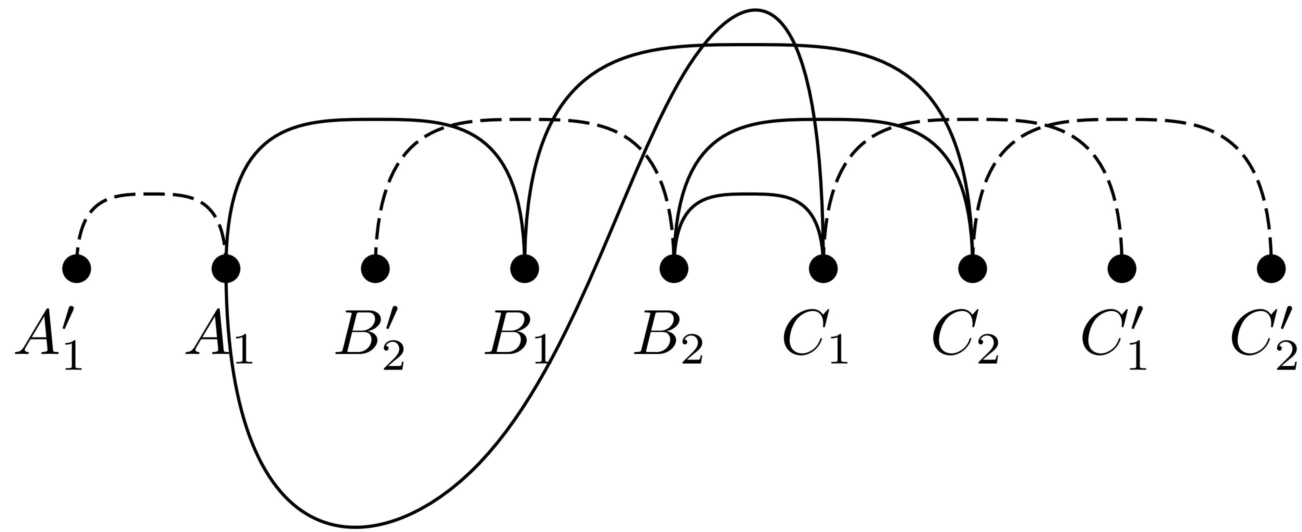
$$x_{\overline{A_1B_1},B_2'} = 0$$
$$x_{\overline{A_1C_1},B_1} = 1$$
$$x_{\overline{A_1C_1},B_2'} = 1$$
$$x_{\overline{A_1C_1},B_2} = 1$$
$$x_{\overline{B_1C_2},C_1} = 0$$
$$x_{\overline{B_1C_2},B_2} = 0$$
$$x_{\overline{B_1C_2},C_1} = 0$$
$$x_{\overline{B_2C_2},C_1} = 0$$
$$x_{\overline{B_2'B_2},B_1} = 1$$
$$x_{\overline{C_1C_1'},C_2} = 1$$
$$x_{\overline{C_2,C_2'},C_1'} = 0$$



- move edge $\overline{A_1C_1}$ over B_1

Solve Equation System

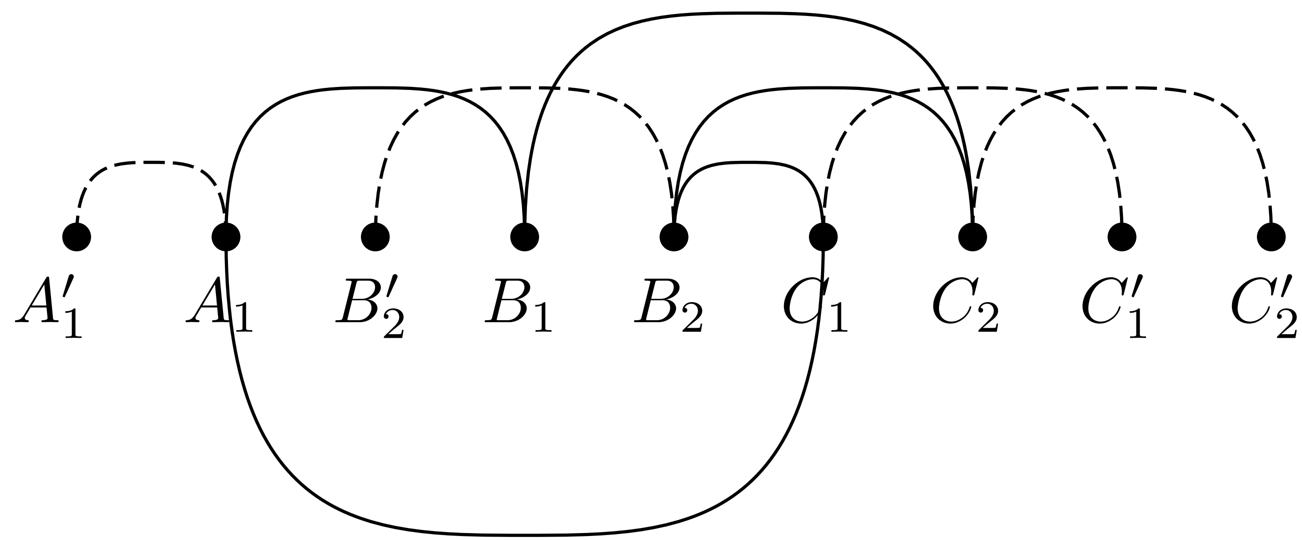
$$x_{\overline{A_1B_1},B_2'} = 0$$
$$x_{\overline{A_1C_1},B_1} = 1$$
$$x_{\overline{A_1C_1},B_2'} = 1$$
$$x_{\overline{A_1C_1},B_2} = 1$$
$$x_{\overline{B_1C_2},C_1} = 0$$
$$x_{\overline{B_1C_2},B_2} = 0$$
$$x_{\overline{B_1C_2},C_1} = 0$$
$$x_{\overline{B_2C_2},C_1} = 0$$
$$x_{\overline{B_2'B_2},B_1} = 1$$
$$x_{\overline{C_1C_1'},C_2} = 1$$
$$x_{\overline{C_2,C_2'},C_1'} = 0$$



- move edge $\overline{A_1C_1}$ over B_1
- move edge $\overline{A_1C_1}$ over B_2'

Solve Equation System

$$x_{\overline{A_1B_1},B'_2} = 0$$
$$x_{\overline{A_1C_1},B_1} = 1$$
$$x_{\overline{A_1C_1},B'_2} = 1$$
$$x_{\overline{A_1C_1},B_2} = 1$$
$$x_{\overline{B_1C_2},C_1} = 0$$
$$x_{\overline{B_1C_2},B_2} = 0$$
$$x_{\overline{B_1C_2},C_1} = 0$$
$$x_{\overline{B_2C_2},C_1} = 0$$
$$x_{\overline{B'_2B_2},B_1} = 1$$
$$x_{\overline{C_1C'_1},C_2} = 1$$
$$x_{\overline{C_2,C'_2},C'_1} = 0$$



- move edge $\overline{A_1C_1}$ over B_1
- move edge $\overline{A_1C_1}$ over B'_2
- move edge $\overline{A_1C_1}$ over B_2

Solve Equation System

$$x_{\overline{A_1B_1},B_2'} = 0$$
$$x_{\overline{A_1C_1},B_1} = 1$$
$$x_{\overline{A_1C_1},B_2'} = 1$$
$$x_{\overline{A_1C_1},B_2} = 1$$
$$x_{\overline{B_1C_2},C_1} = 0$$
$$x_{\overline{B_1C_2},B_2} = 0$$
$$x_{\overline{B_1C_2},C_1} = 0$$
$$x_{\overline{B_2C_2},C_1} = 0$$
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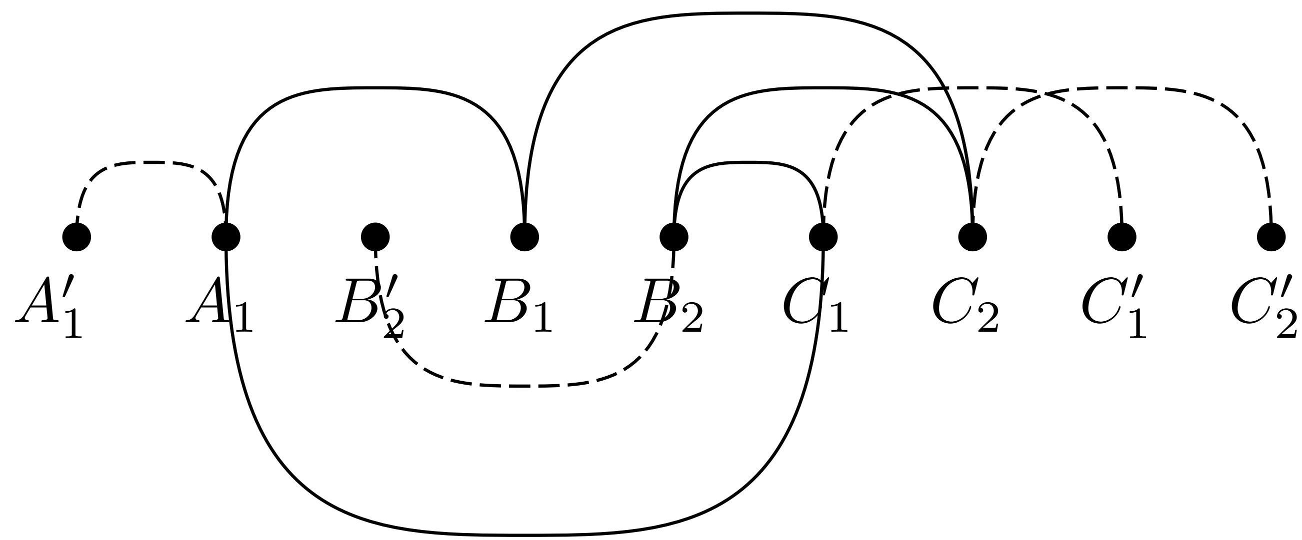

- move edge $\overline{A_1C_1}$ over B_1
- move edge $\overline{A_1C_1}$ over B_2'
- move edge $\overline{A_1C_1}$ over B_2
- move edge $\overline{B_2'B_2}$ over B_1

Solve Equation System

$$x_{\overline{A_1B_1},B_2'} = 0$$
$$x_{\overline{A_1C_1},B_1} = 1$$
$$x_{\overline{A_1C_1},B_2'} = 1$$
$$x_{\overline{A_1C_1},B_2} = 1$$
$$x_{\overline{B_1C_2},C_1} = 0$$
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$$x_{\overline{C_1C_1'},C_2} = 1$$
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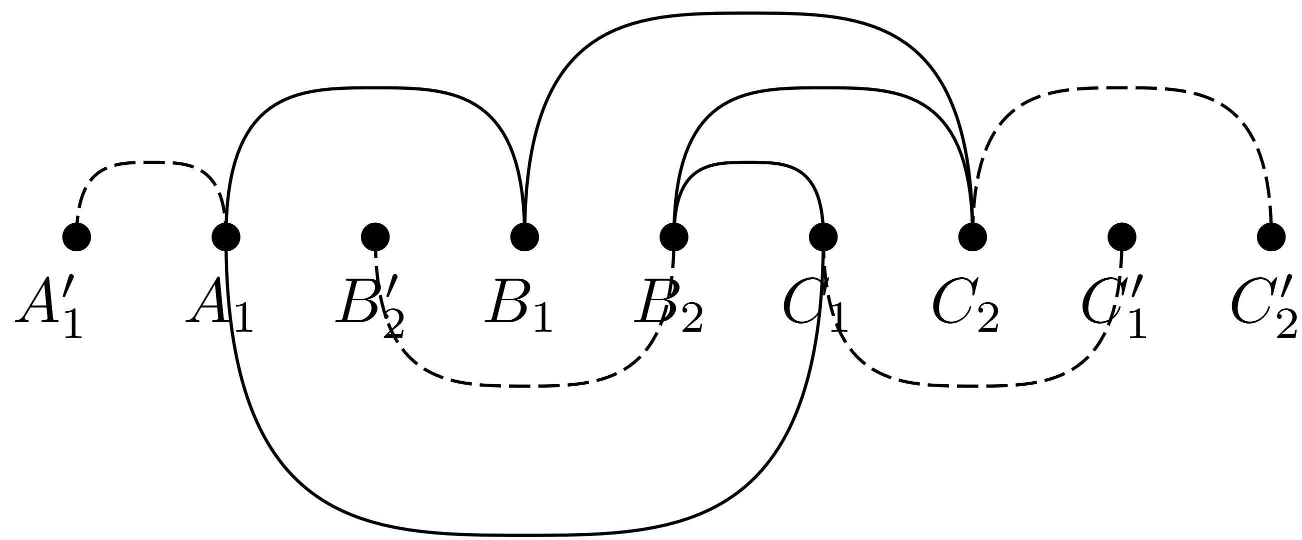
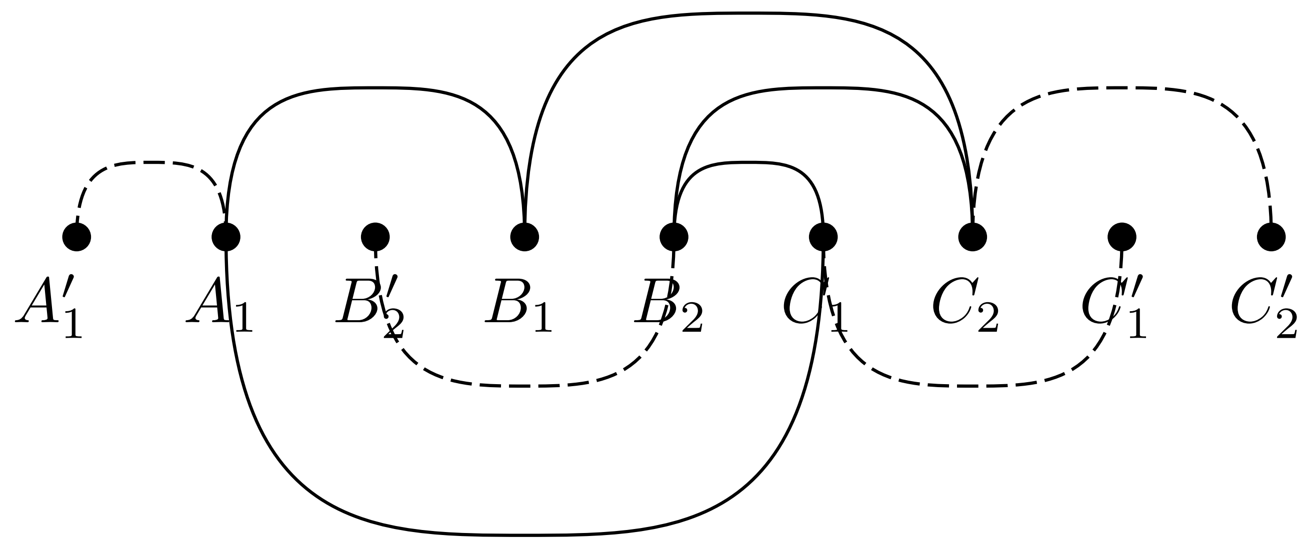
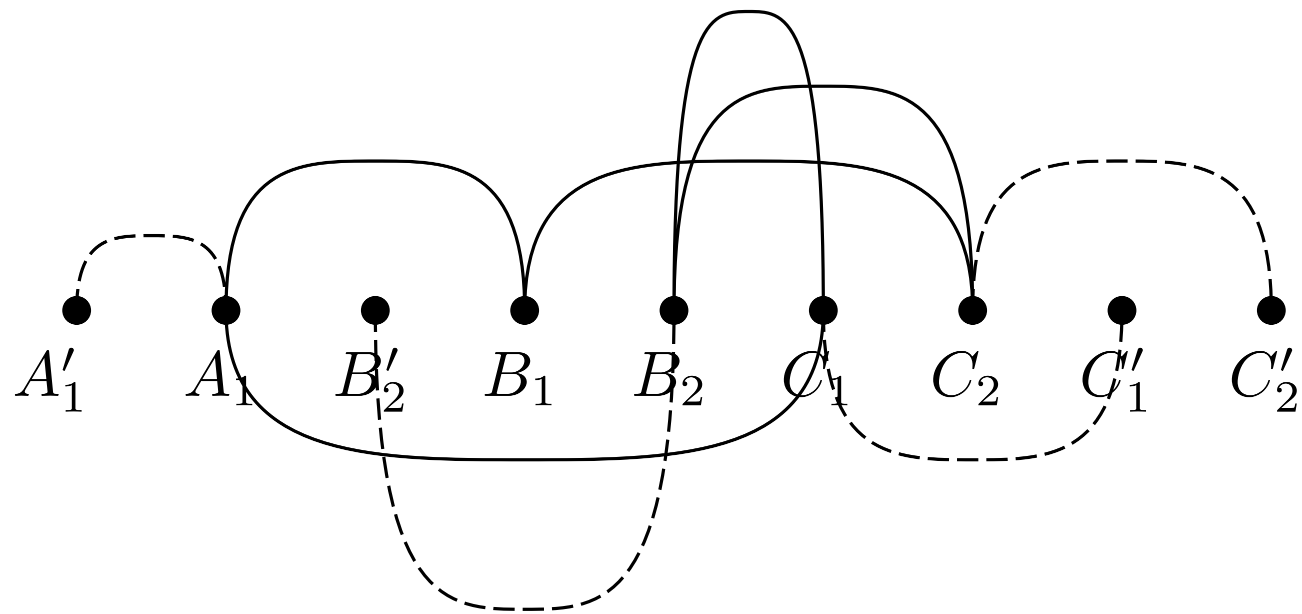


- move edge $\overline{A_1C_1}$ over B_1
- move edge $\overline{A_1C_1}$ over B_2'
- move edge $\overline{A_1C_1}$ over B_2
- move edge $\overline{B_2'B_2}$ over B_1
- move edge $\overline{C_1C_1'}$ over C_2

Solve Equation System

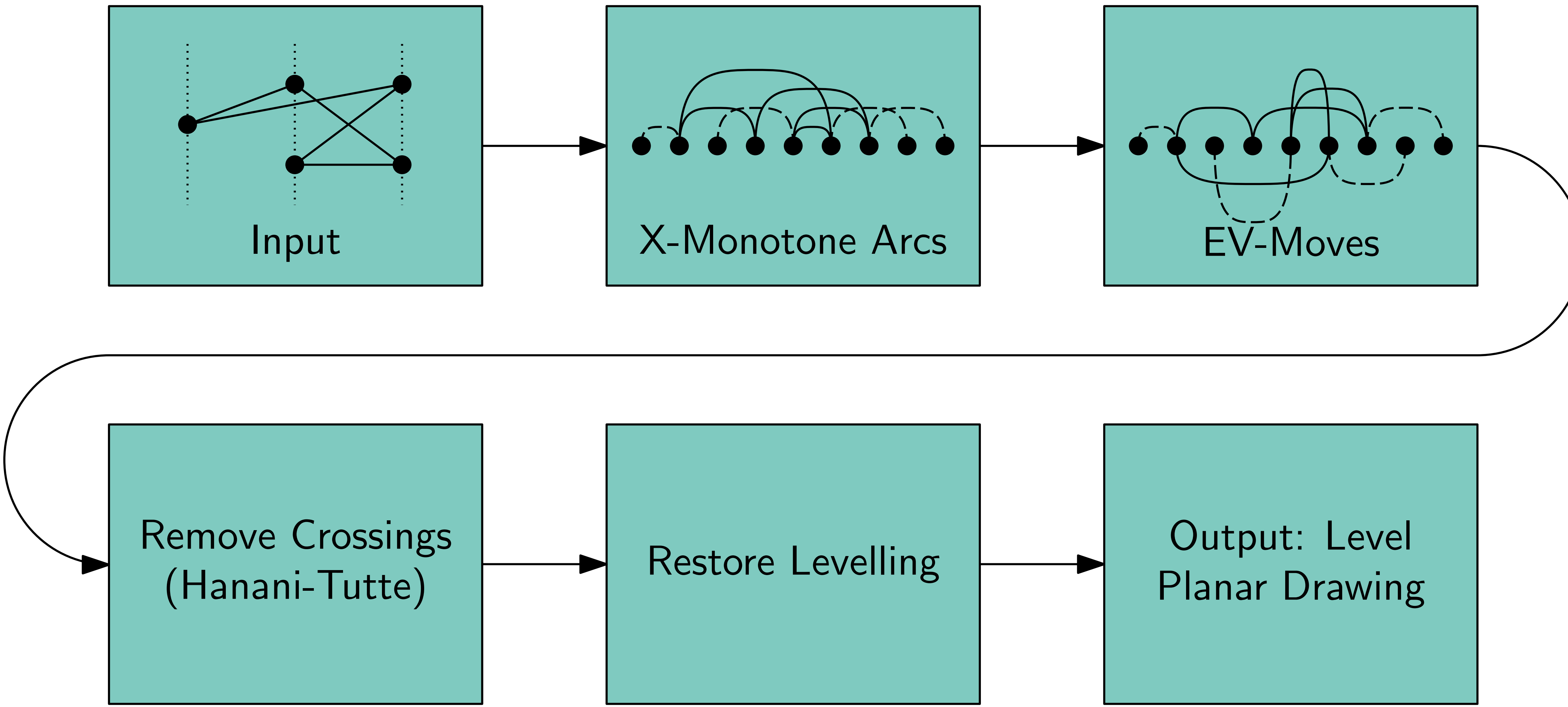


Solve Equation System



Leveled Embedding Construction

Algorithm



Hanani-Tutte

Hanani-Tutte

Weak Monotone Hanani-Tutte Theorem: *If G is a multigraph that has an x-monotone drawing in which all edges are even, then G has an x-monotone embedding with the same vertex locations and rotation system.*

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Proof by Induction on $|V|$

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Proof by Induction on $|V|$

Base Case

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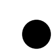
Proof by Induction on $|V|$

Base Case



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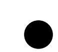
Proof by Induction on $|V|$

Base Case $\Rightarrow$ trivial.



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Proof by Induction on $|V|$

Induction-Step:

Case 1

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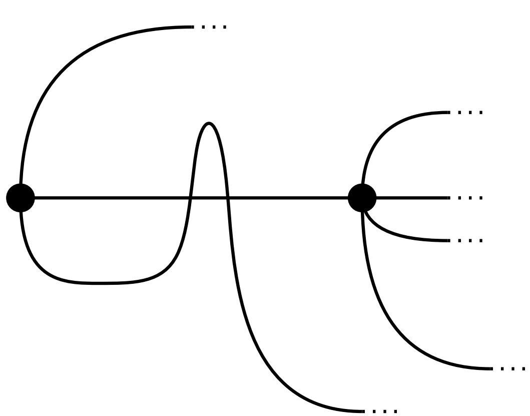

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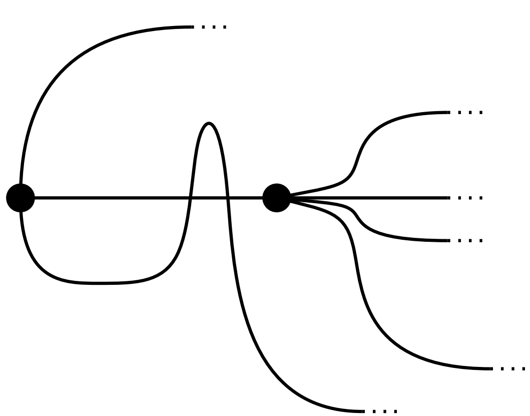

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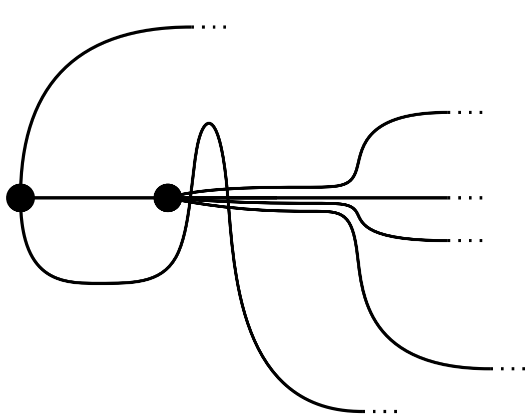

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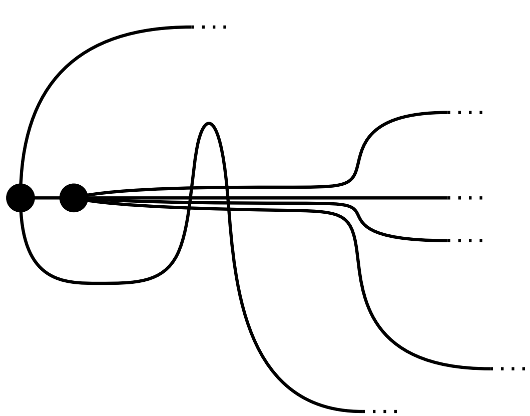

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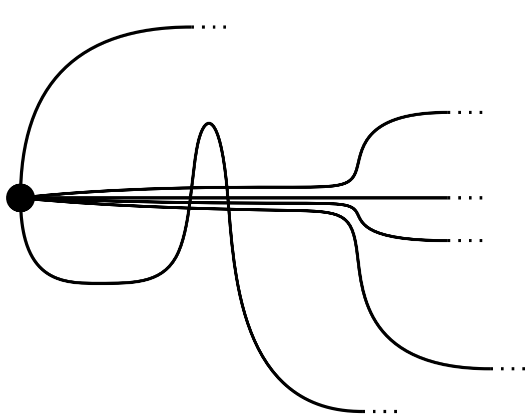

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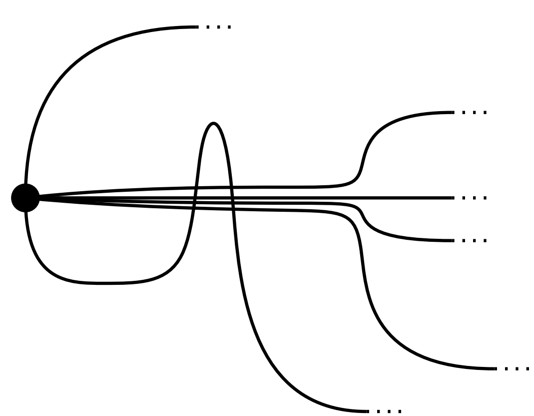

Induction Hypothesis!

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Proof by Induction on $|V|$

Induction-Step:

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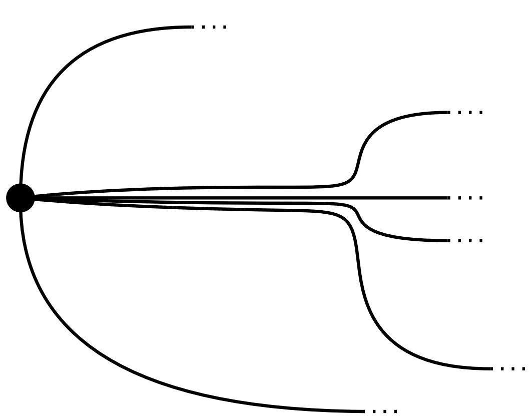

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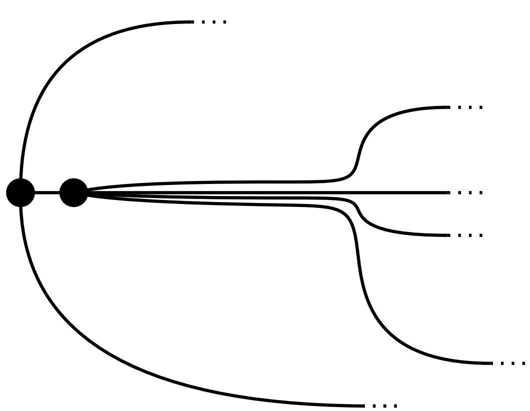

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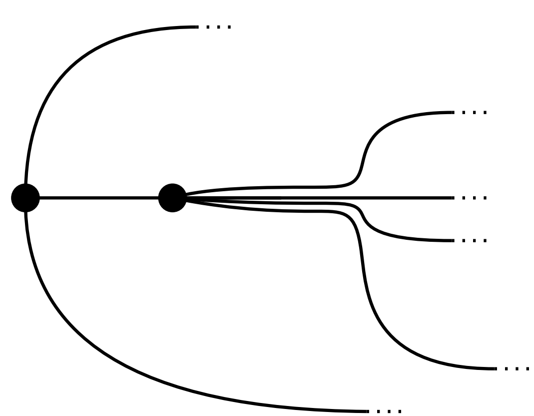

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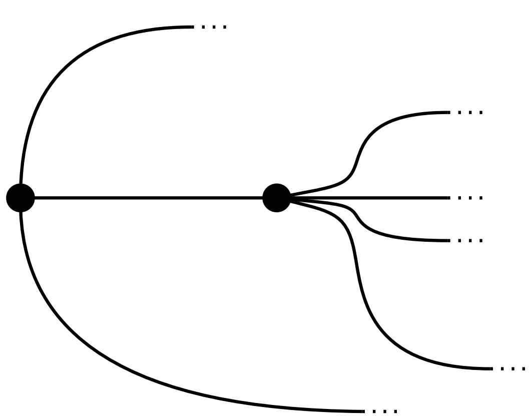

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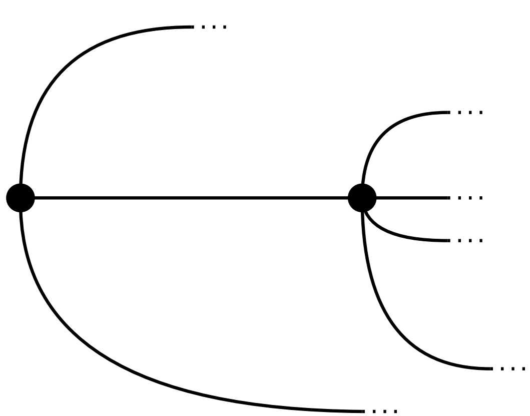

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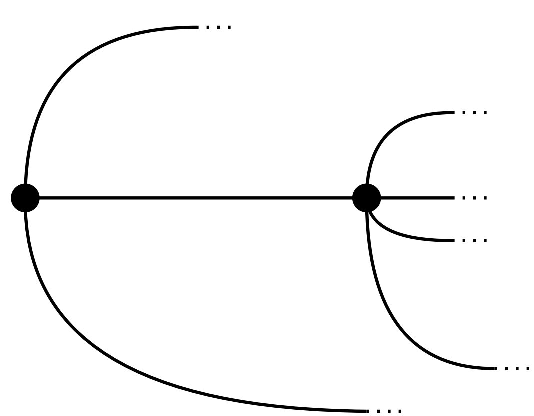
Proof by Induction on $|V|$

Induction-Step:

Case 1

Case 2



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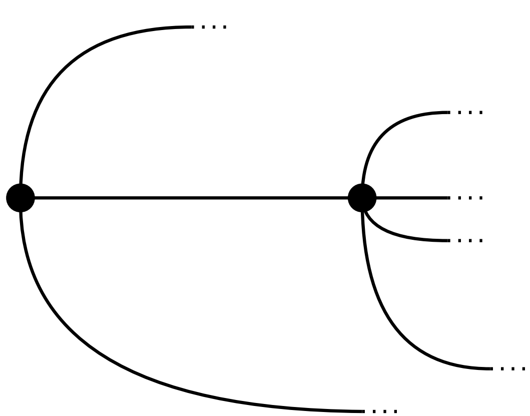
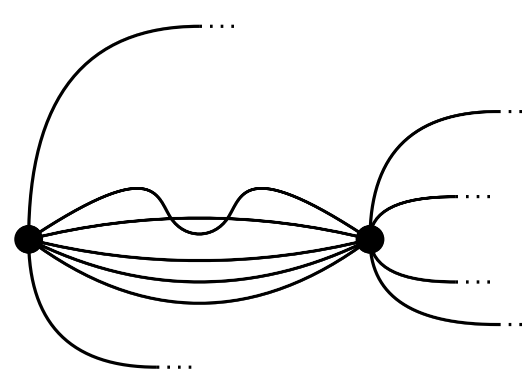


Case 2



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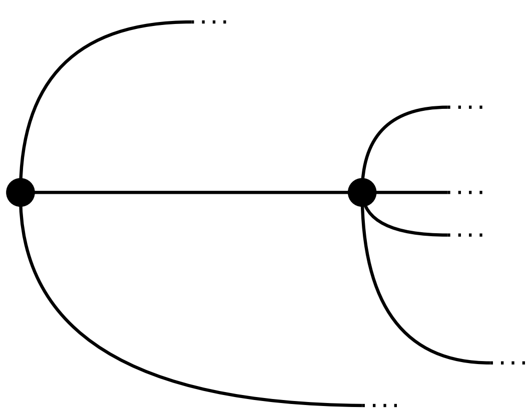

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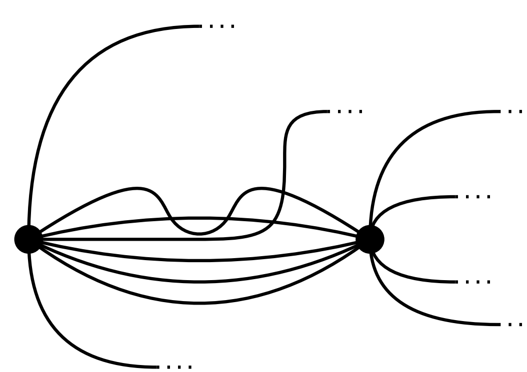

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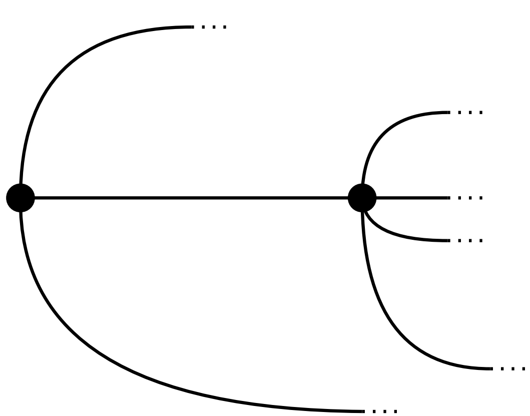

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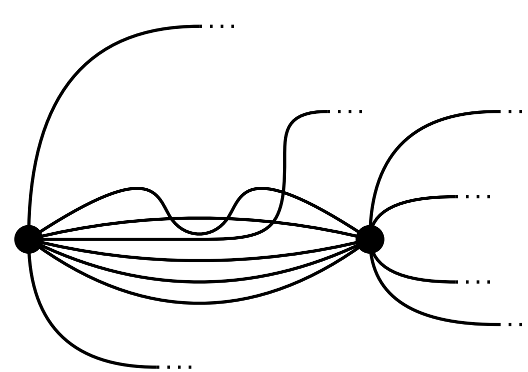

Contradiction!

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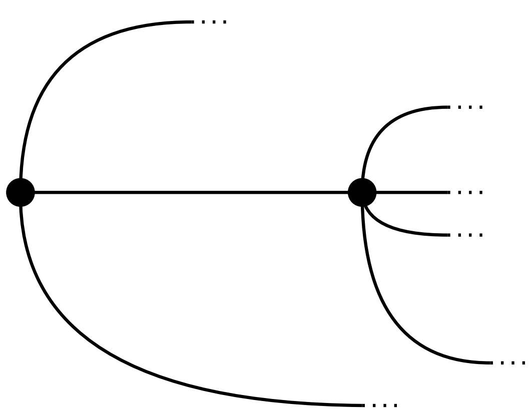

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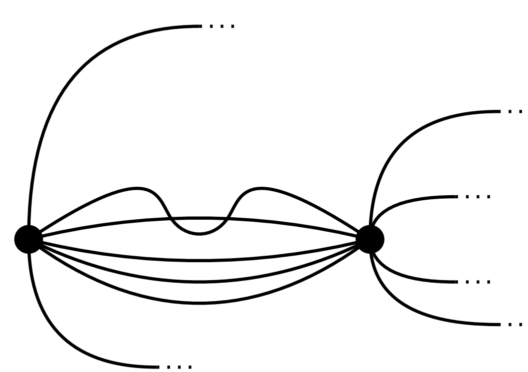

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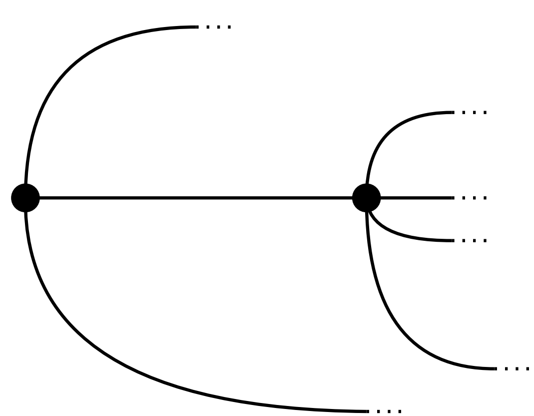

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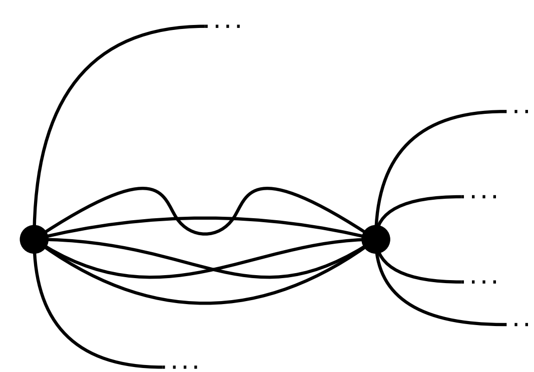

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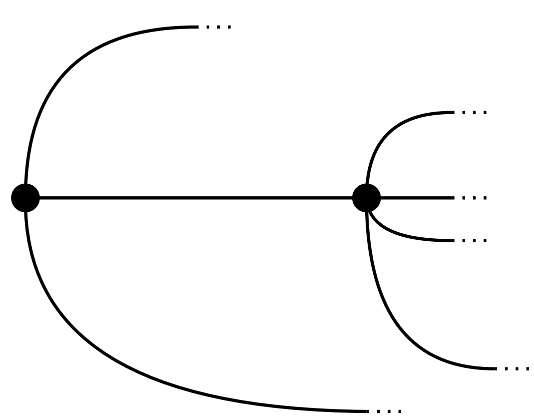

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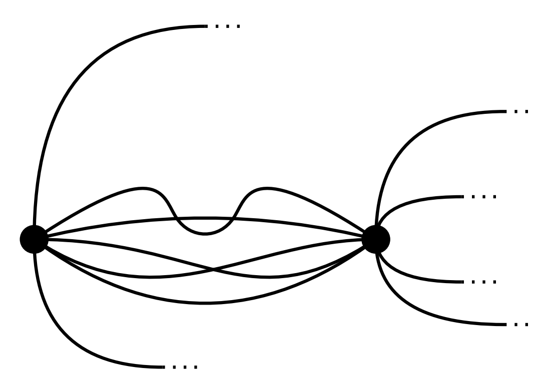

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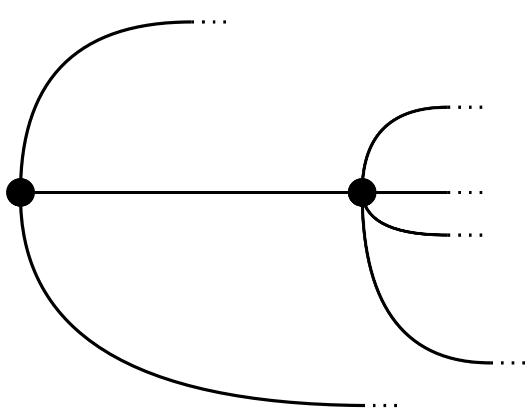

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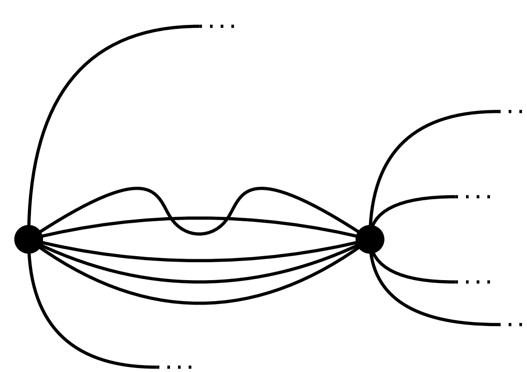

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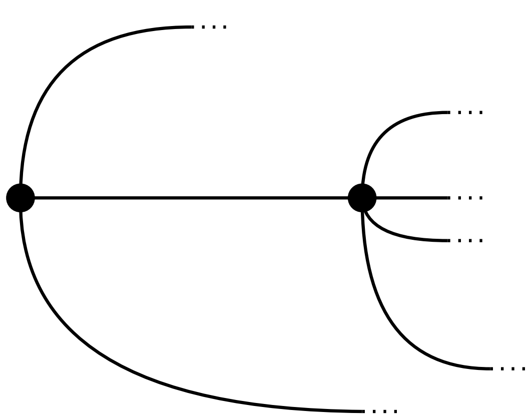

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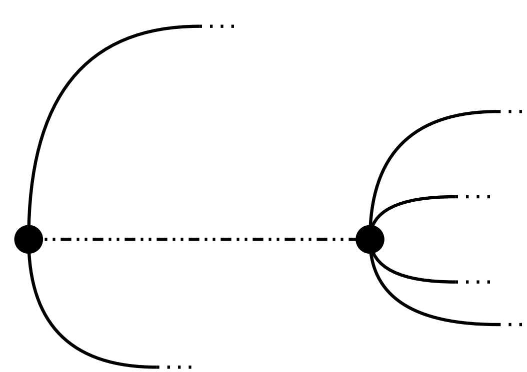

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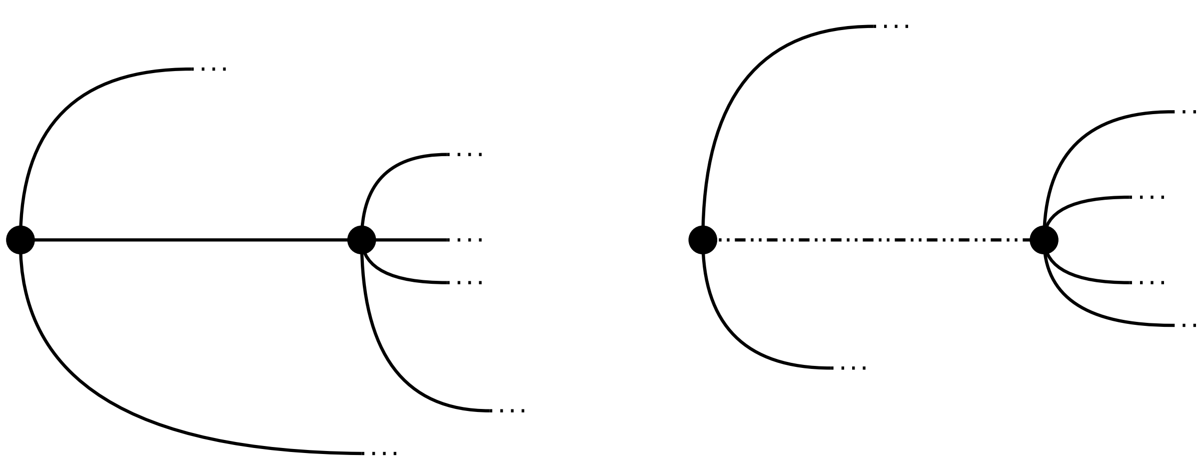

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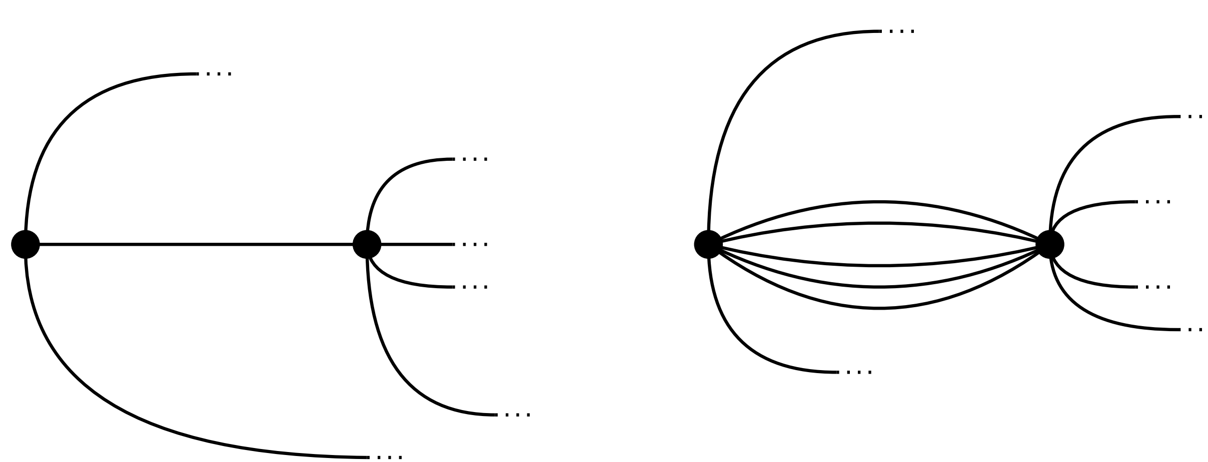

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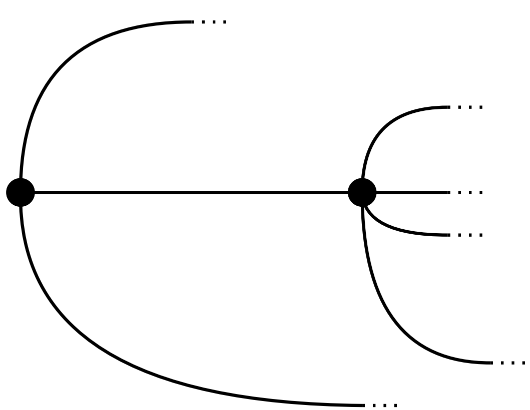
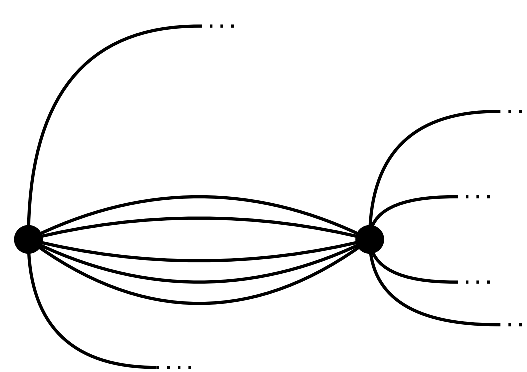


Case 2



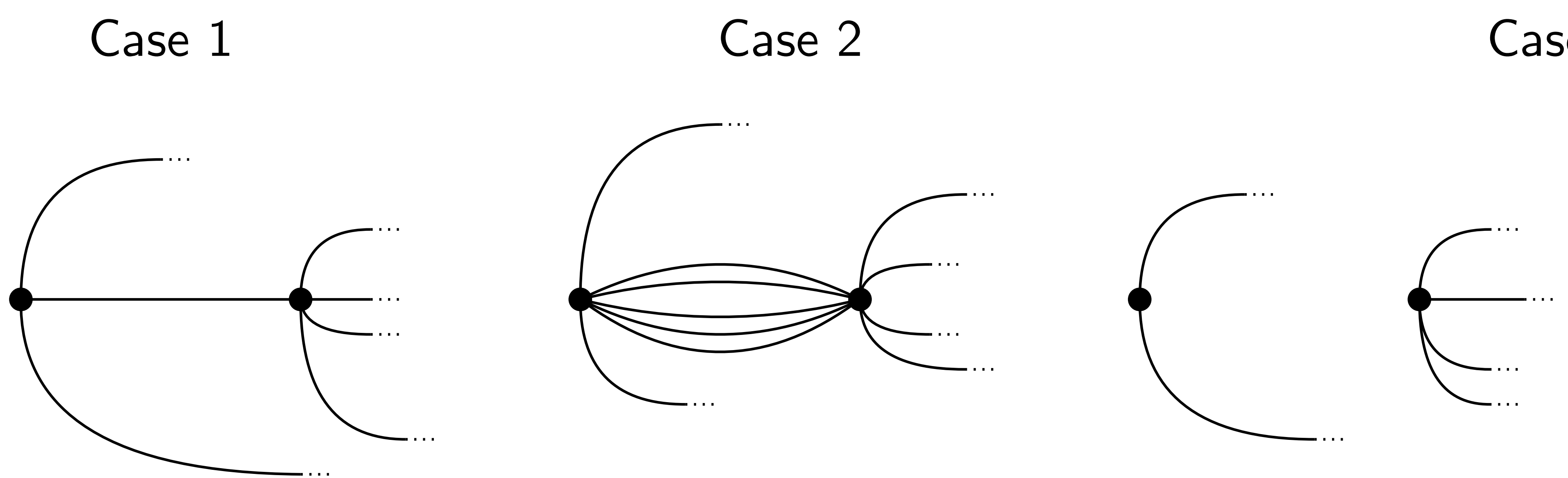
Case 3

Hanani-Tutte

Weak Monotone Hanani-Tutte Theorem: *If G is a multigraph that has an x-monotone drawing in which all edges are even, then G has an x-monotone embedding with the same vertex locations and rotation system.*

Proof by Induction on $|V|$

Induction-Step:



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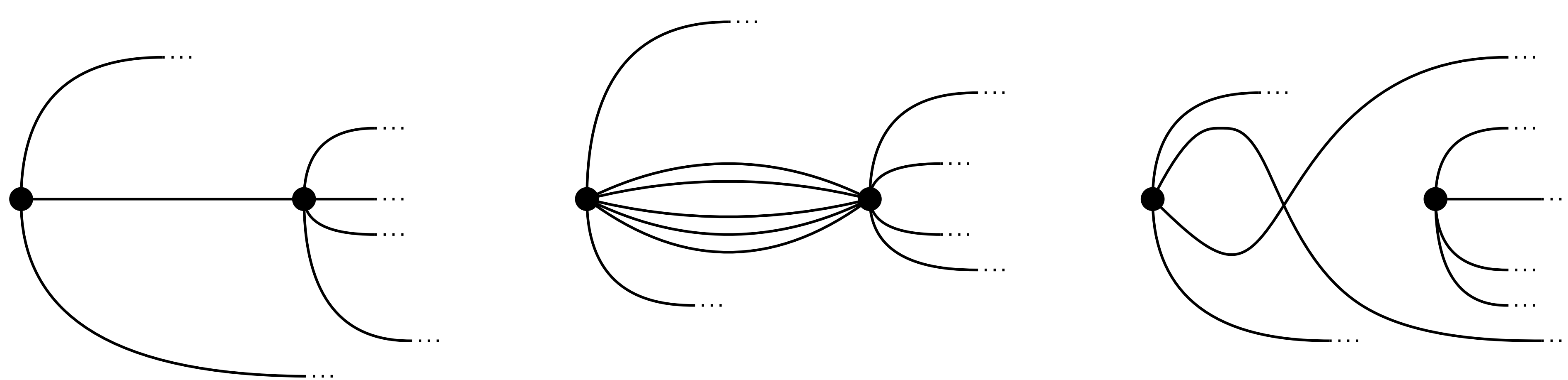

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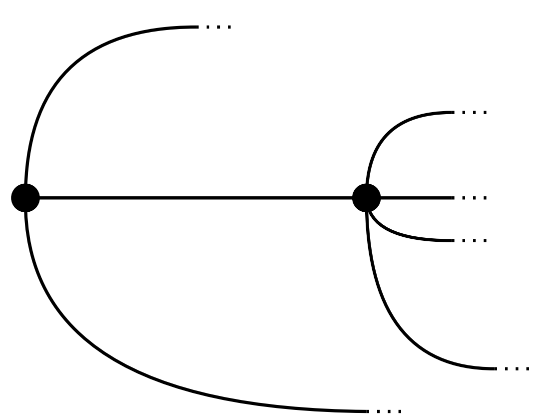

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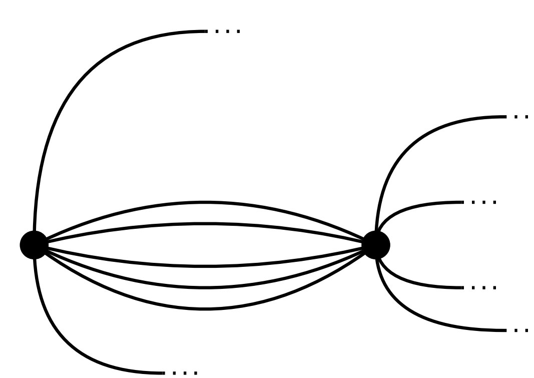

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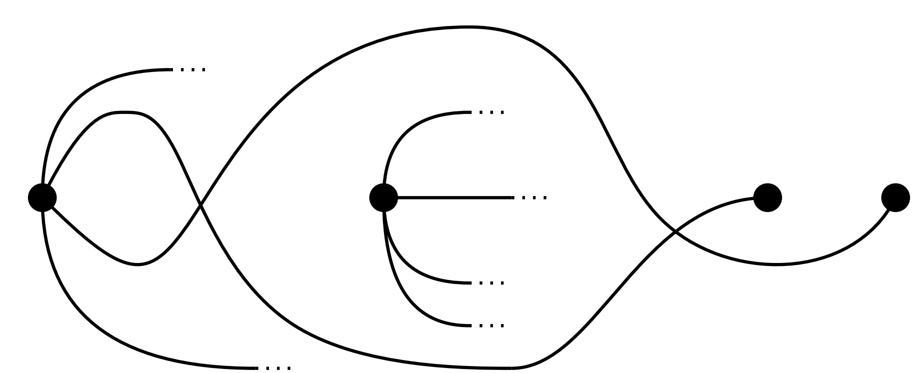

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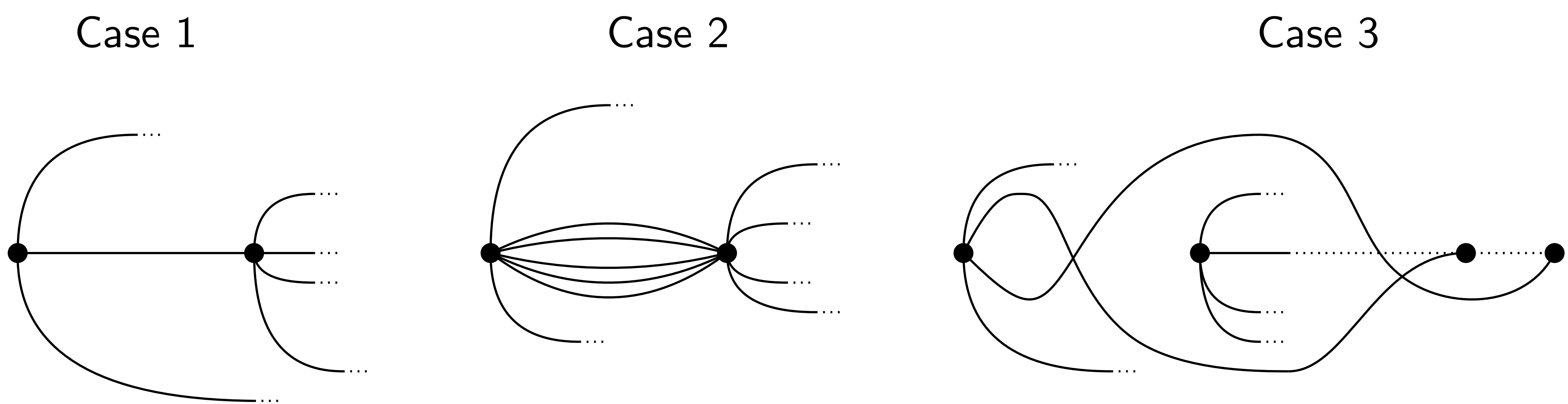

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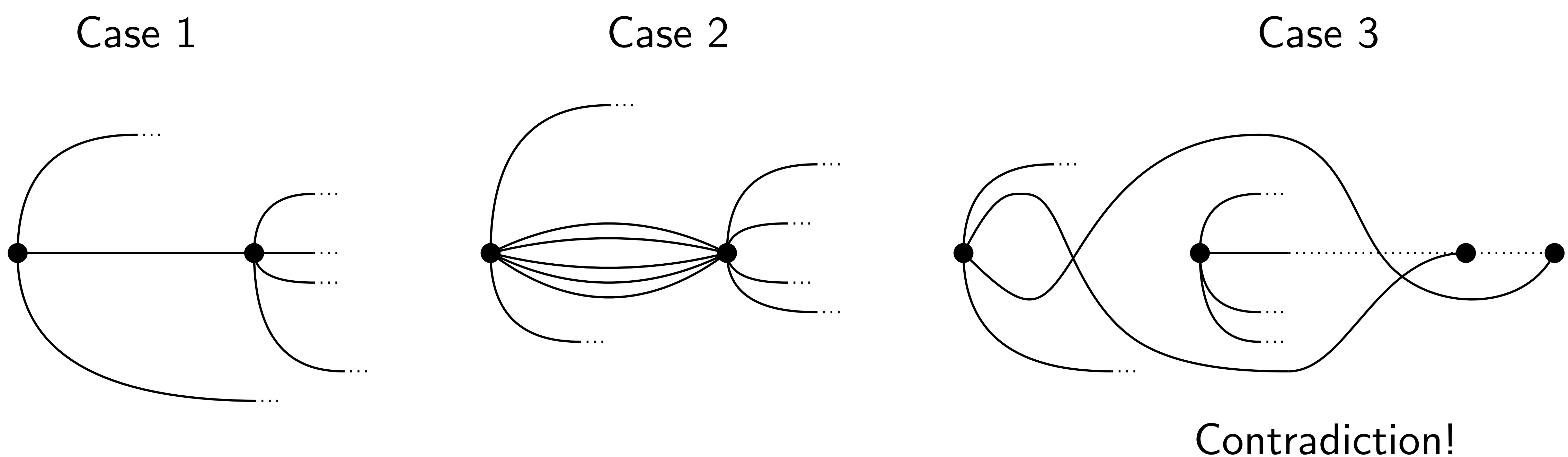

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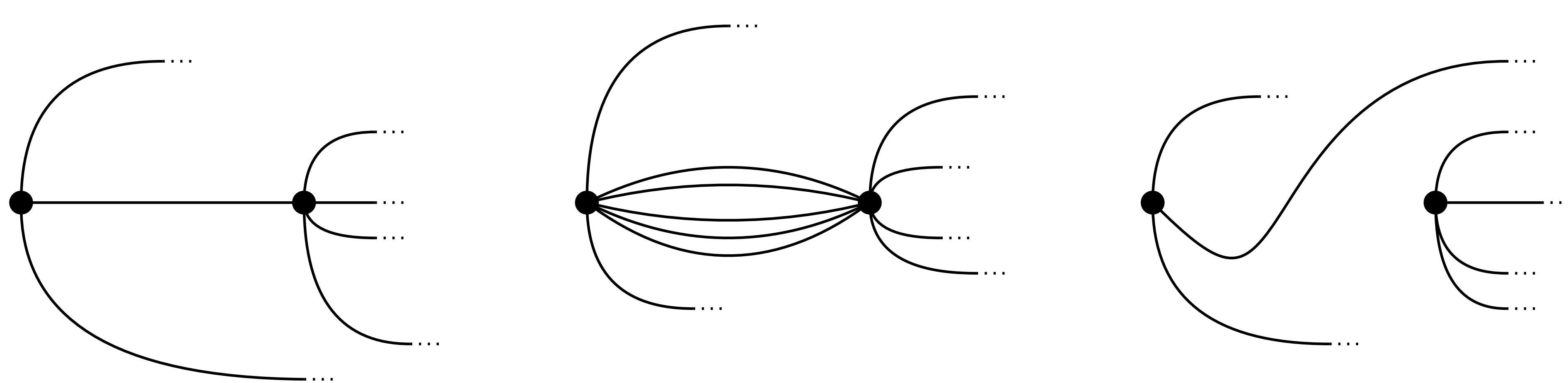

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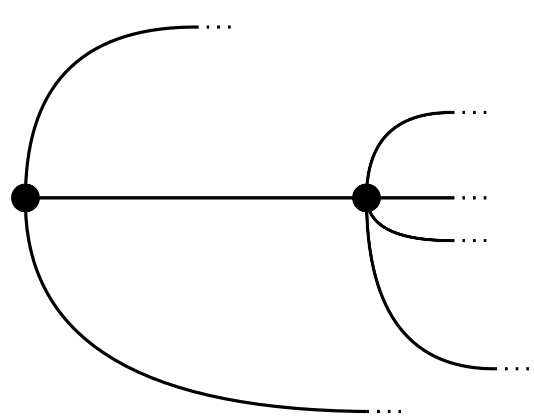

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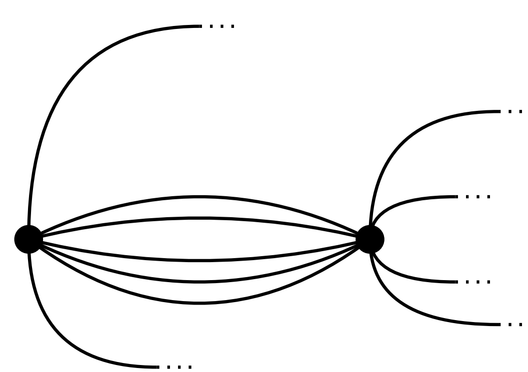

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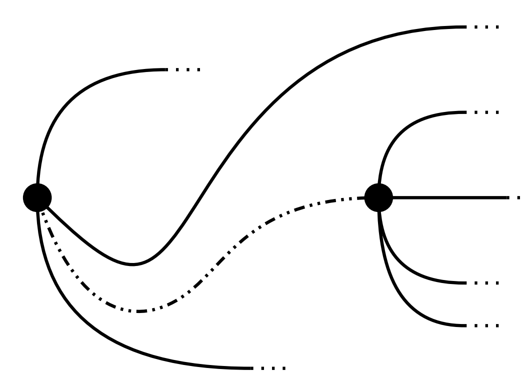

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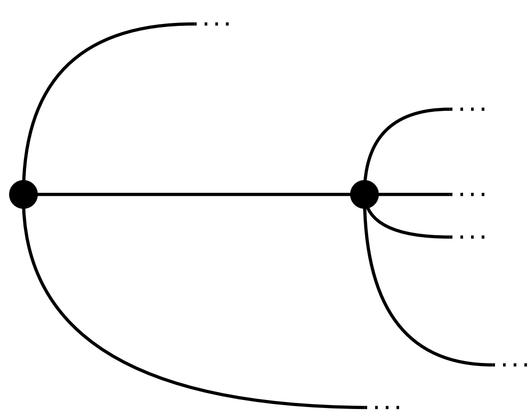

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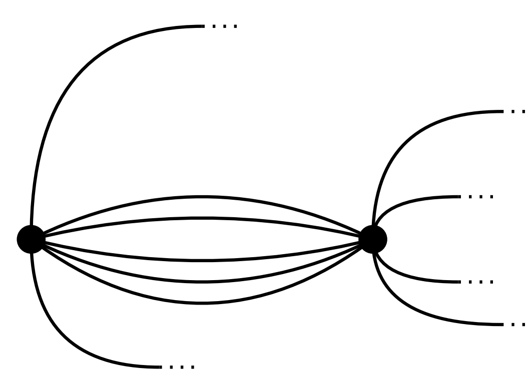

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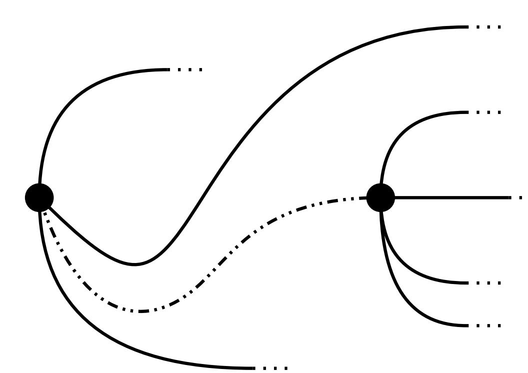

Case 1!

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Proof by Induction on nr of odd edges

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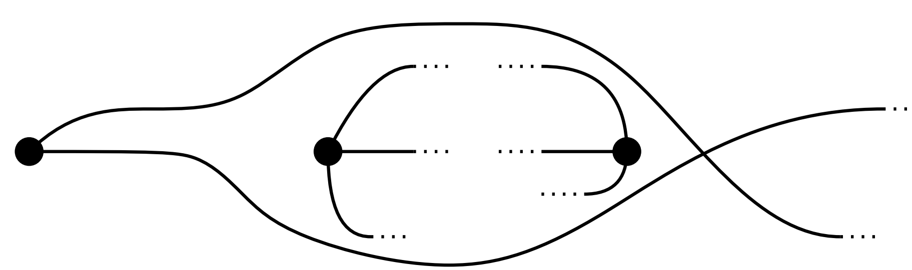

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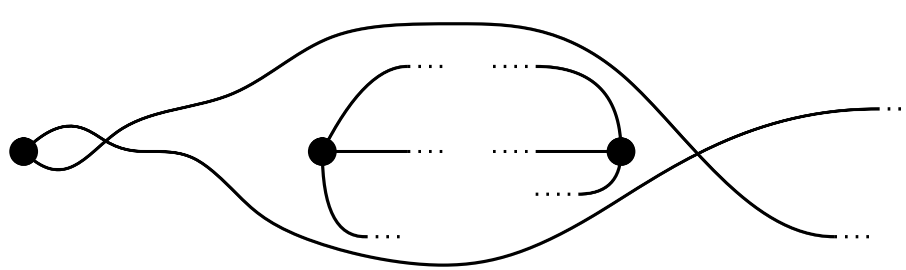

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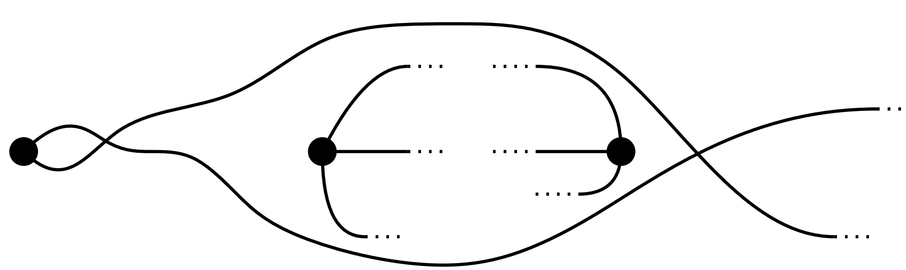

Induction Hypothesis

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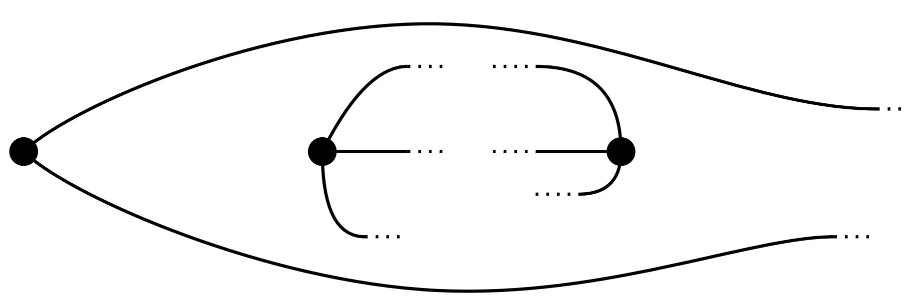

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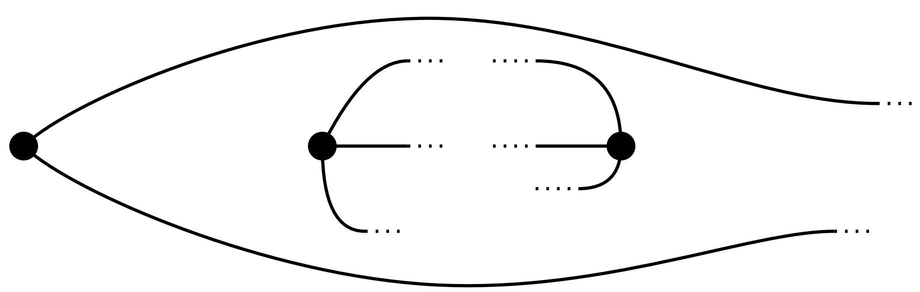

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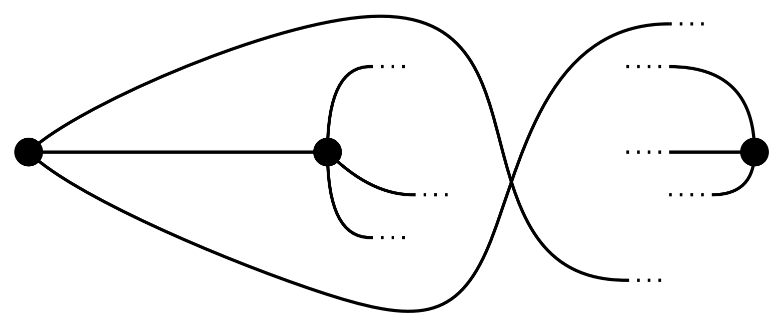

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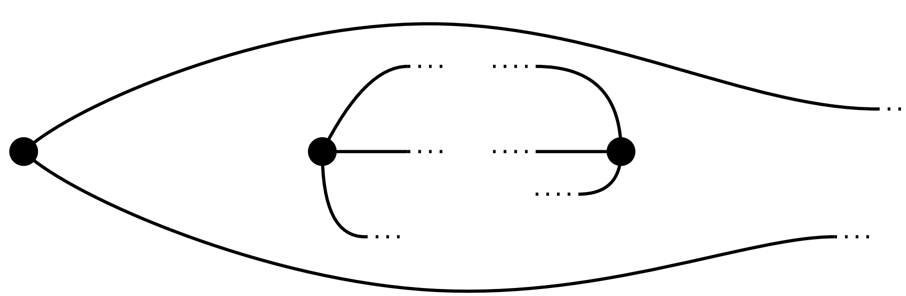

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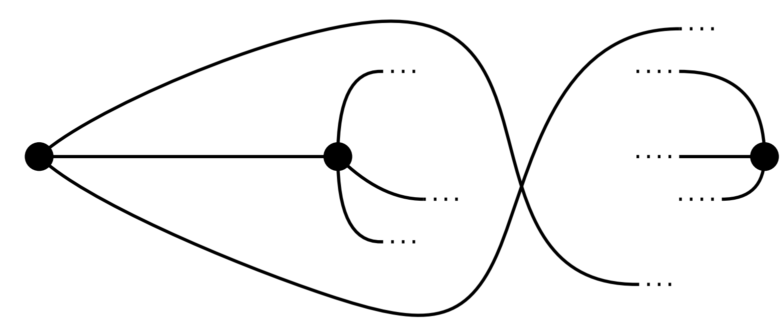

Contradiction

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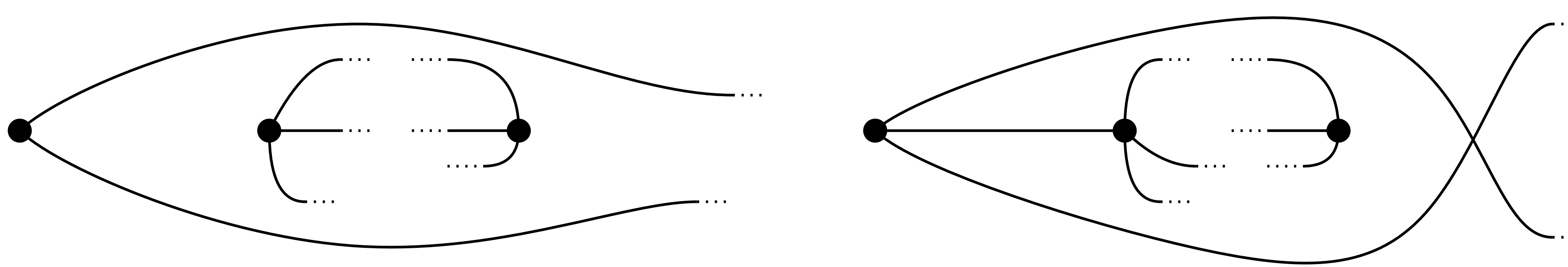

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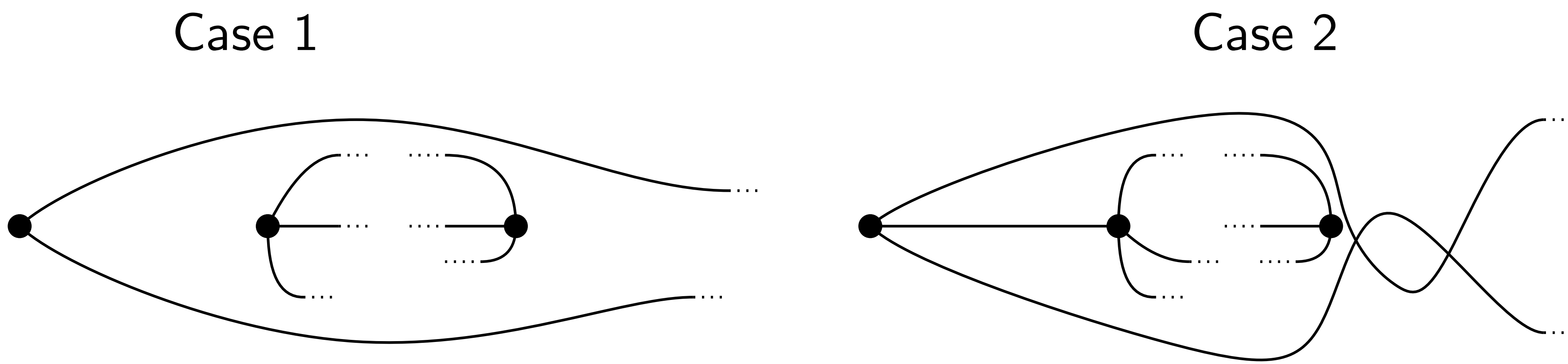

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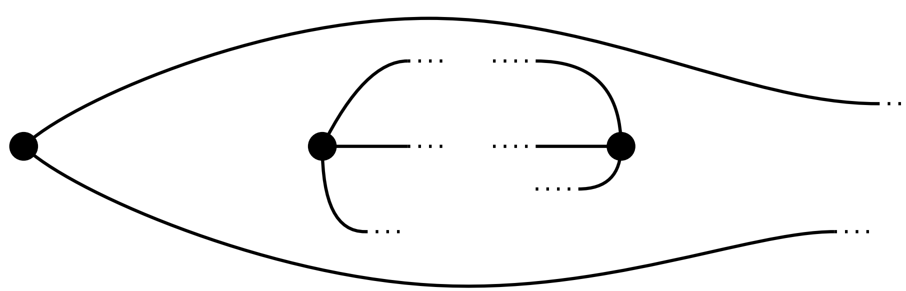

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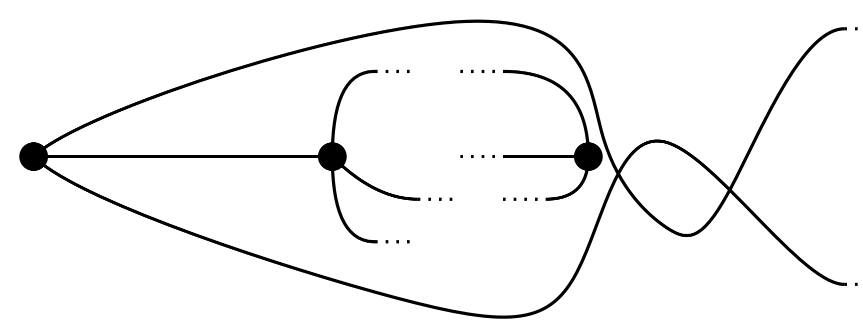

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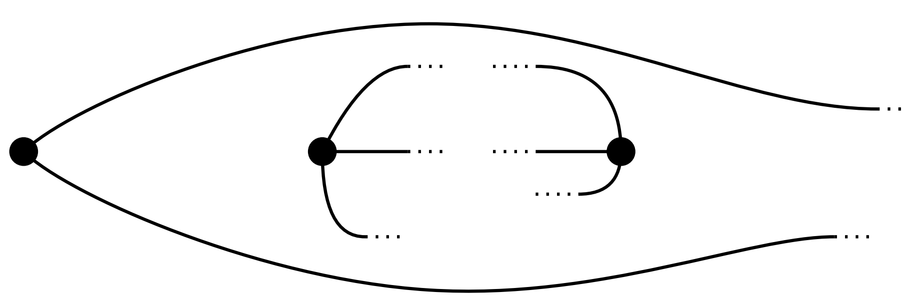

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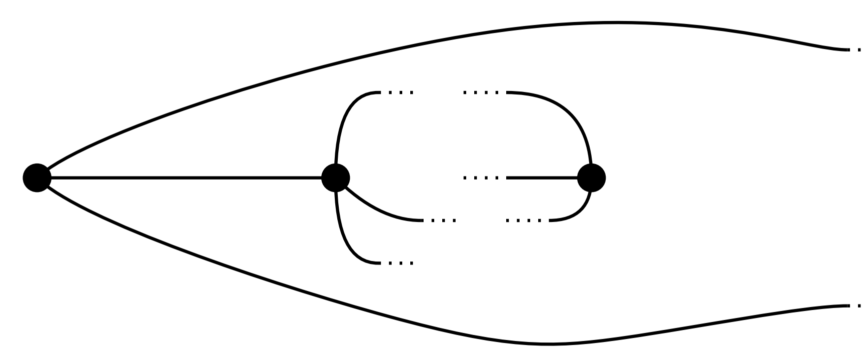
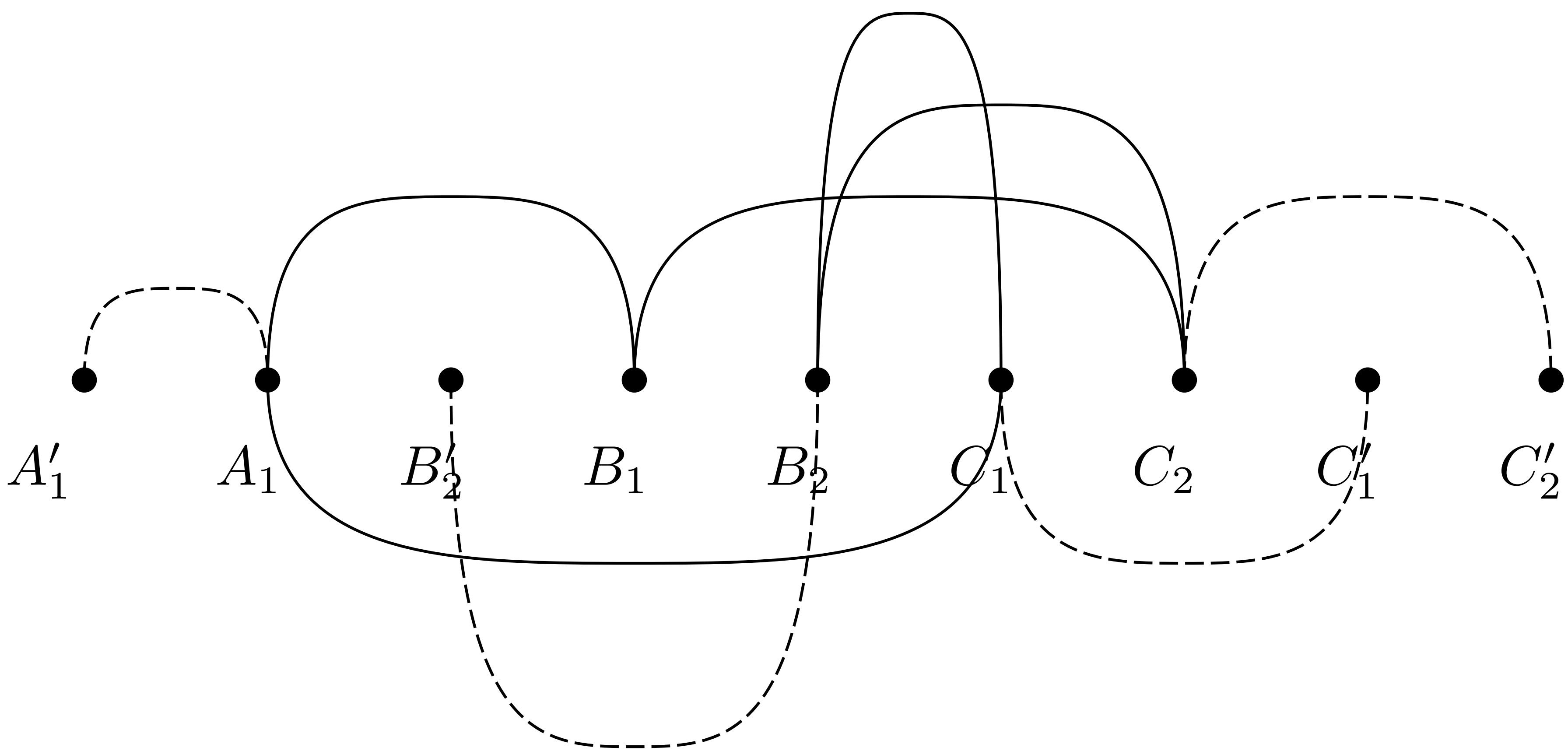


Induction Hypothesis

Remove Crossings



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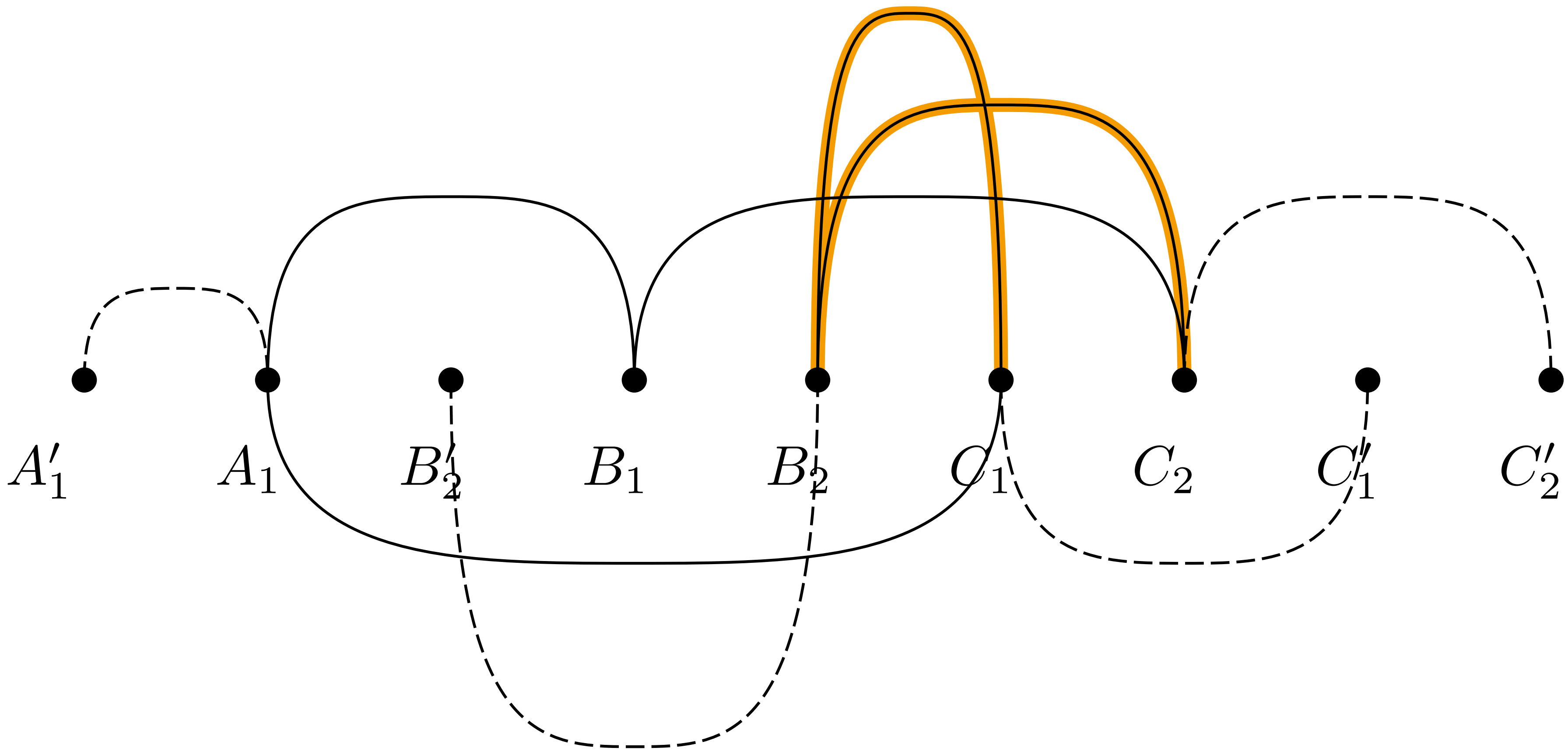

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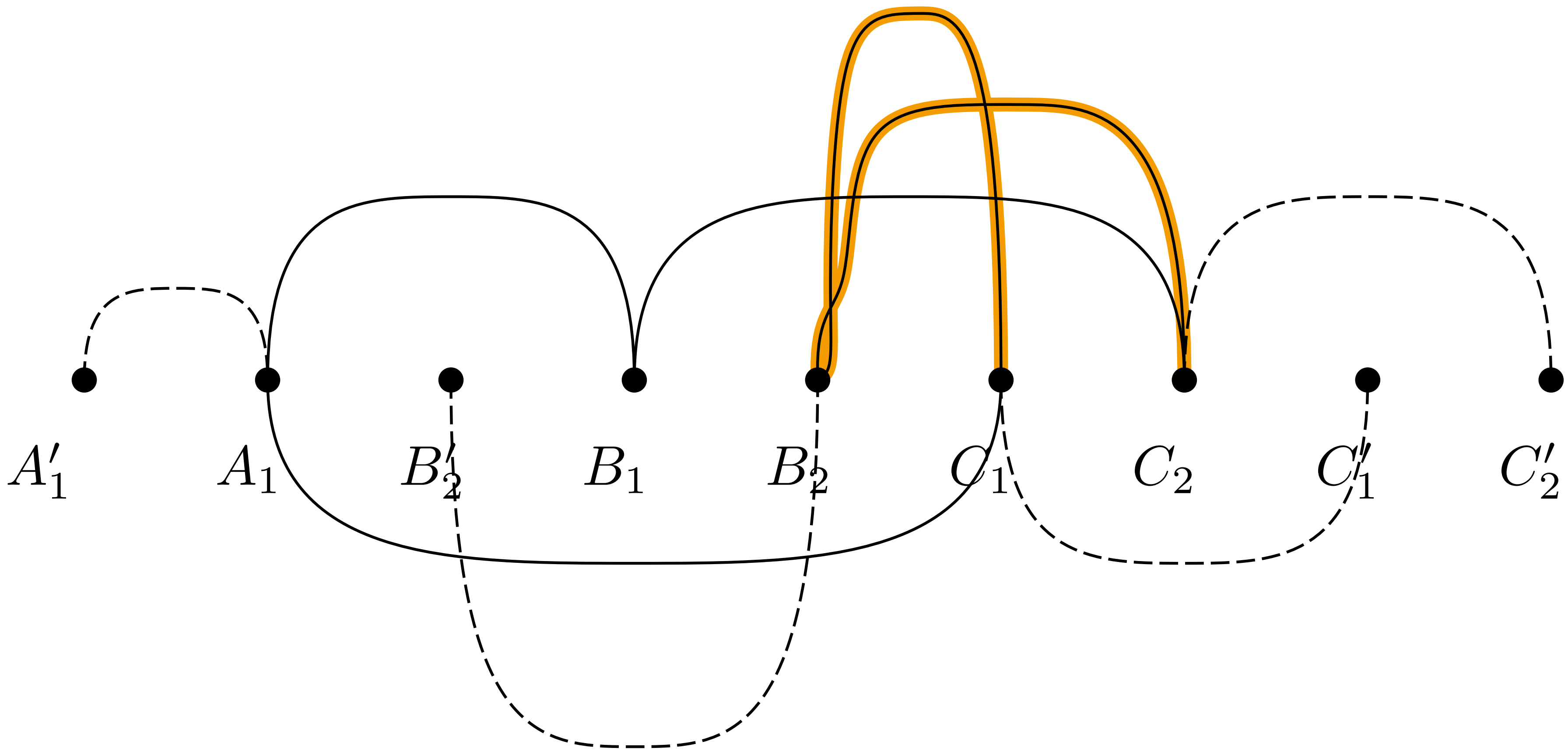

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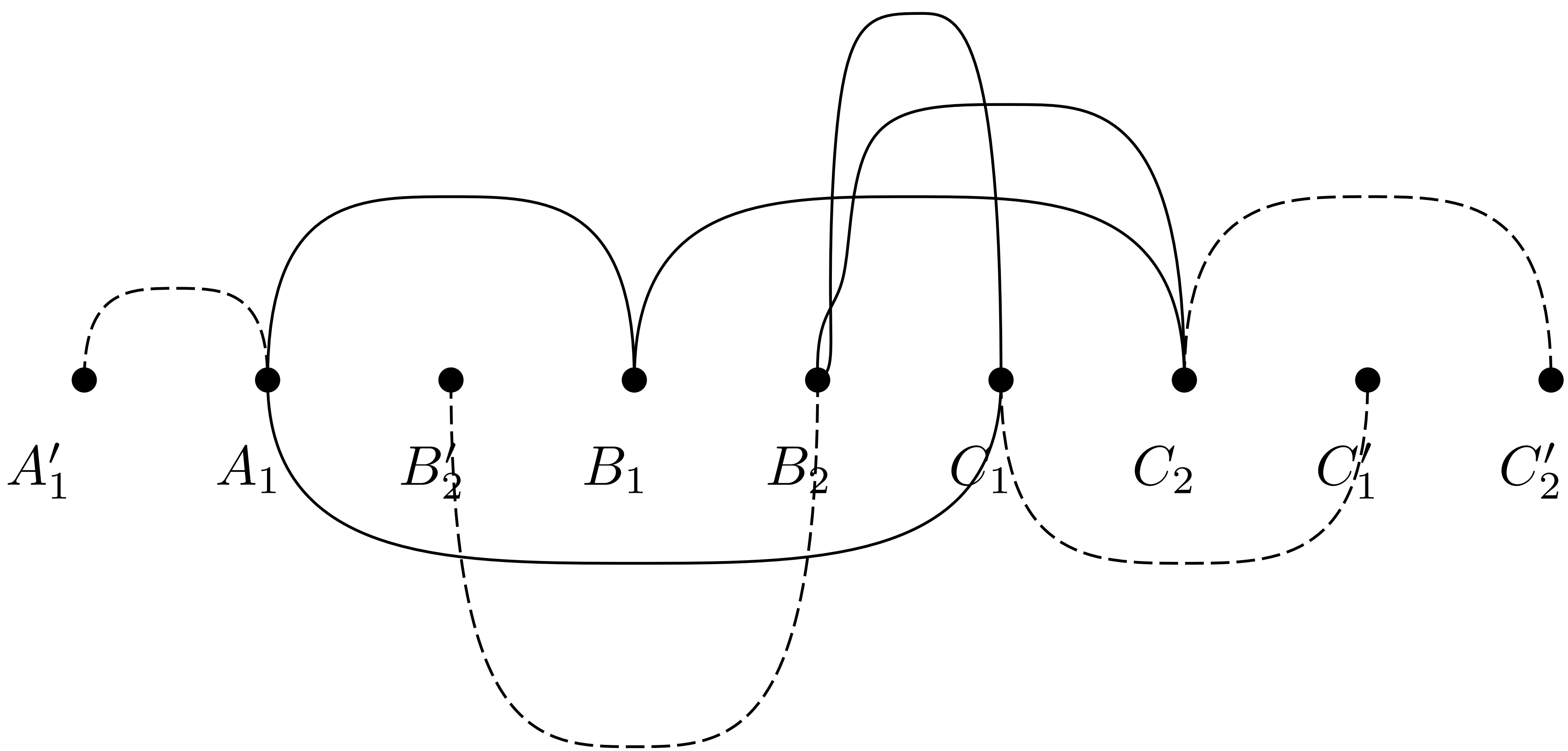

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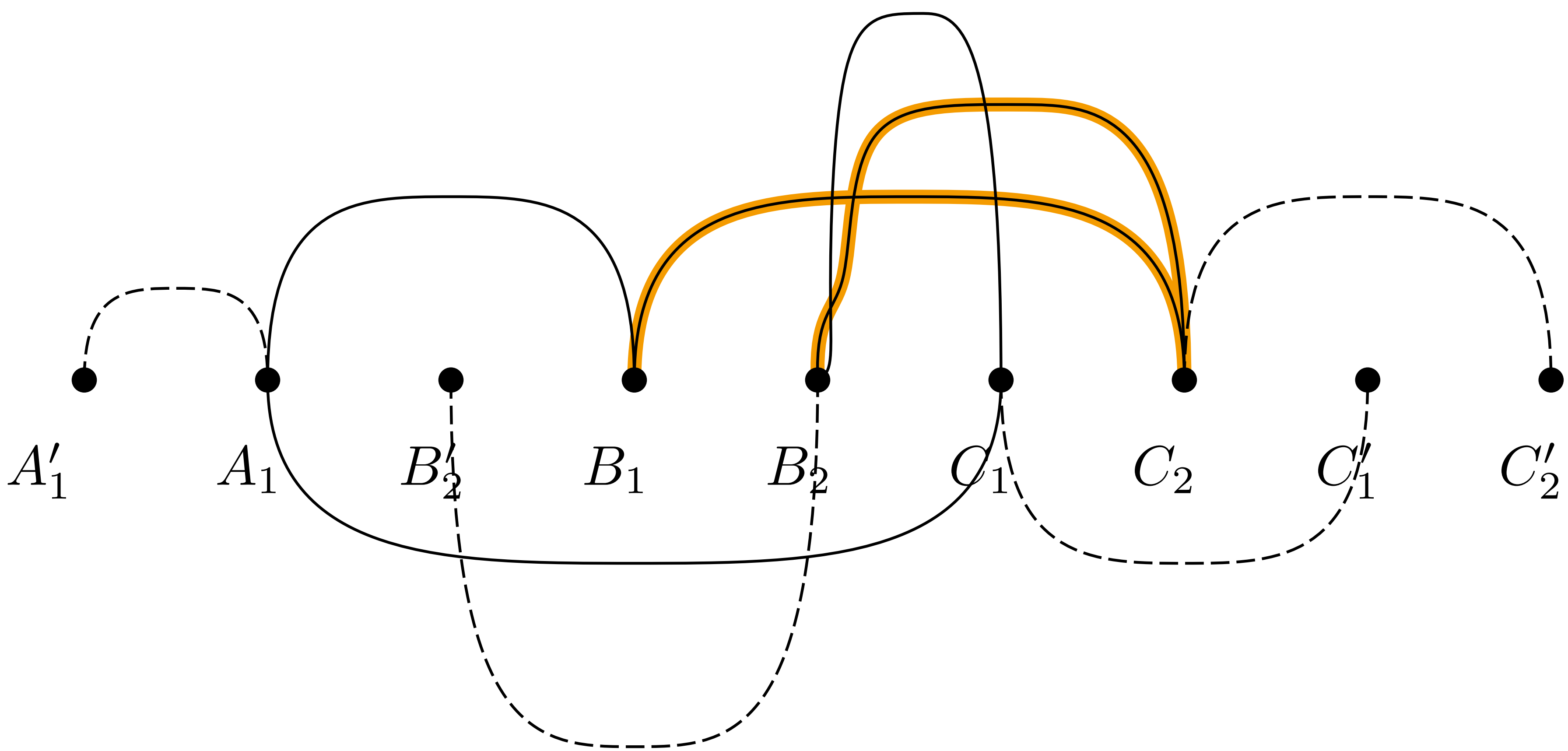

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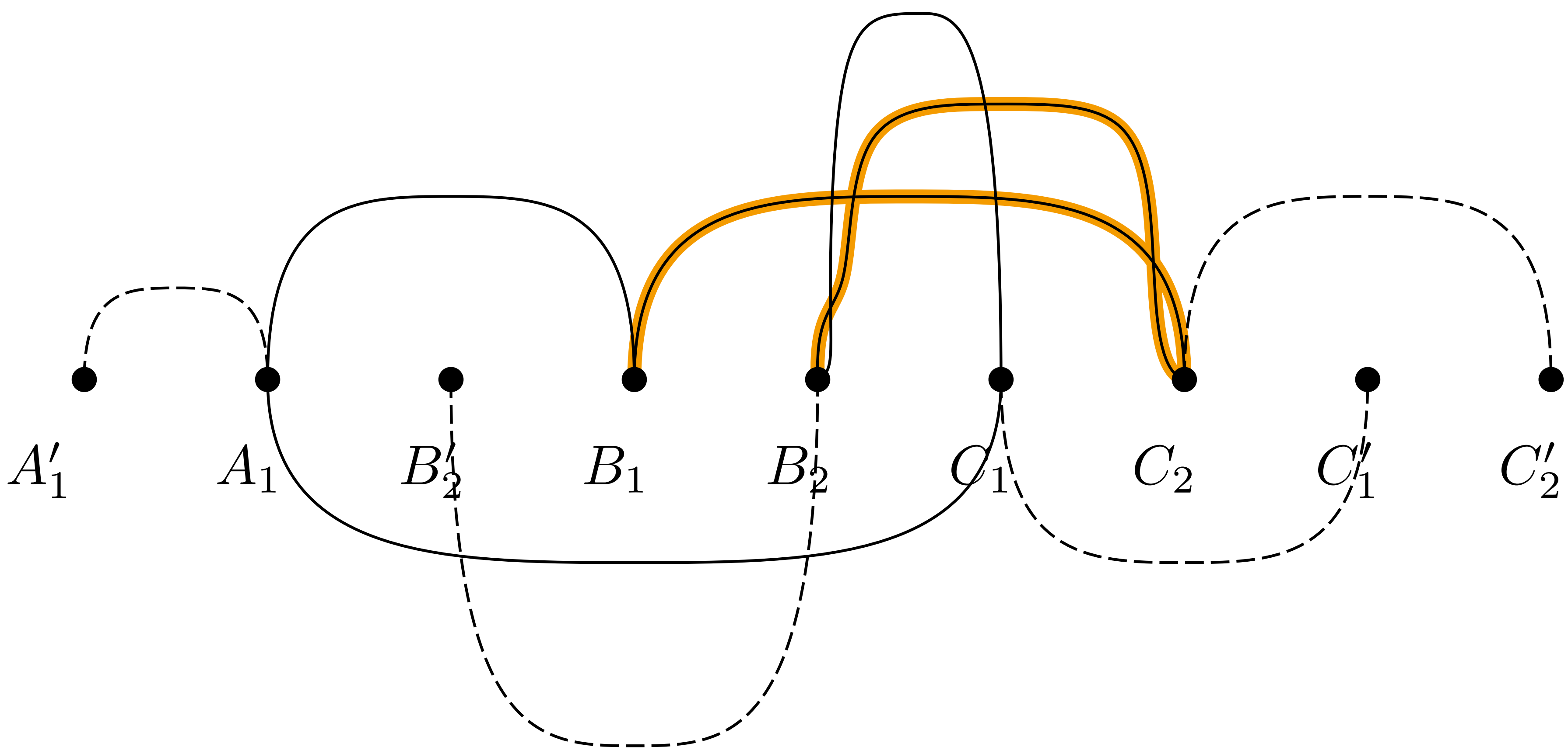

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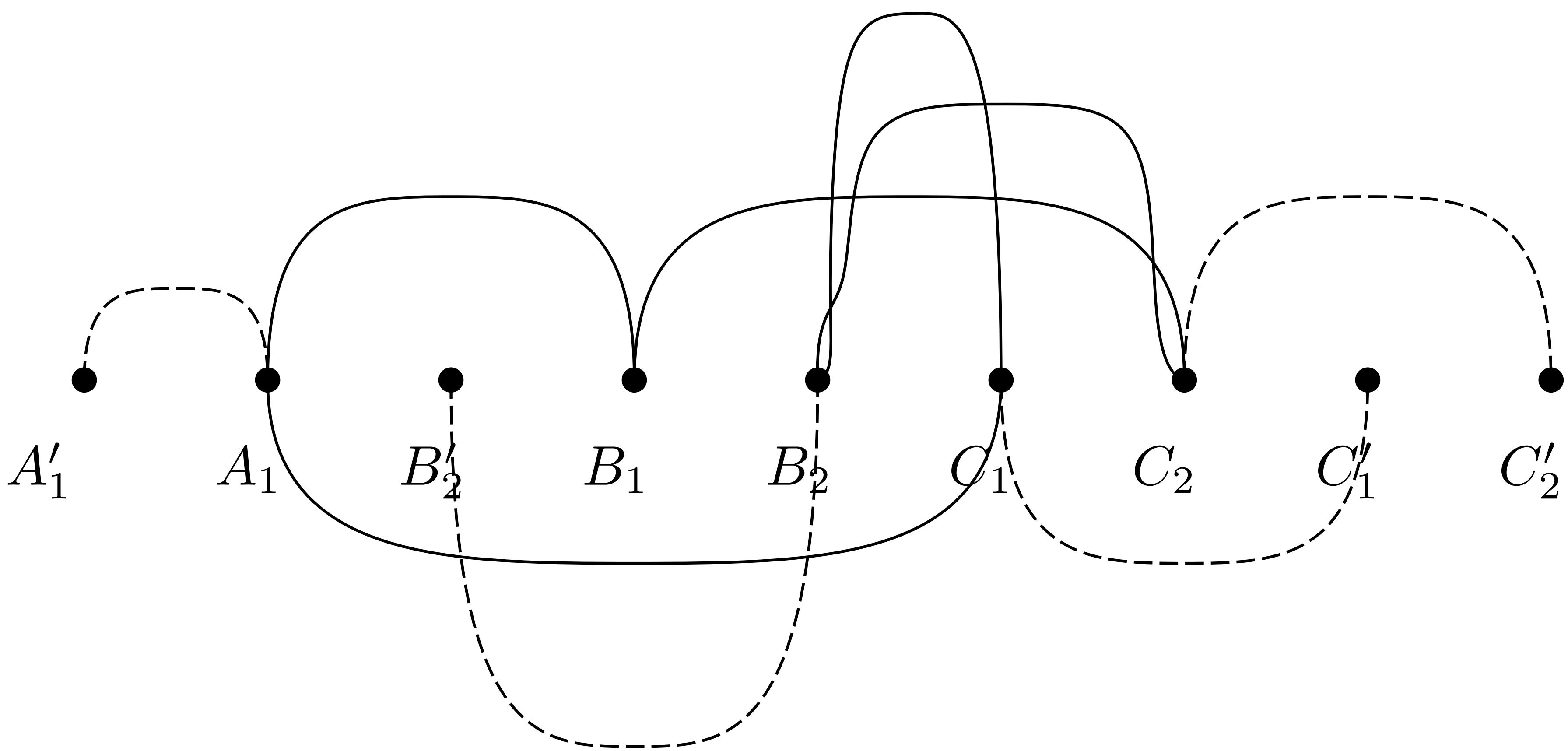

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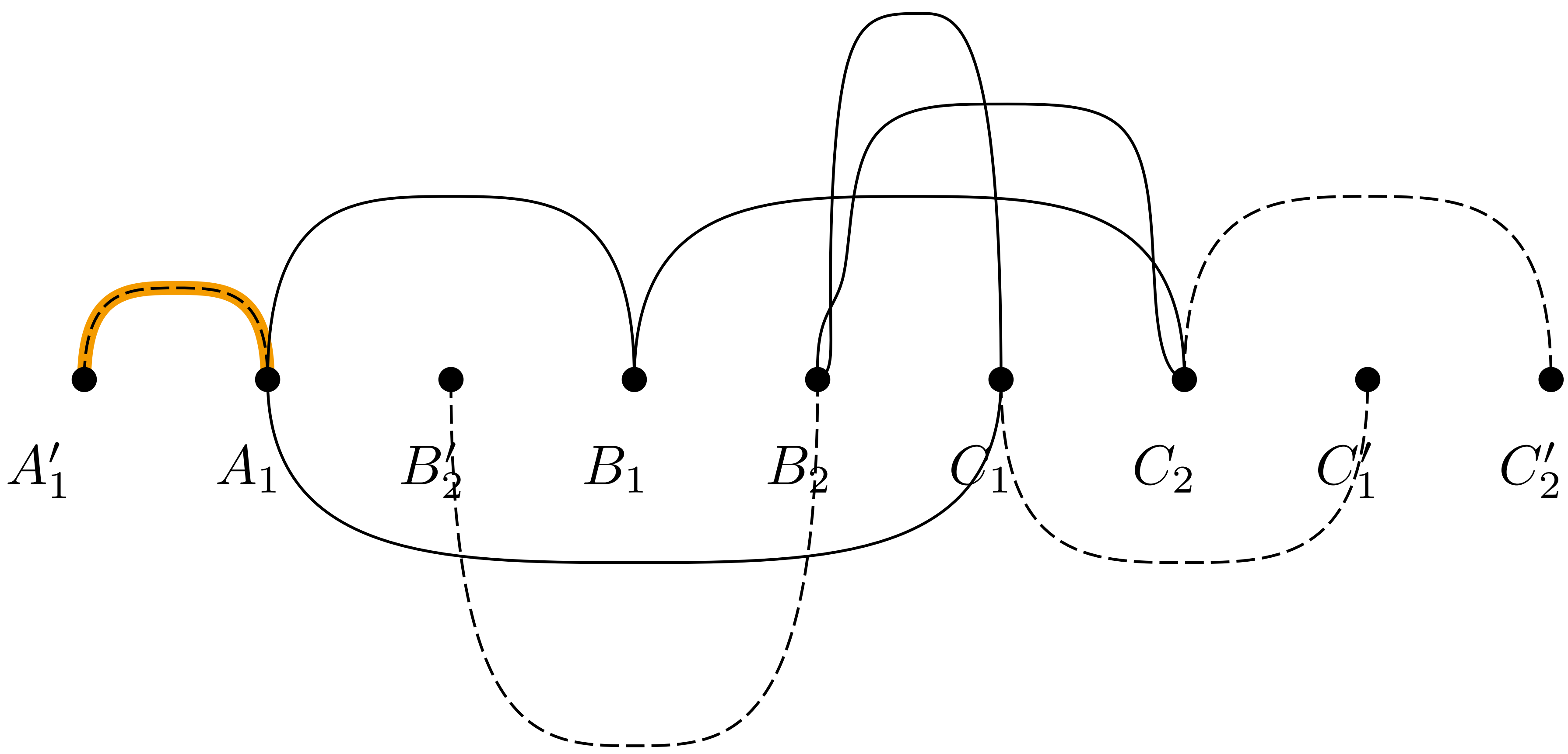

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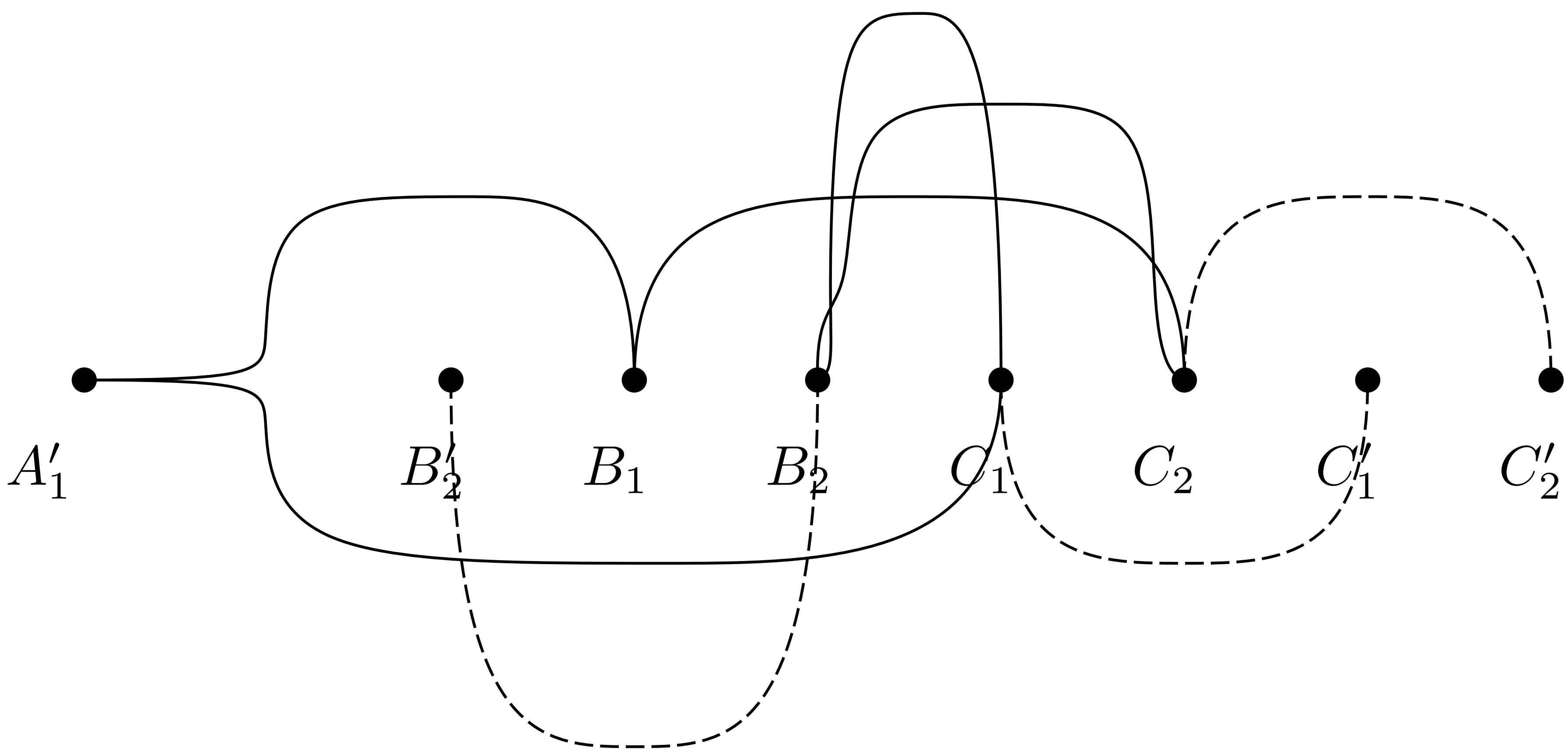

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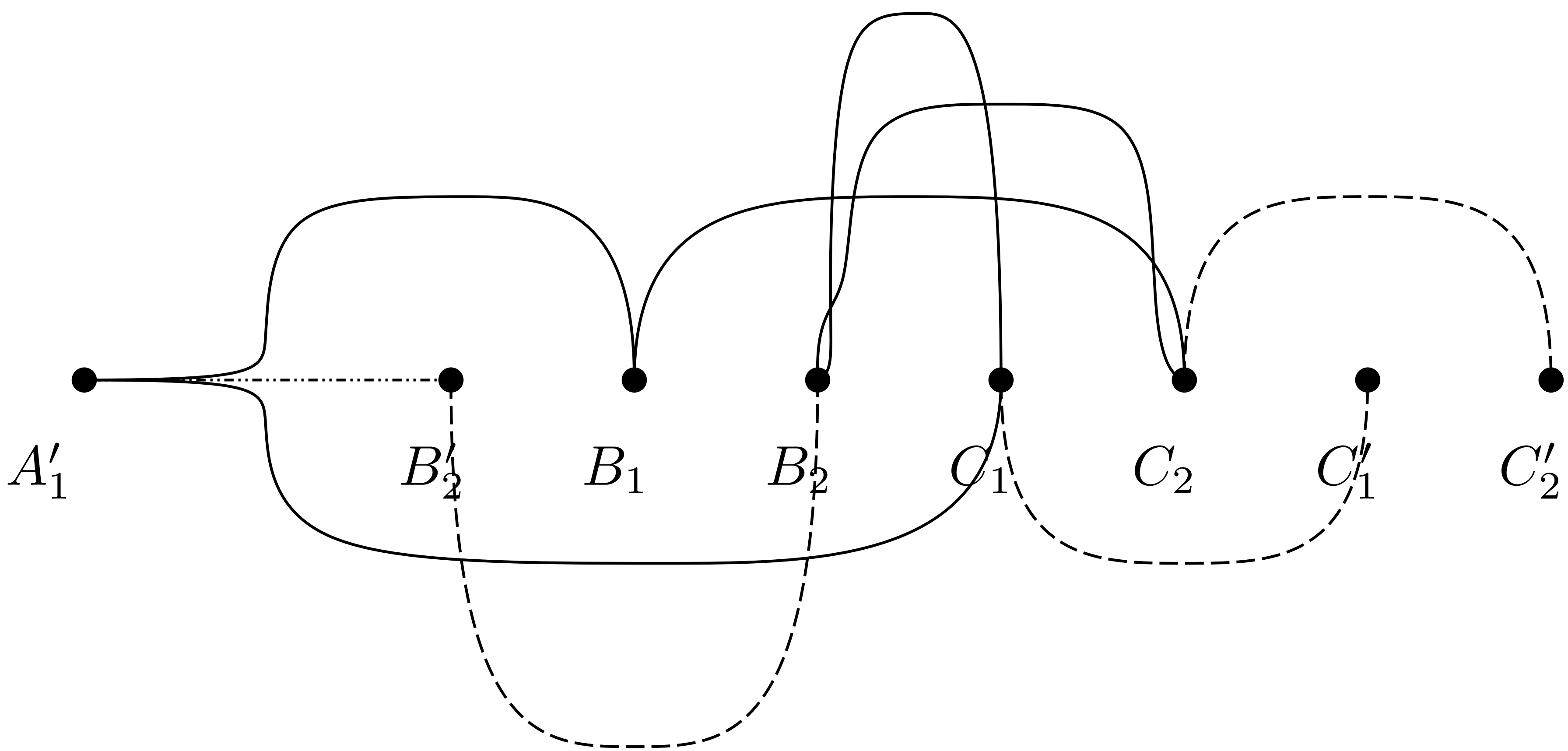

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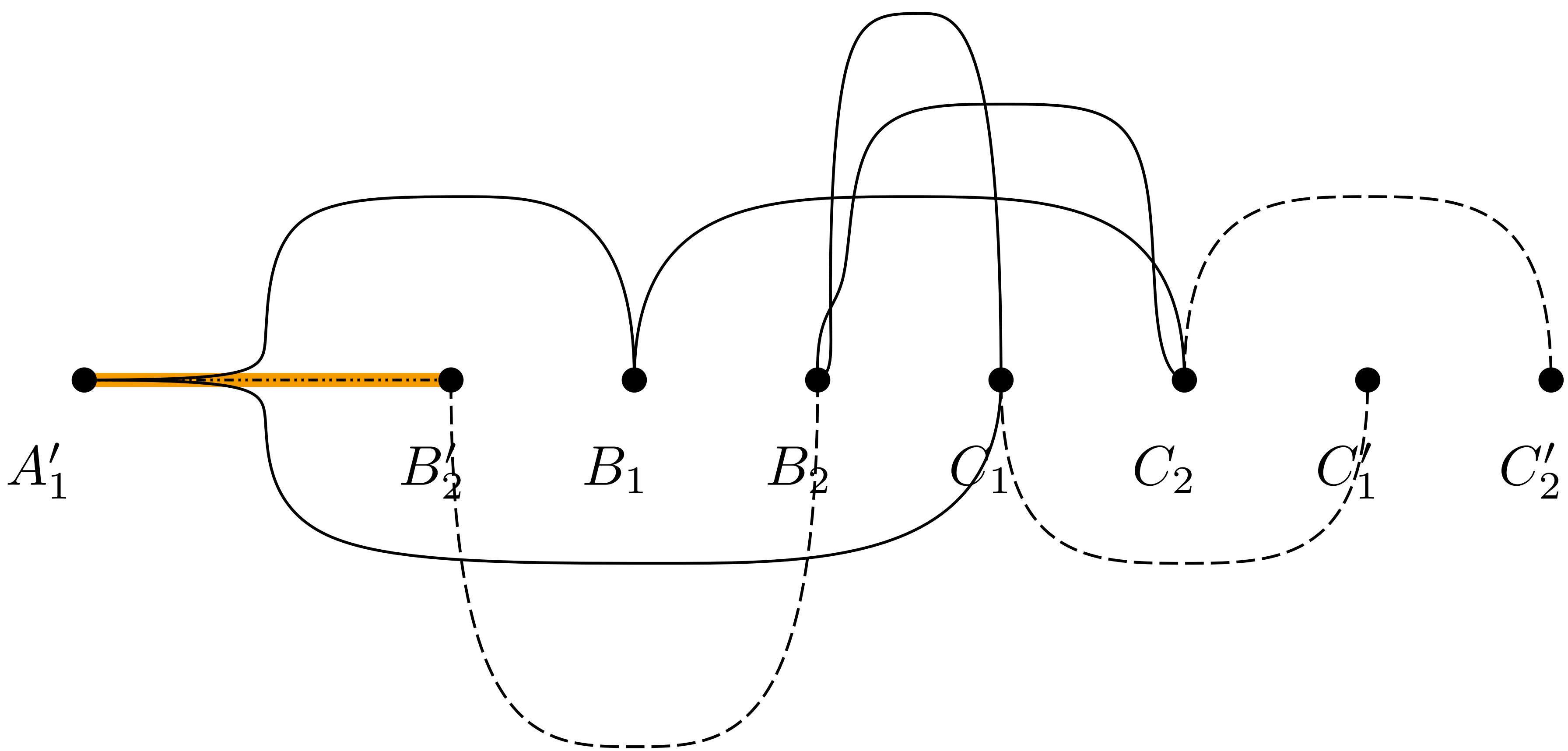

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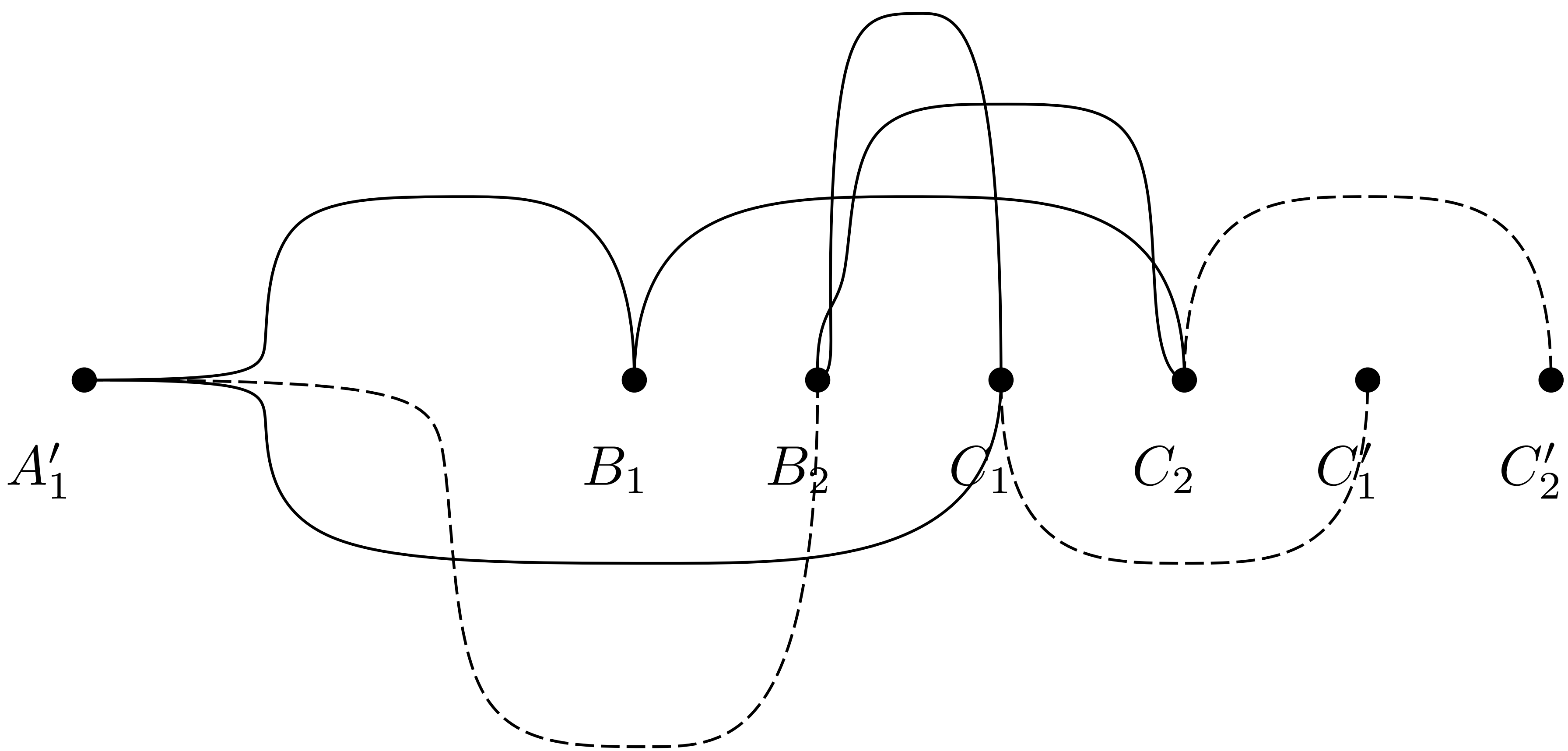

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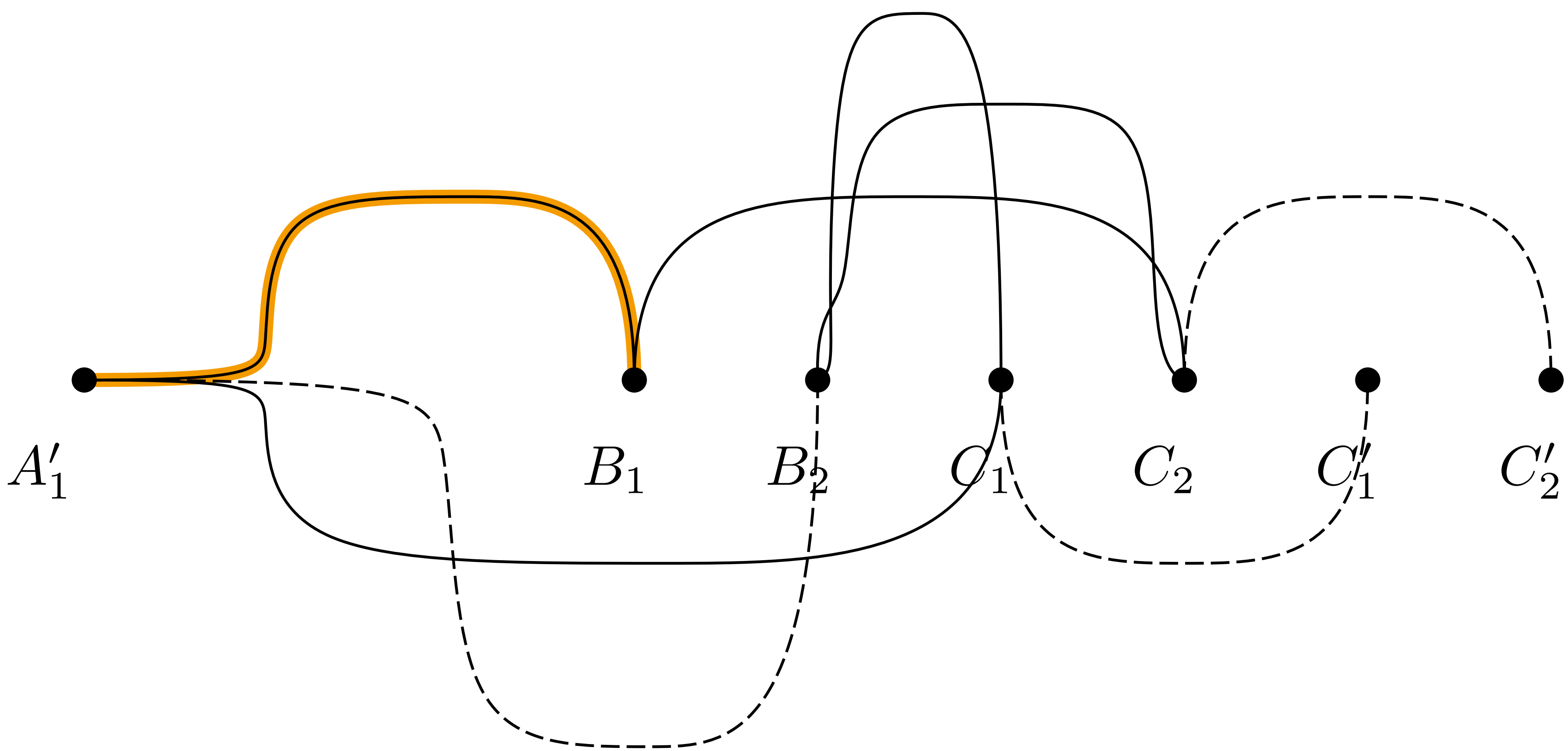

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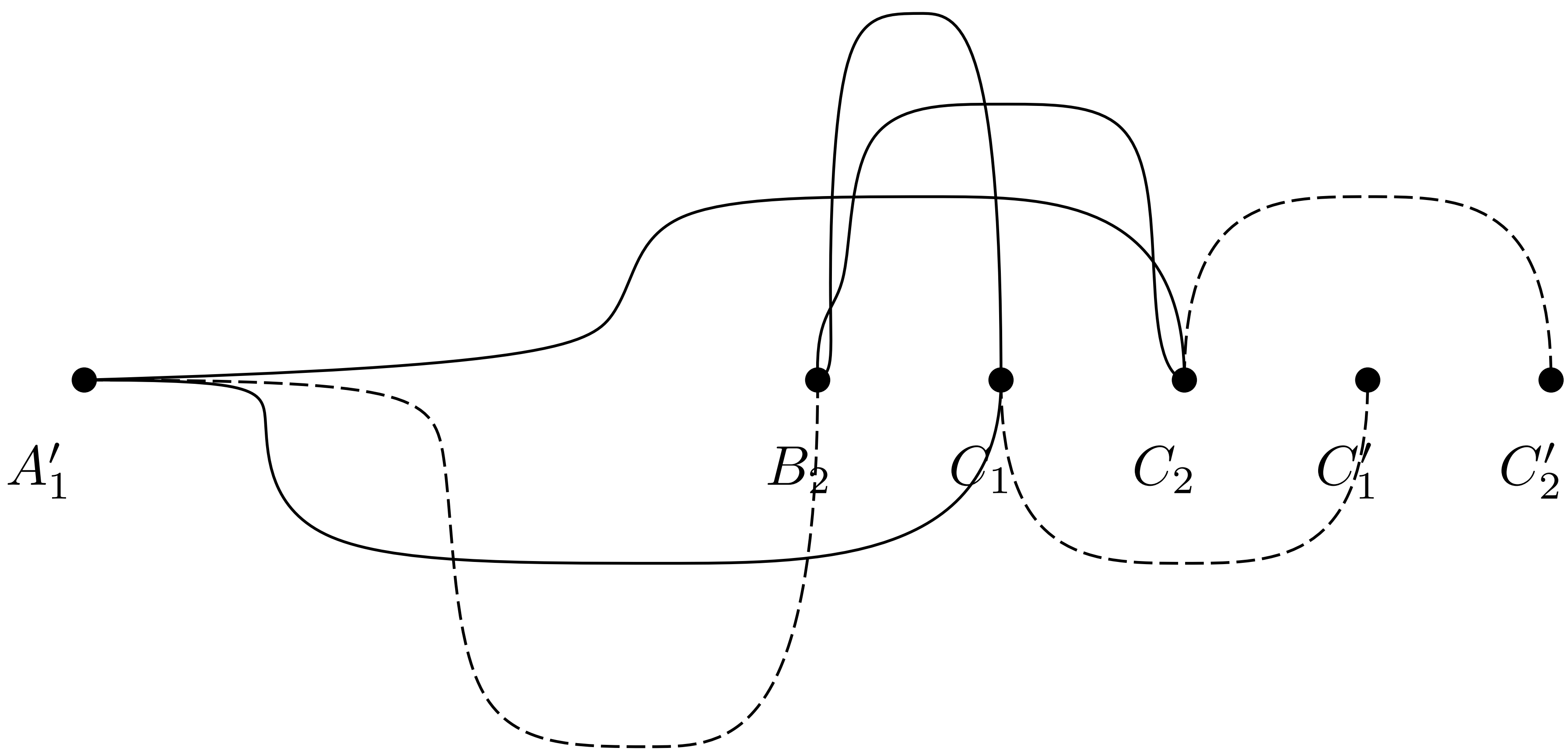

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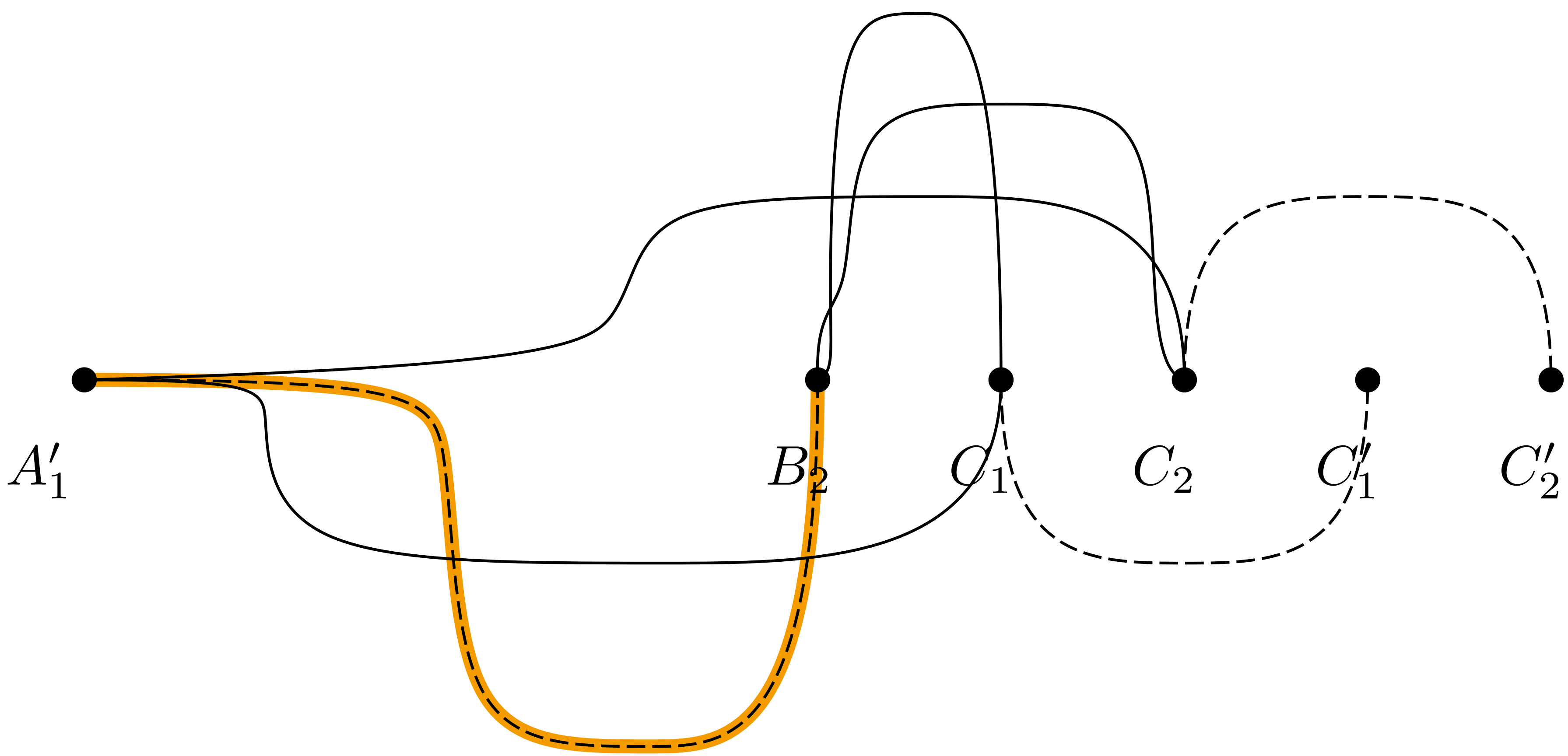

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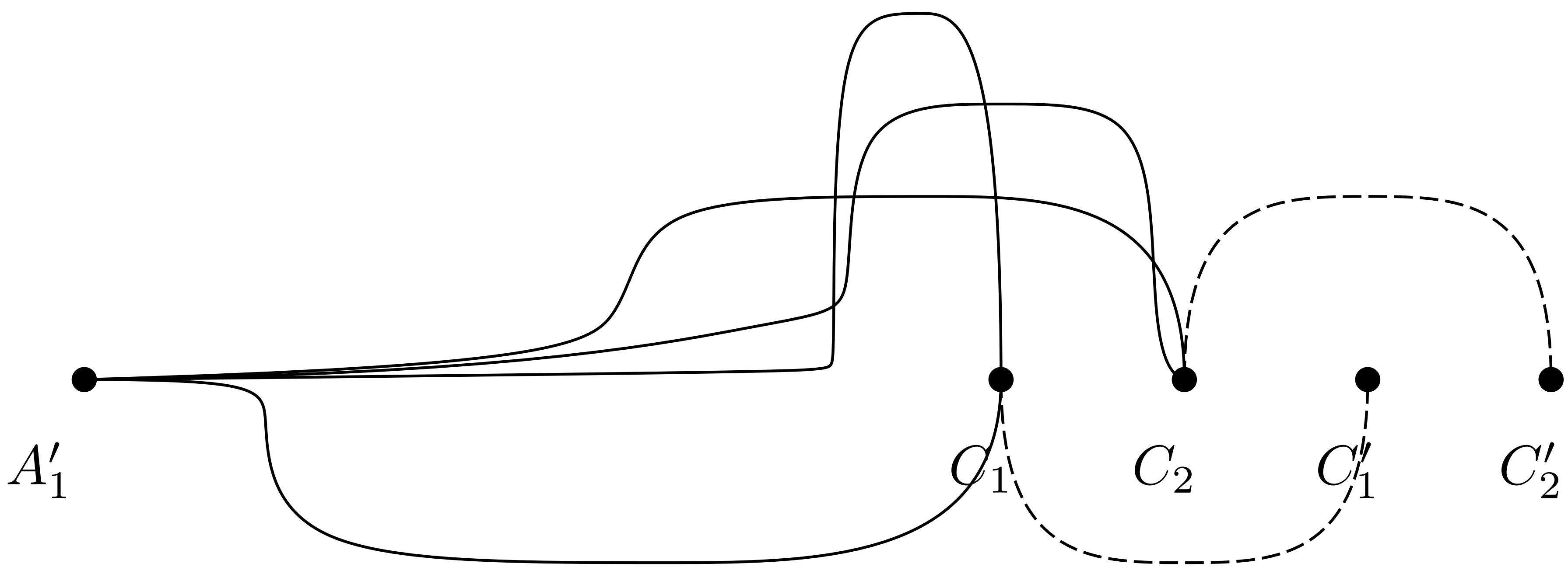

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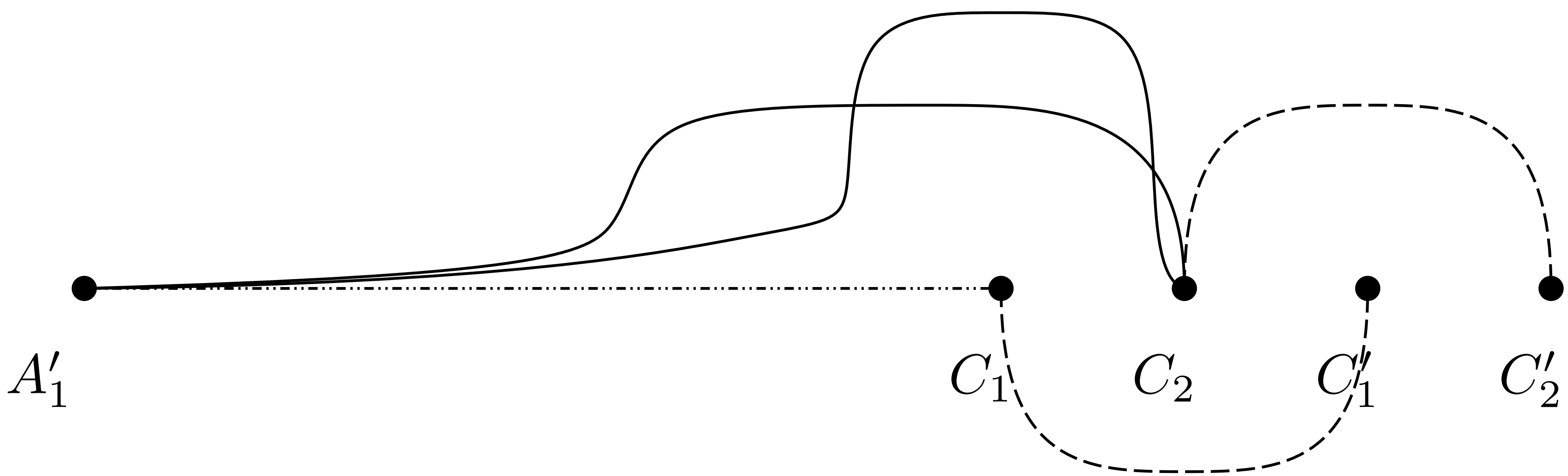

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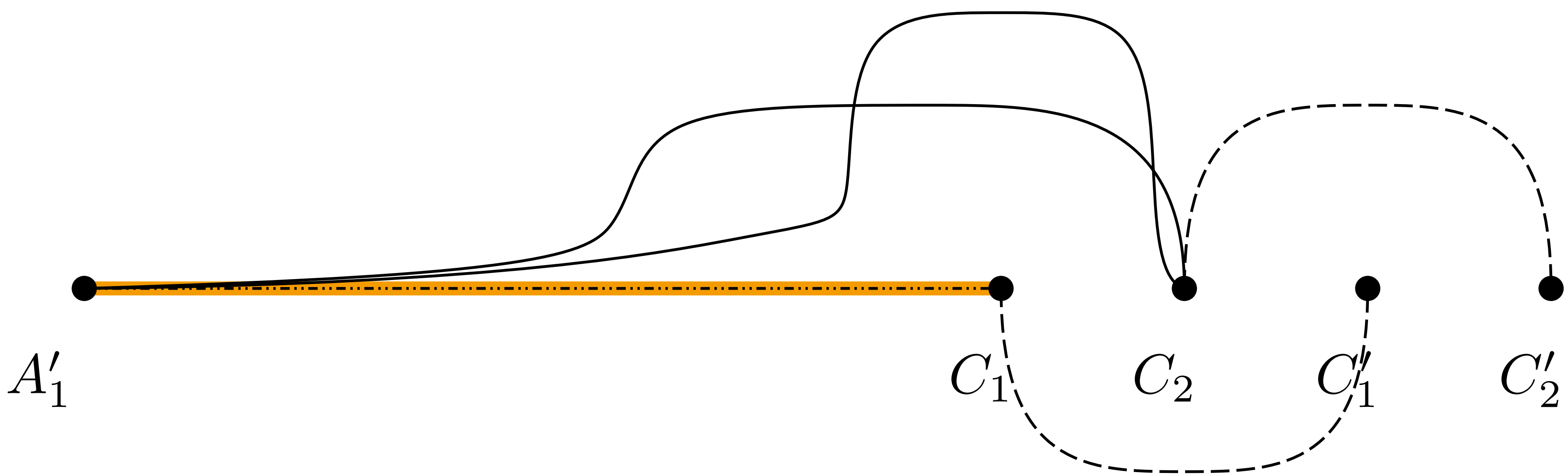

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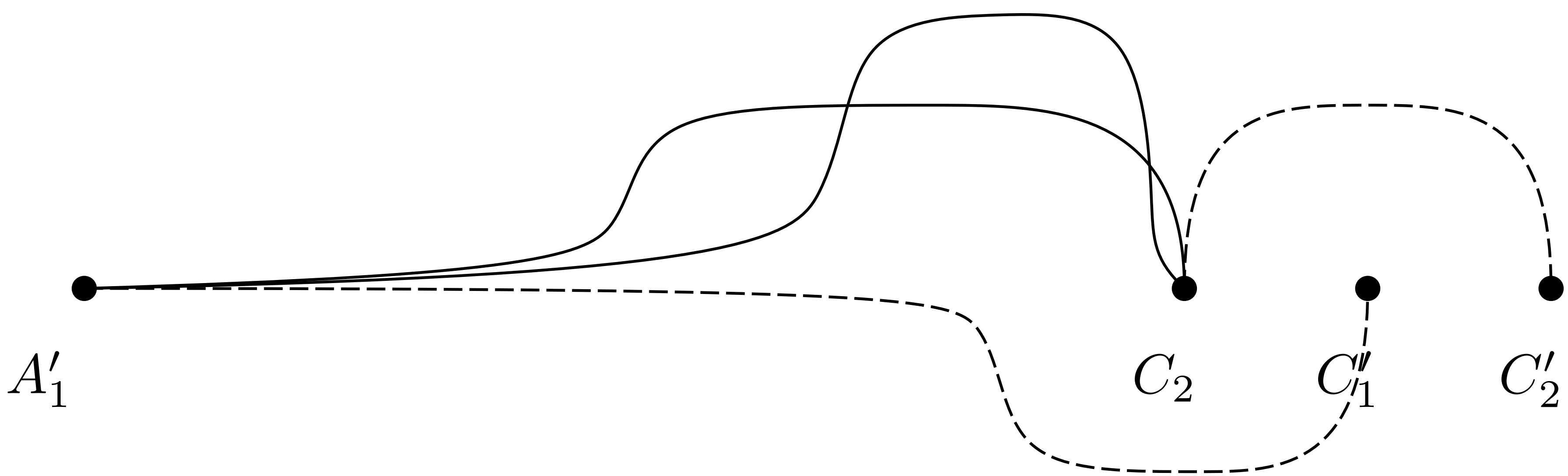

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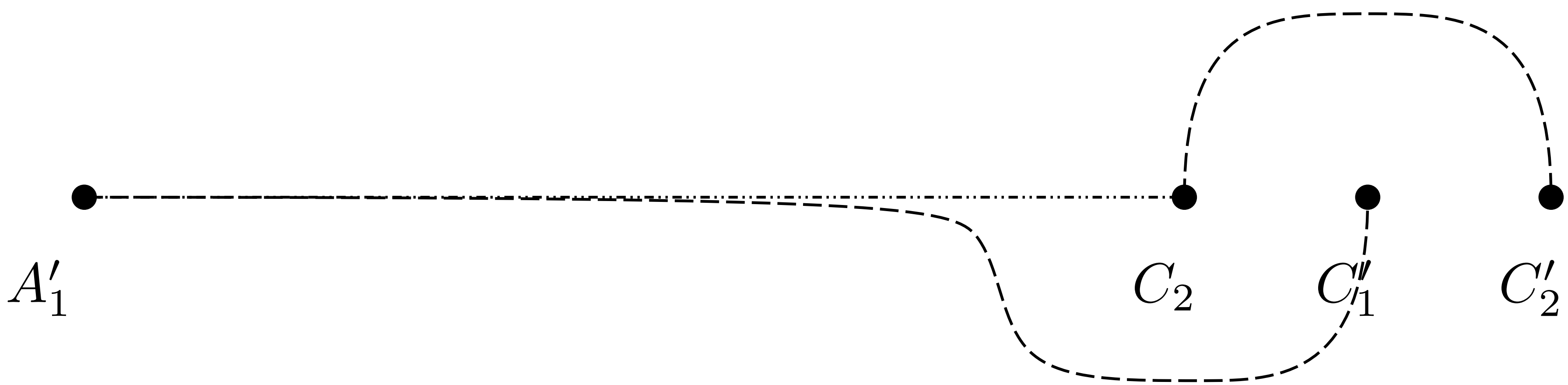

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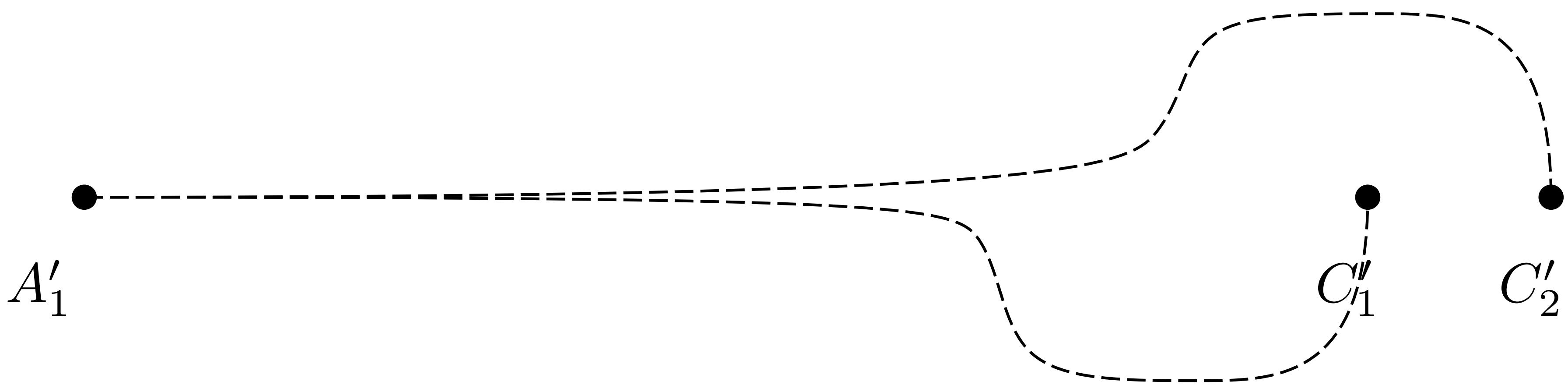

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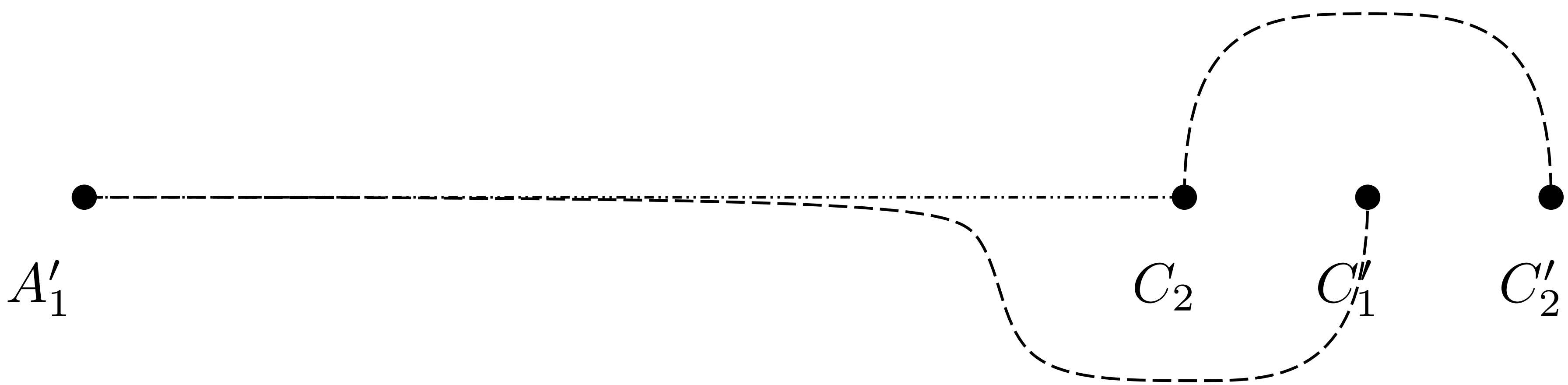

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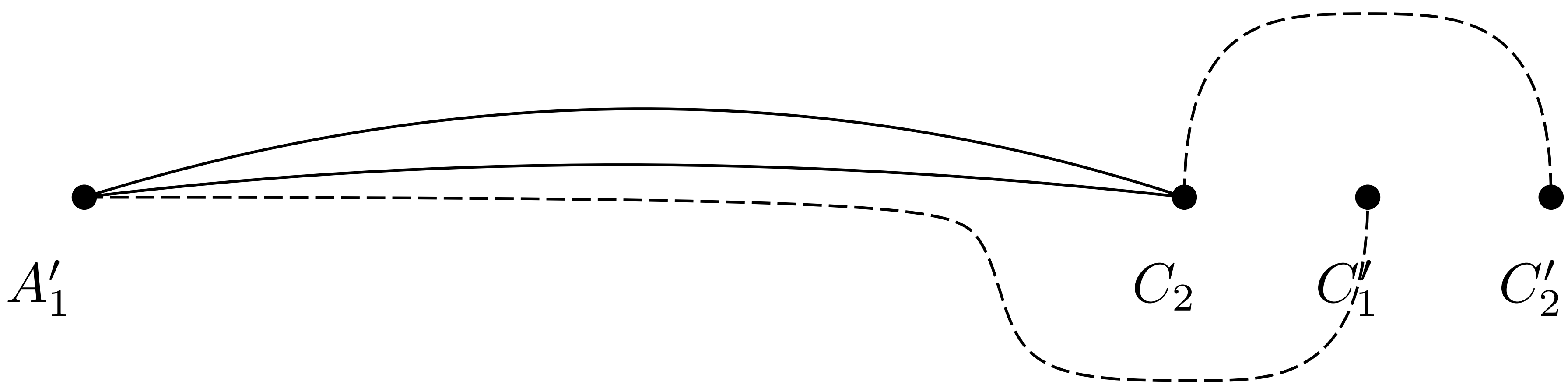

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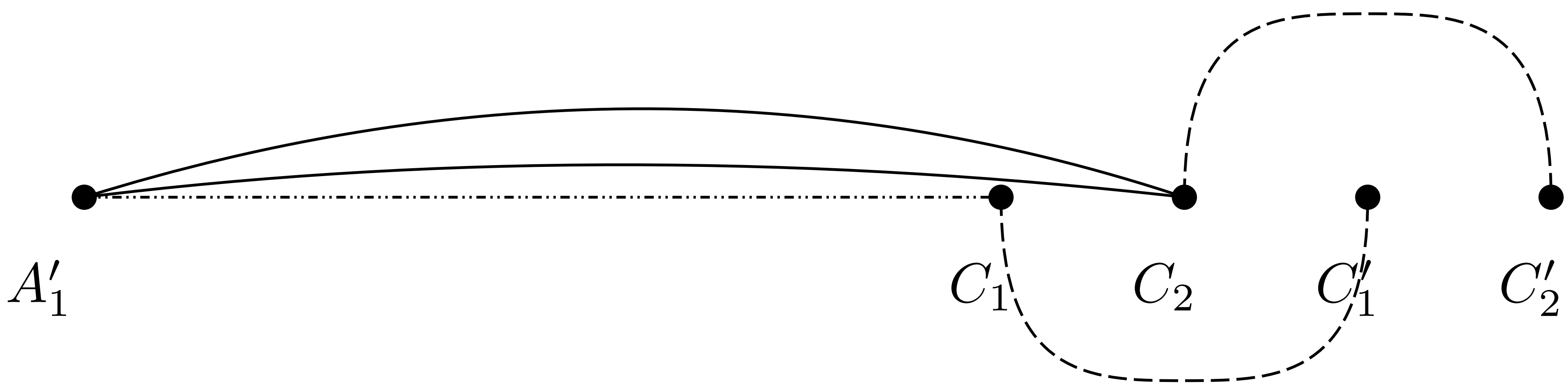

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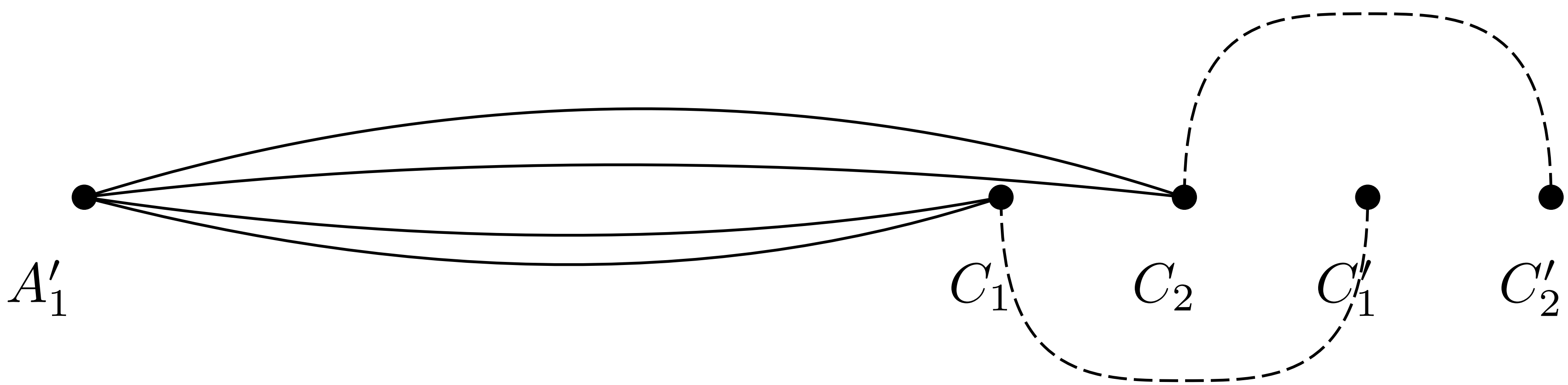

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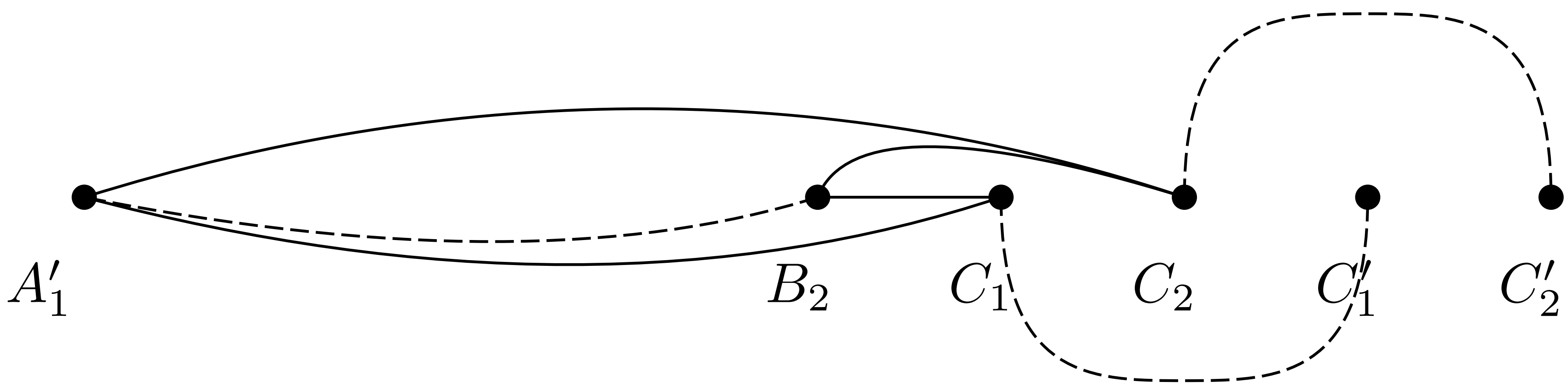
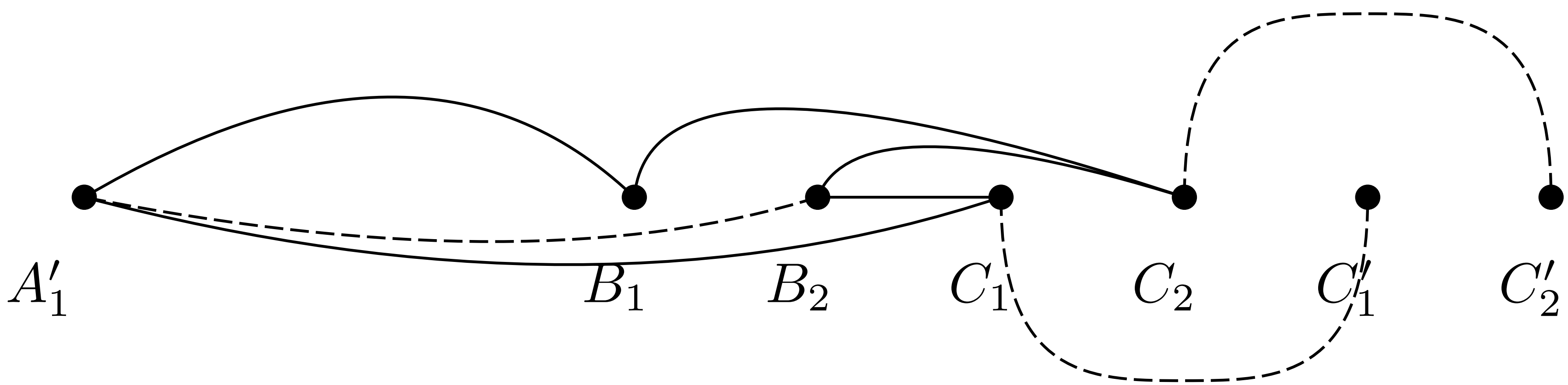


Remove Crossings



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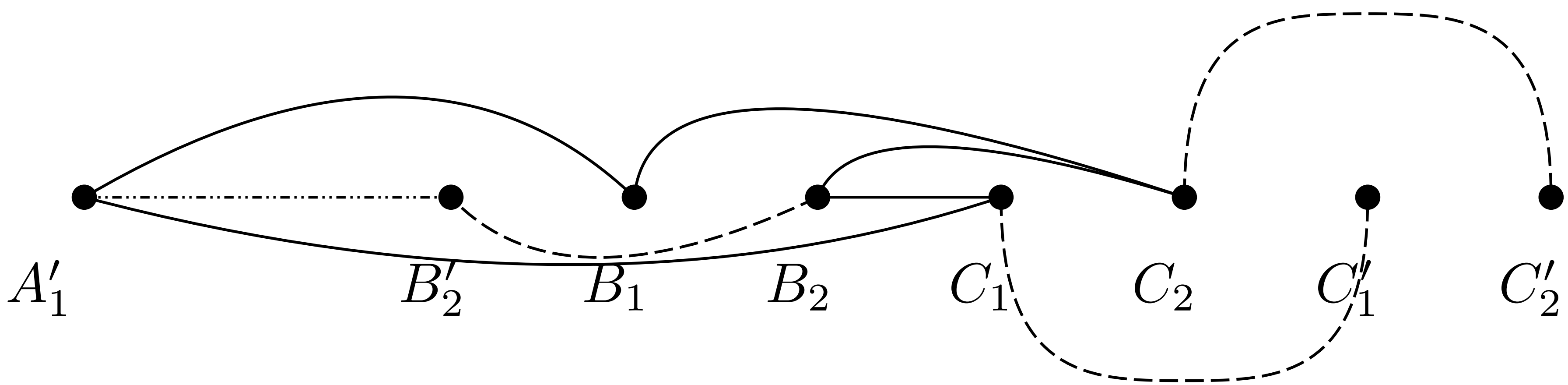
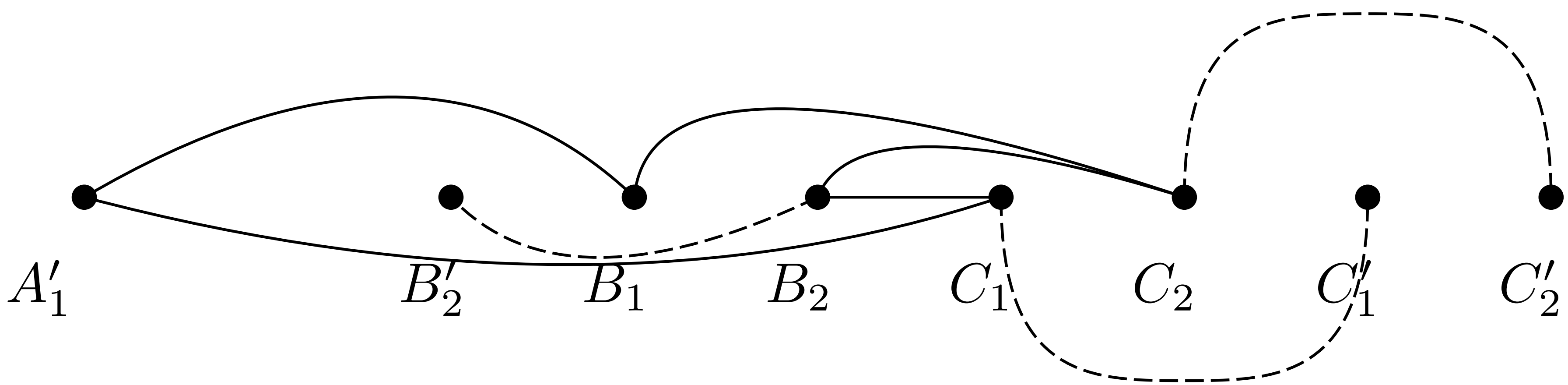
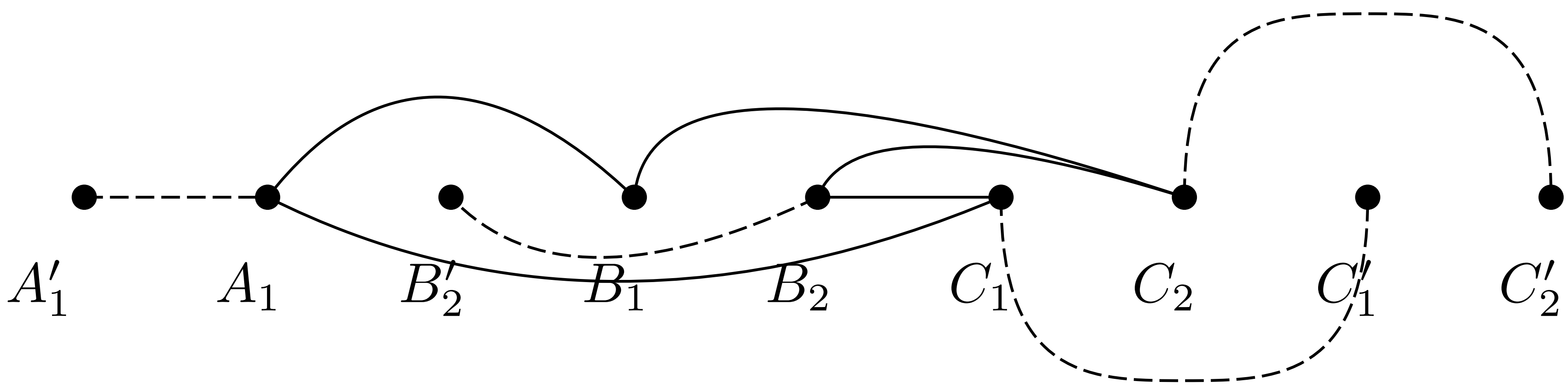


Remove Crossings

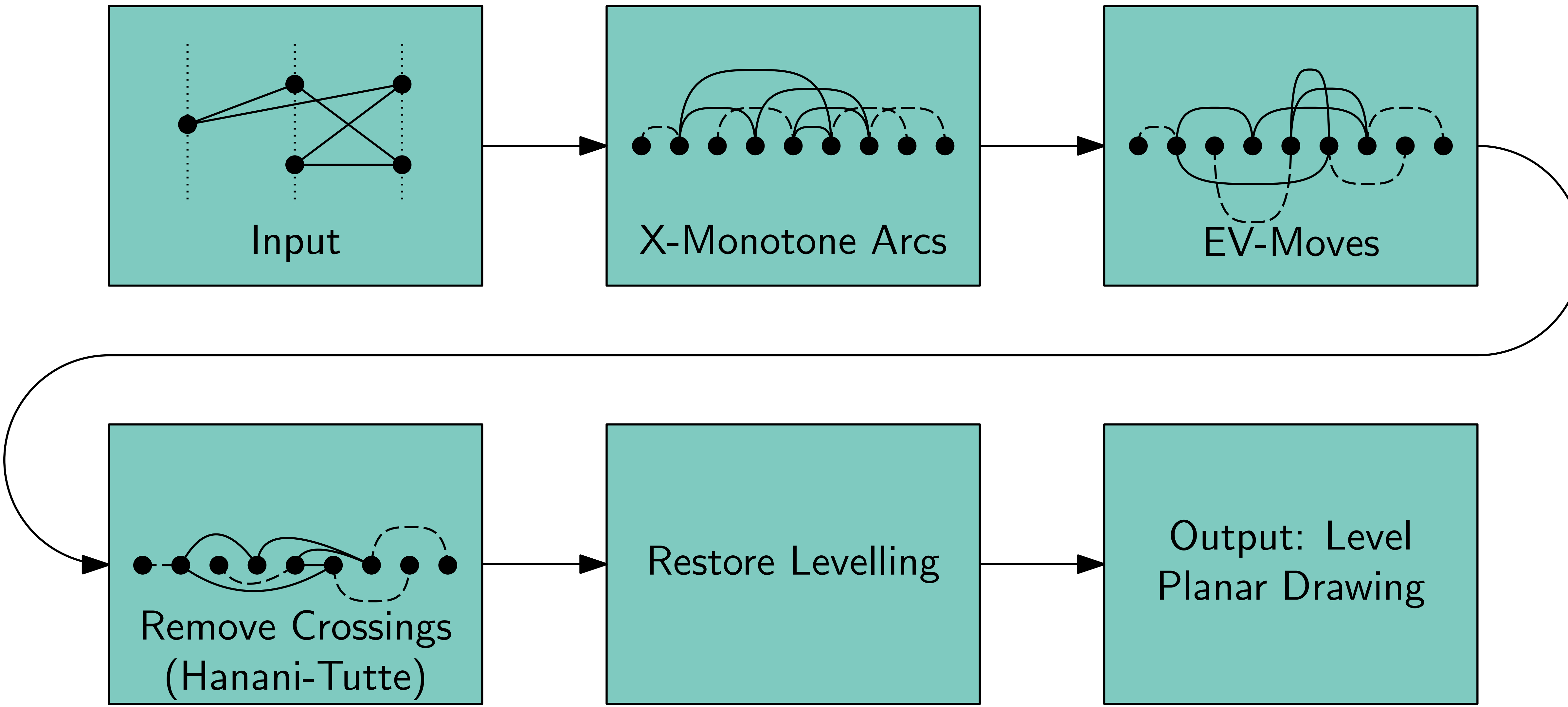


Remove Crossings

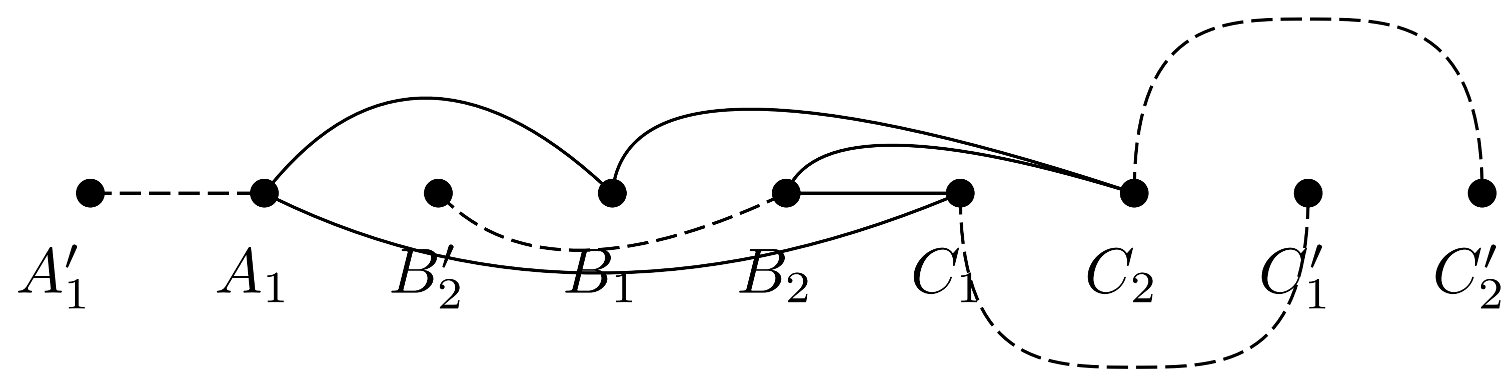


Leveled Embedding Construction

Algorithm

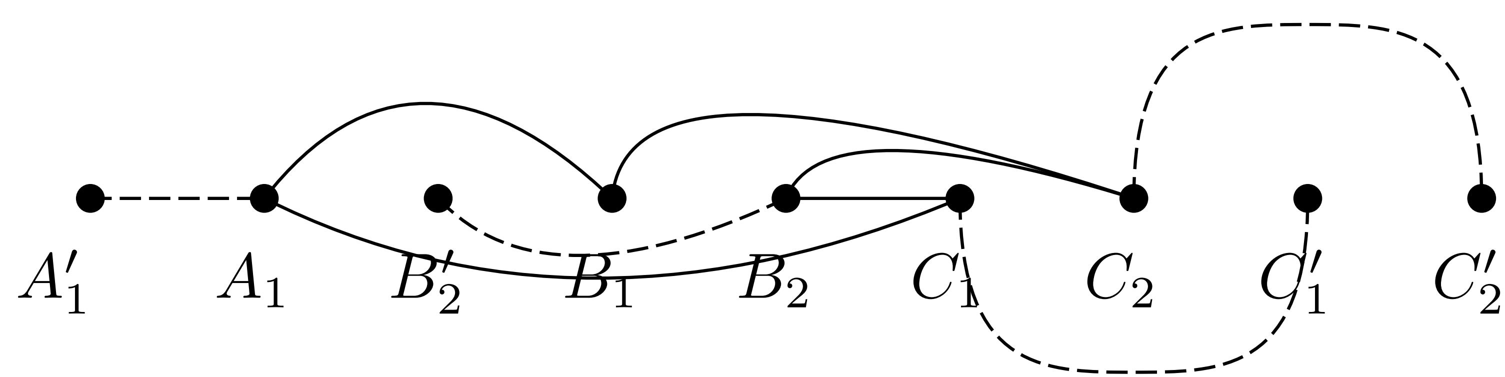


Restore Levelling



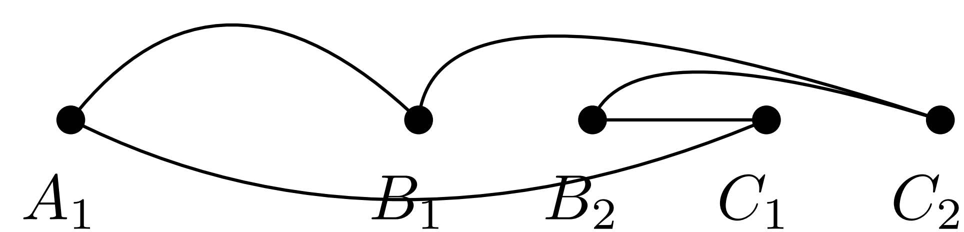
Restore Levelling

- remove helper edges



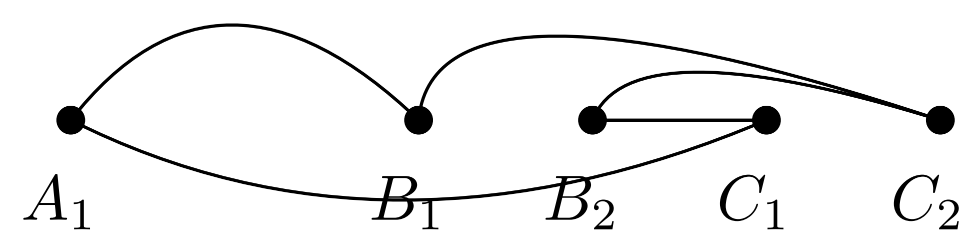
Restore Levelling

- remove helper edges



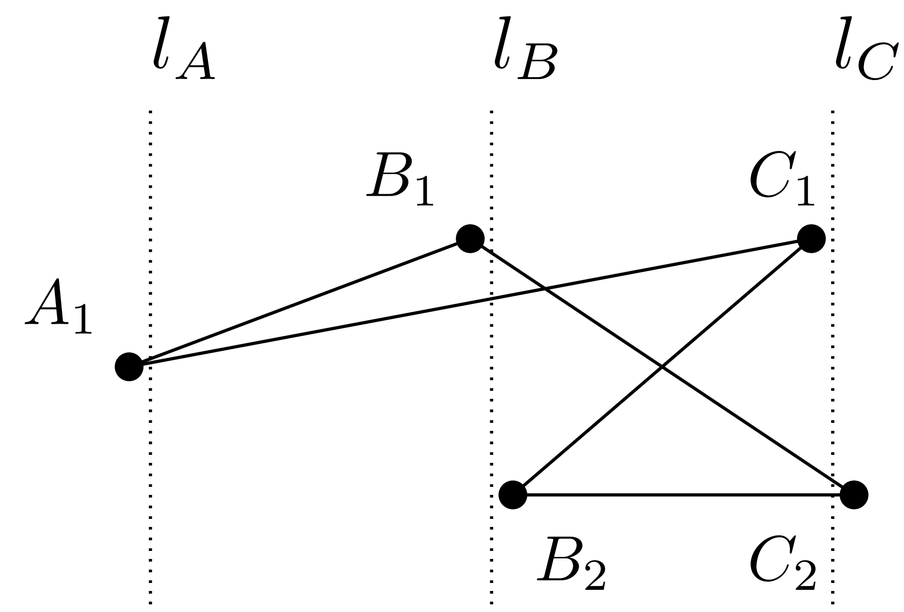
Restore Levelling

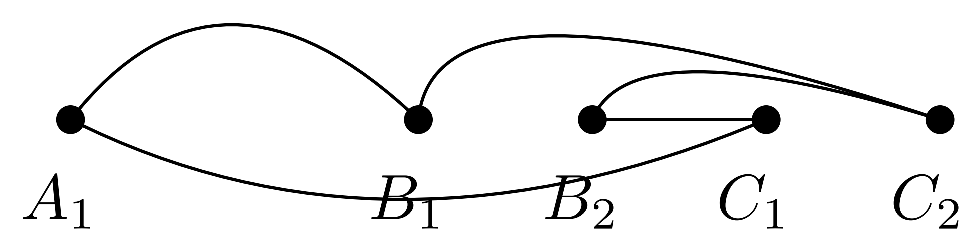
- remove helper edges
- squish and stretch horizontally



Restore Levelling

- remove helper edges
- squish and stretch horizontally





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- remove helper edges
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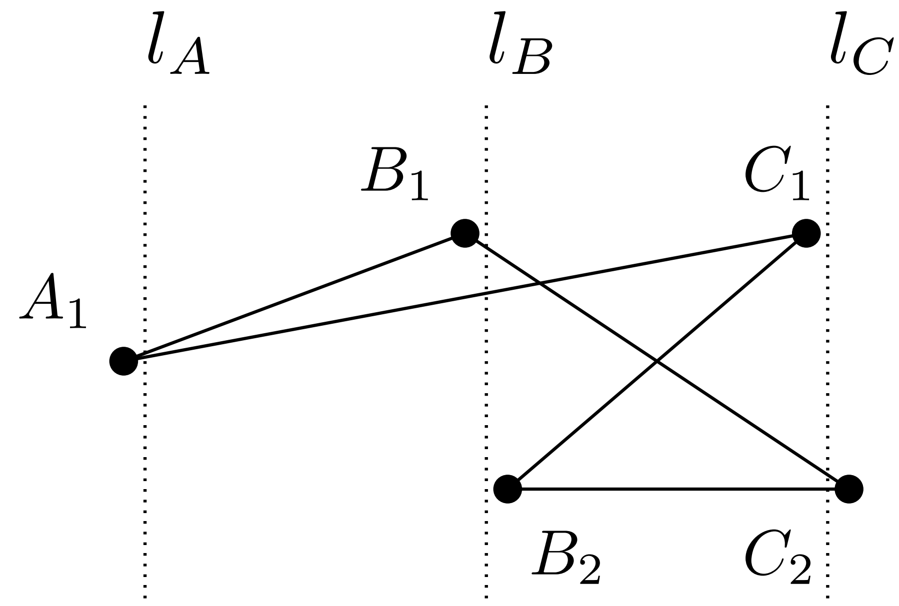
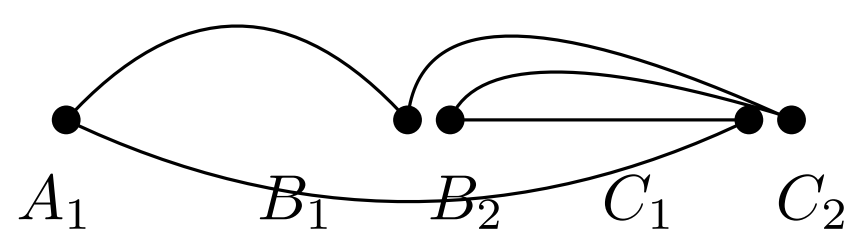
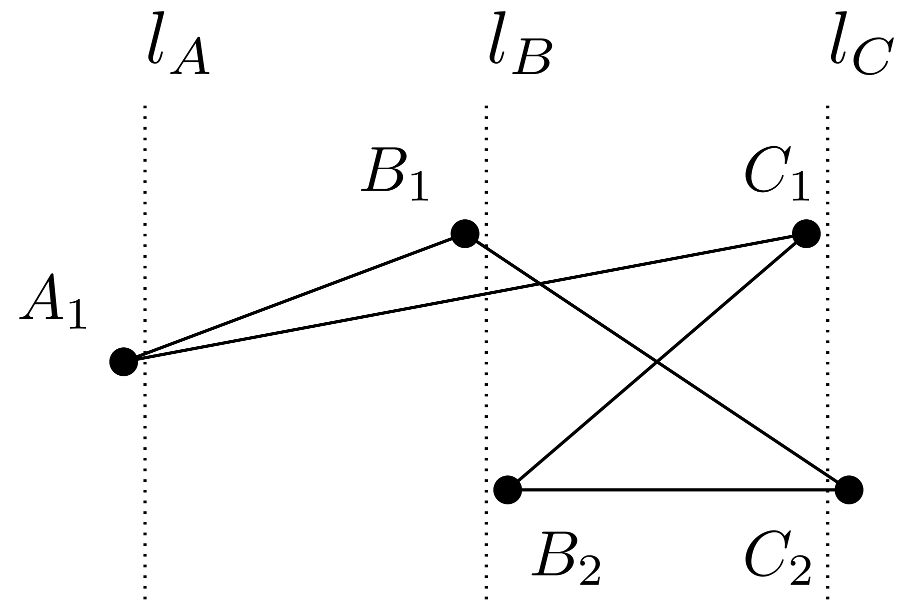
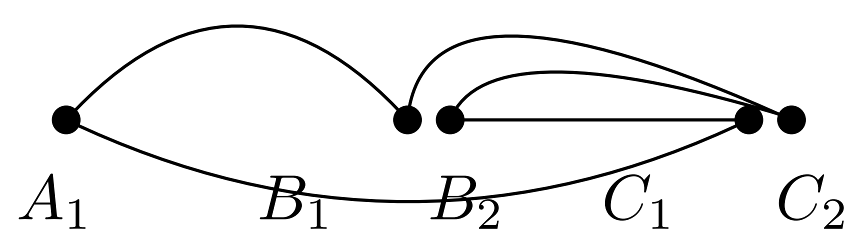




Restore Levelling

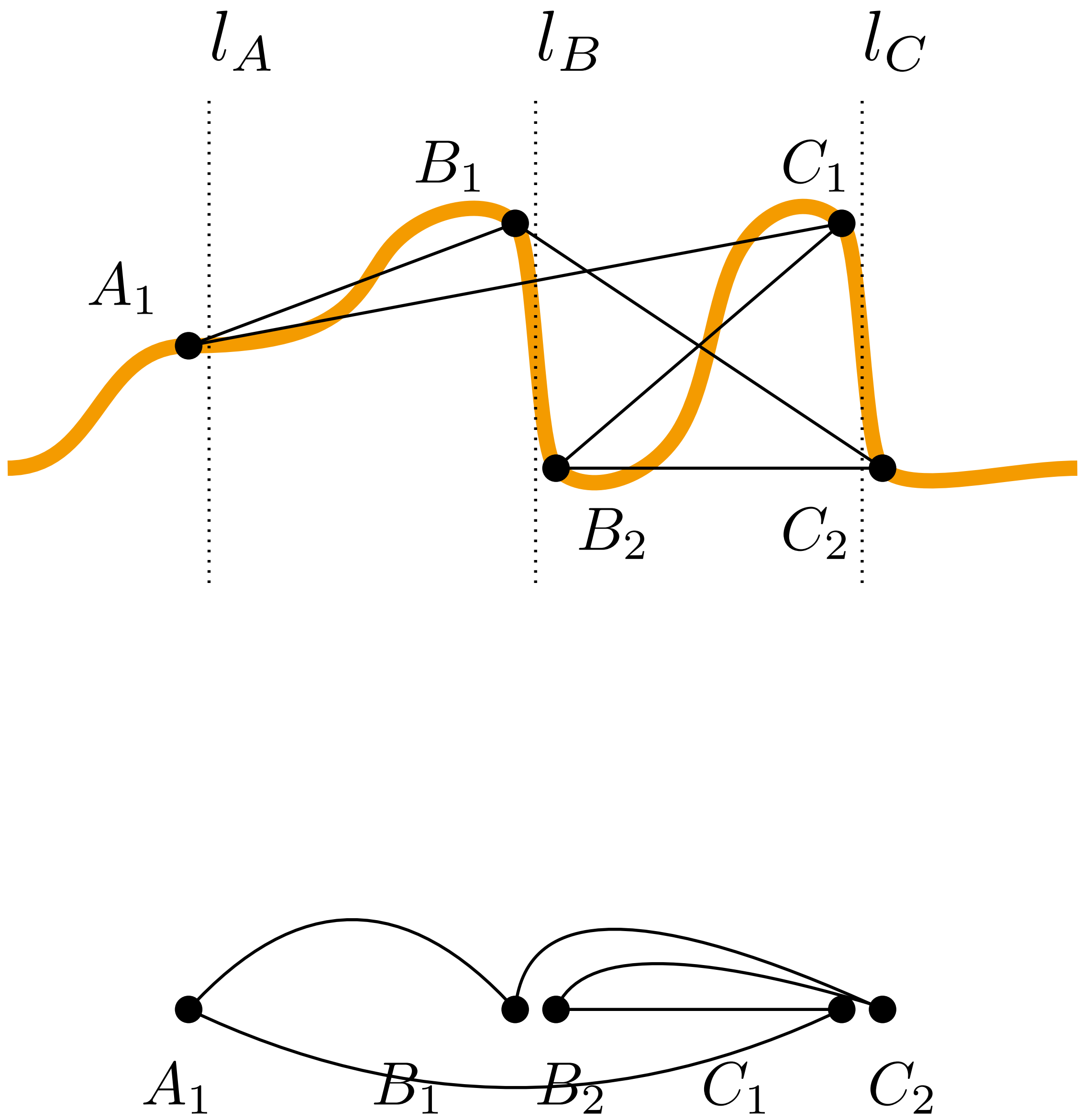
- remove helper edges
- squish and stretch horizontally
- fit x-monotone curve





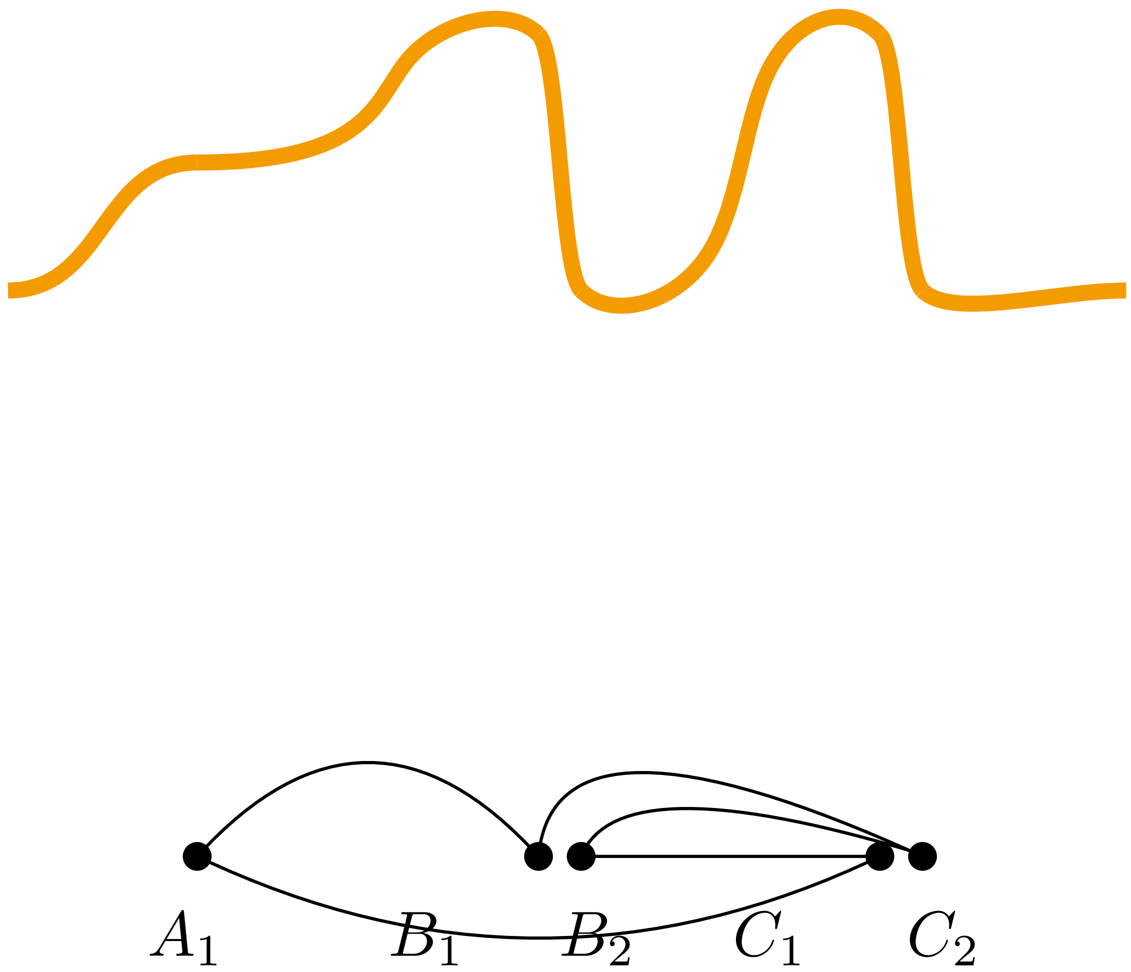
Restore Levelling

- remove helper edges
- squish and stretch horizontally
- fit x-monotone curve



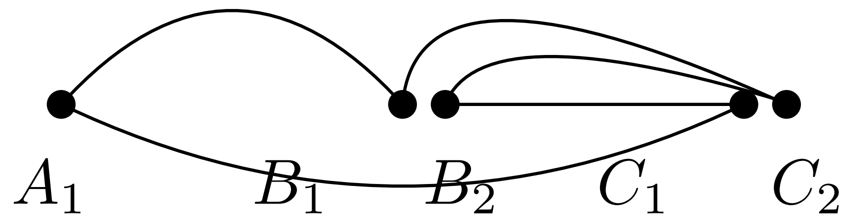
Restore Levelling

- remove helper edges
- squish and stretch horizontally
- fit x-monotone curve



Restore Levelling

- remove helper edges
- squish and stretch horizontally
- fit x-monotone curve
- add curve



Restore Levelling

- remove helper edges
- squish and stretch horizontally
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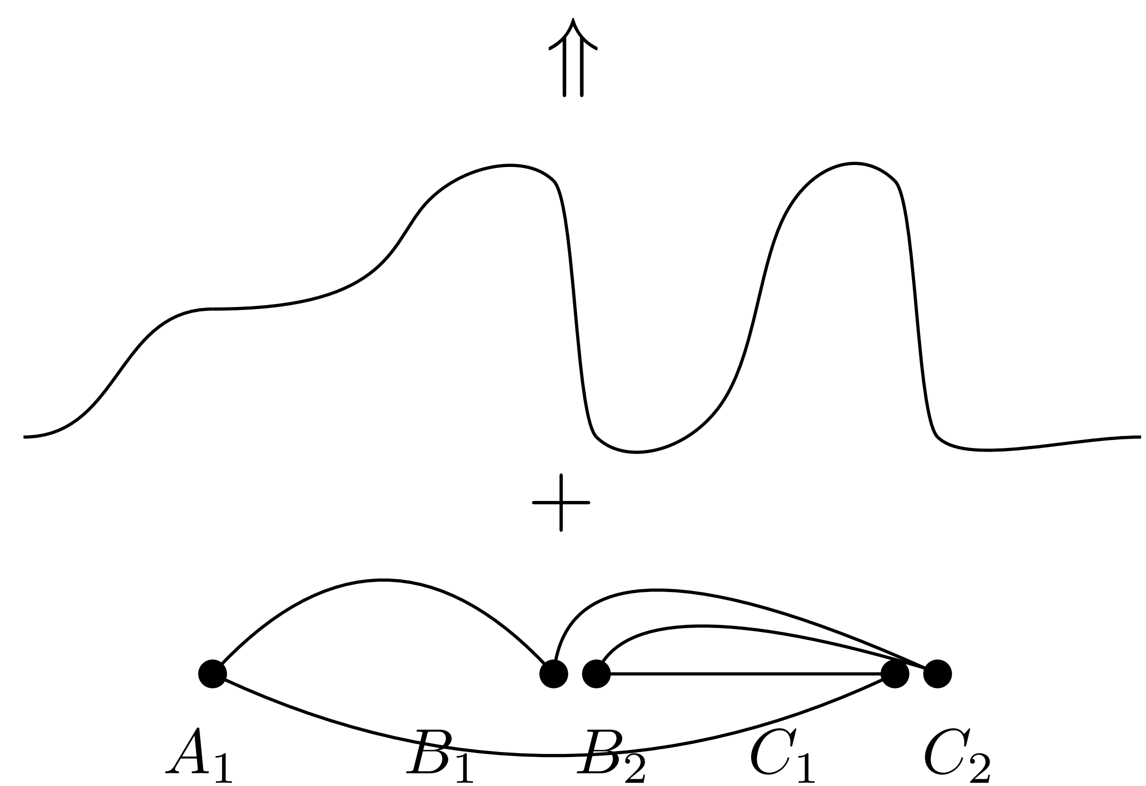
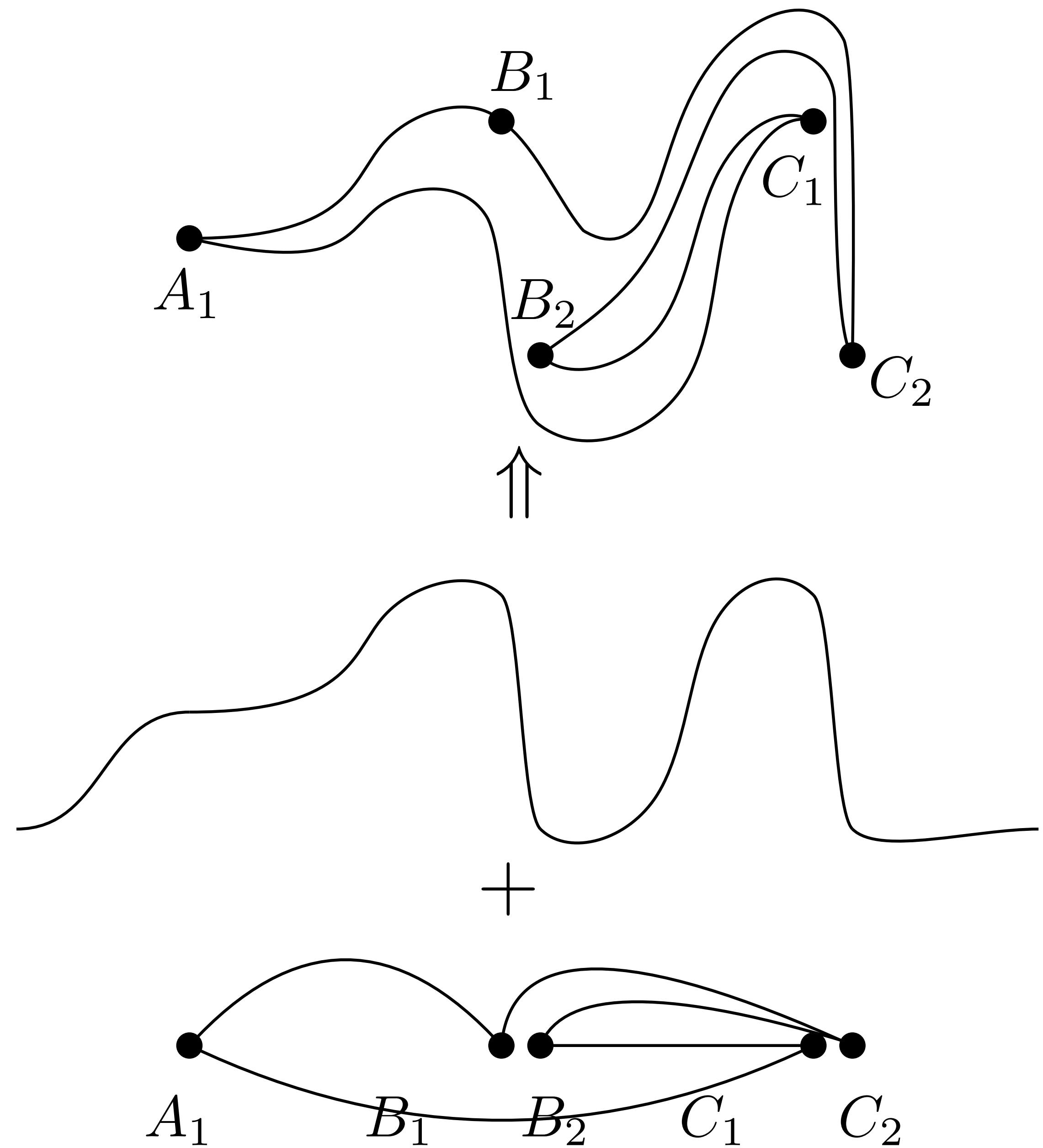
- add curve



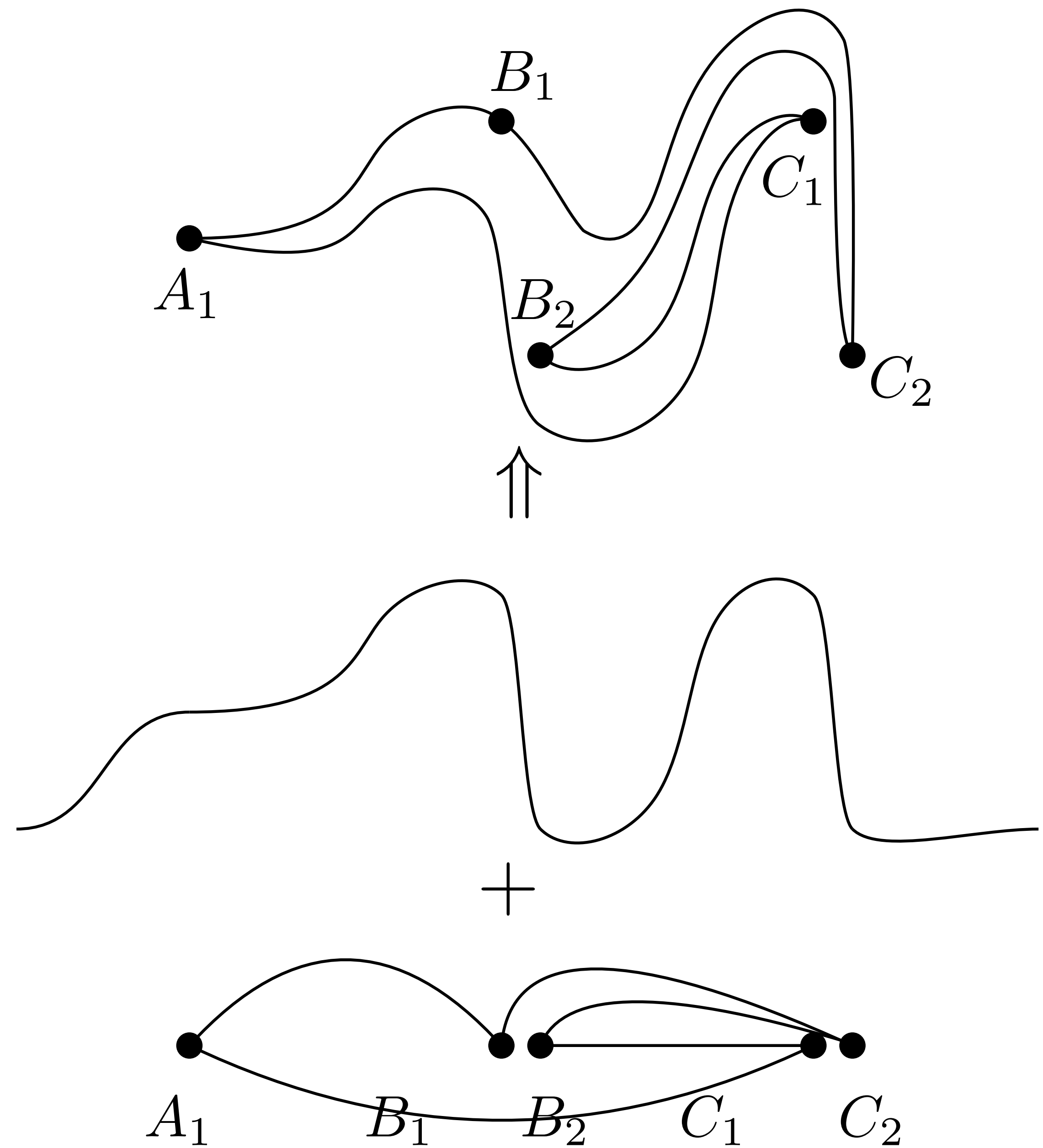
Restore Levelling

- remove helper edges
- squish and stretch horizontally
- fit x-monotone curve
- add curve



Restore Levelling

- remove helper edges
- squish and stretch horizontally
- fit x-monotone curve
- add curve
- reverse perturbation



Restore Levelling

- remove helper edges
- squish and stretch horizontally
- fit x-monotone curve
- add curve
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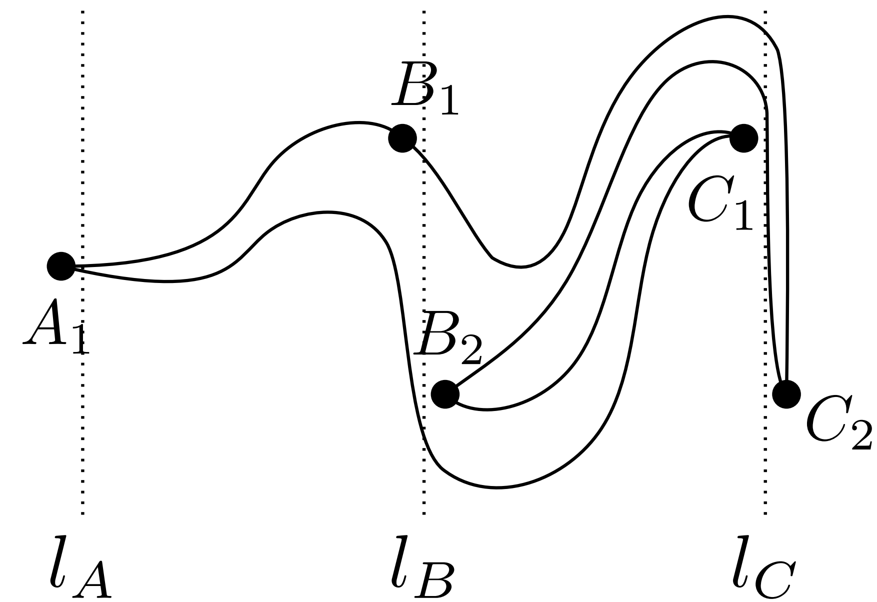

Restore Levelling

- remove helper edges
- squish and stretch horizontally
- fit x-monotone curve
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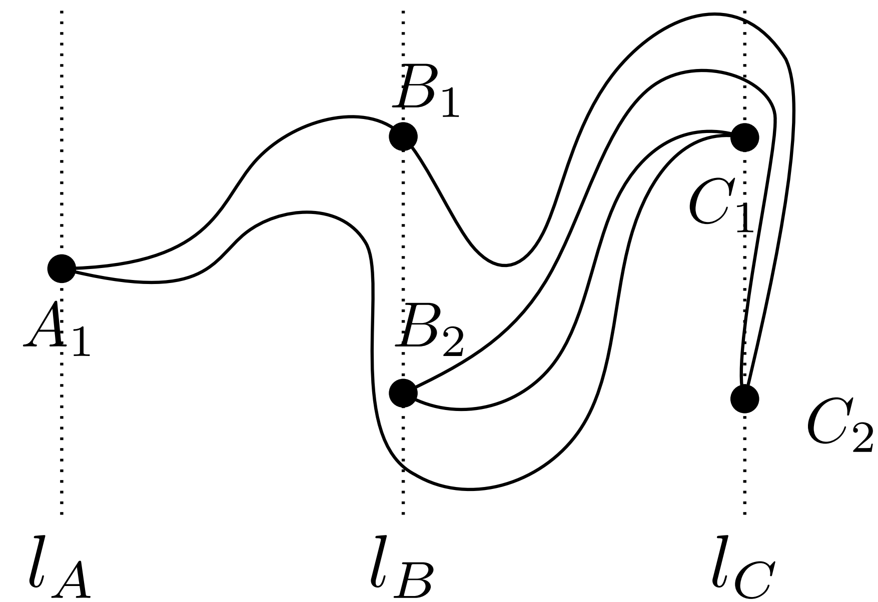
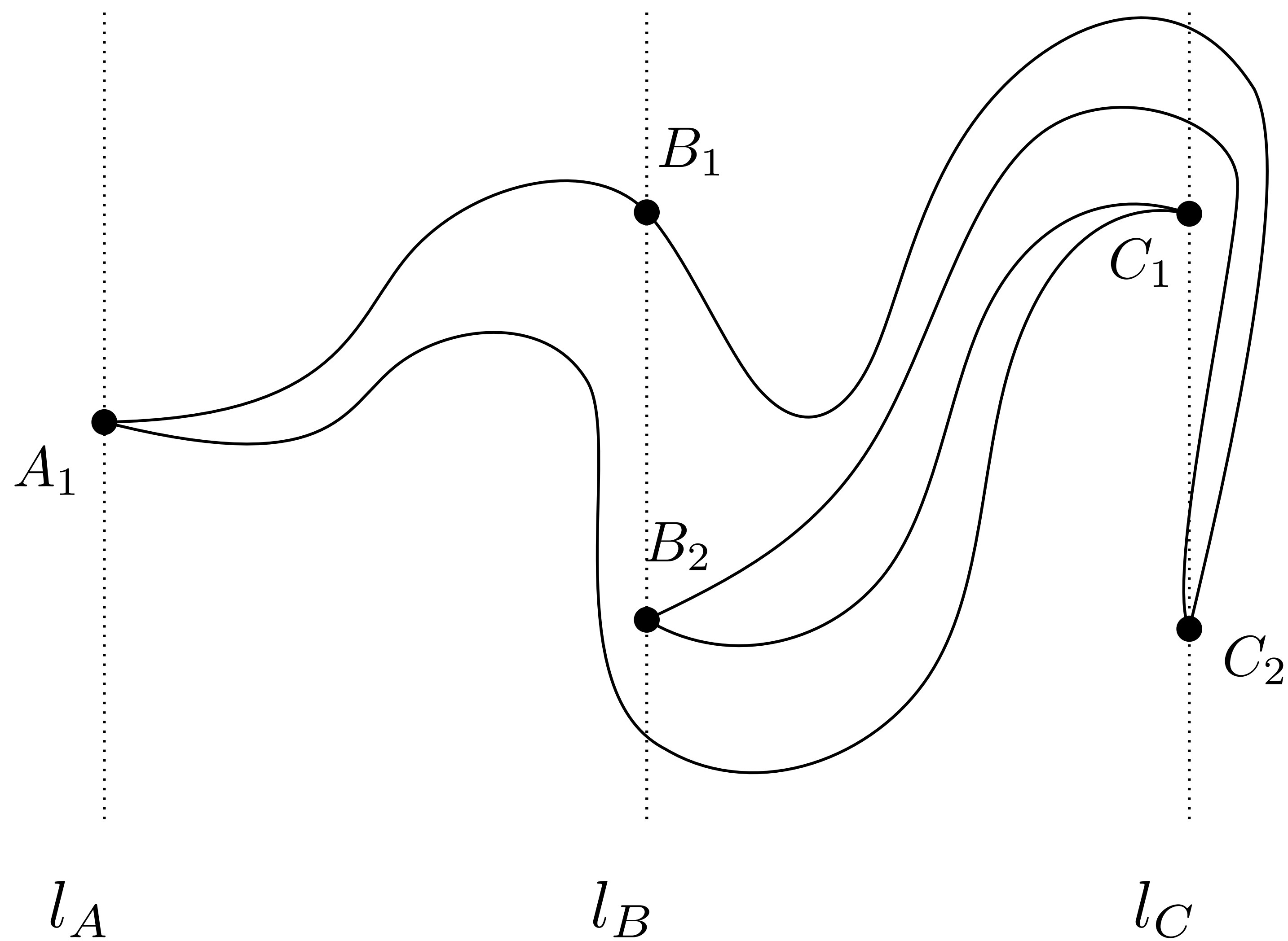
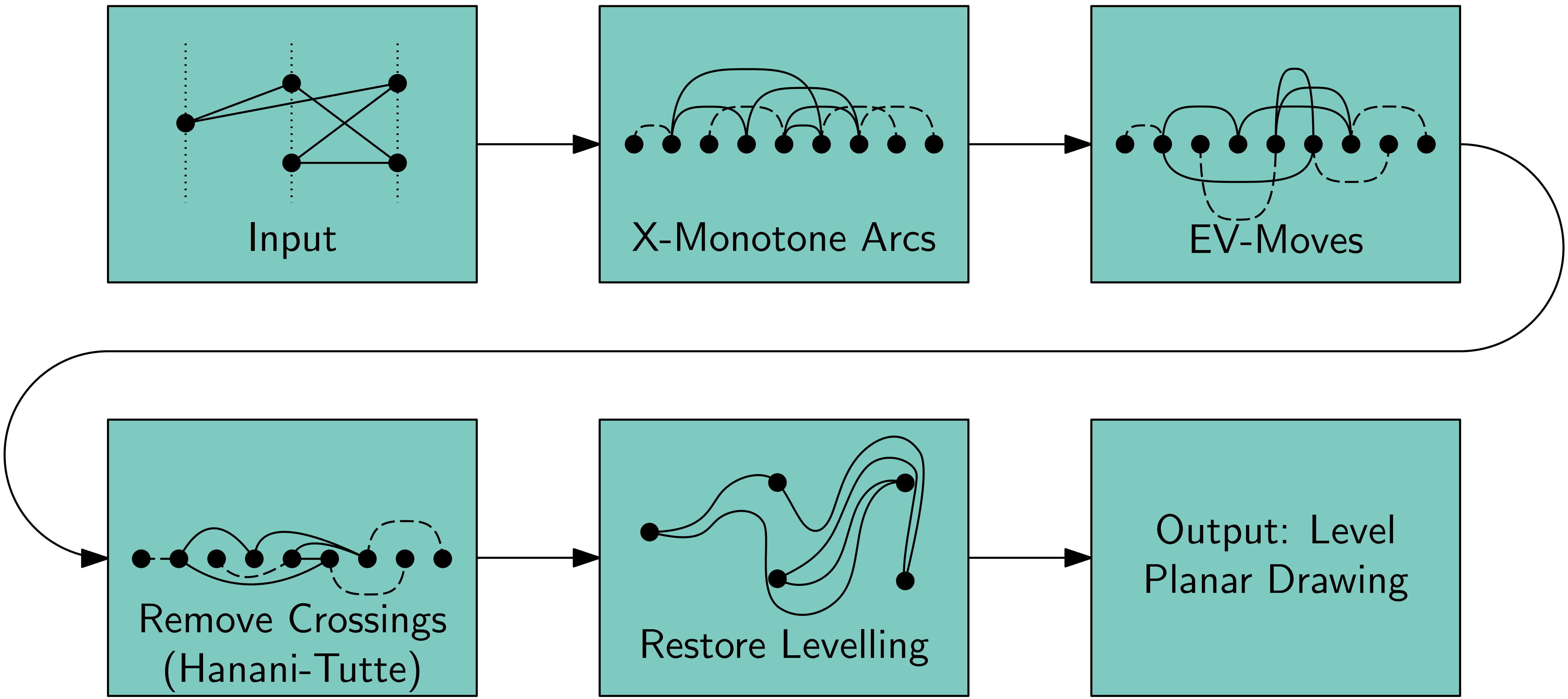


Restore Levelling



Leveled Embedding Construction

Algorithm



Remarks

Remarks

Complexity

Remarks

Complexity

- $O(|V|^2)$

Remarks

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- iteration over pairs of edges

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- for planar graphs $|E| \leq 3|V|$

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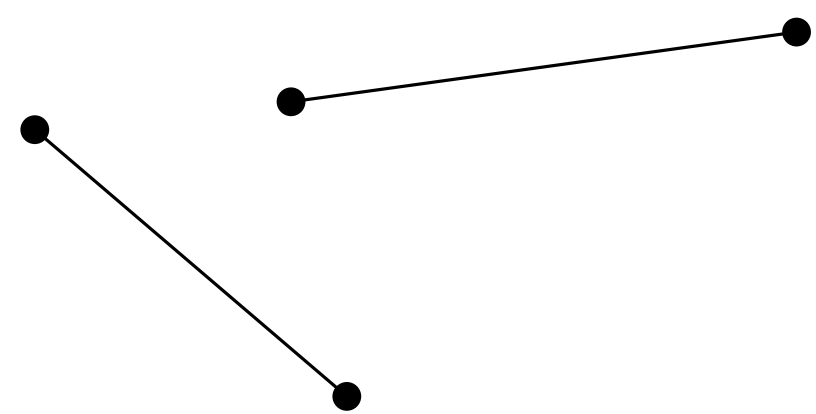
Generalization

Remarks

Complexity

- $O(|V|^2)$
- iteration over pairs of edges
- for planar graphs $|E| \leq 3|V|$
- 2-SAT $\in O(n)$

Generalization



Remarks

Complexity

- $O(|V|^2)$
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- for planar graphs $|E| \leq 3|V|$
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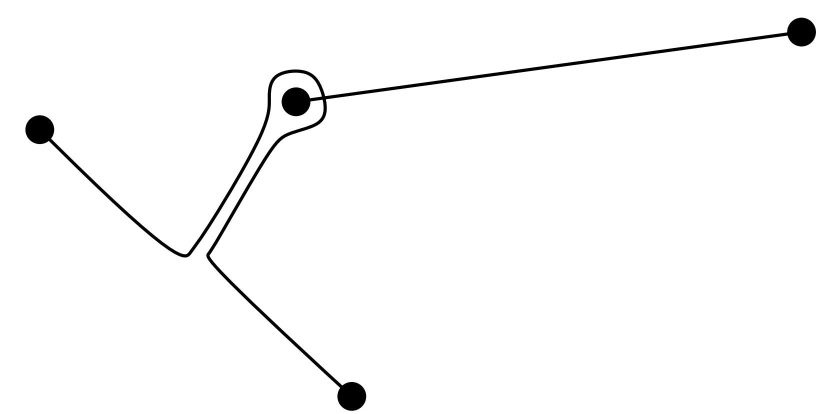
Generalization



Remarks

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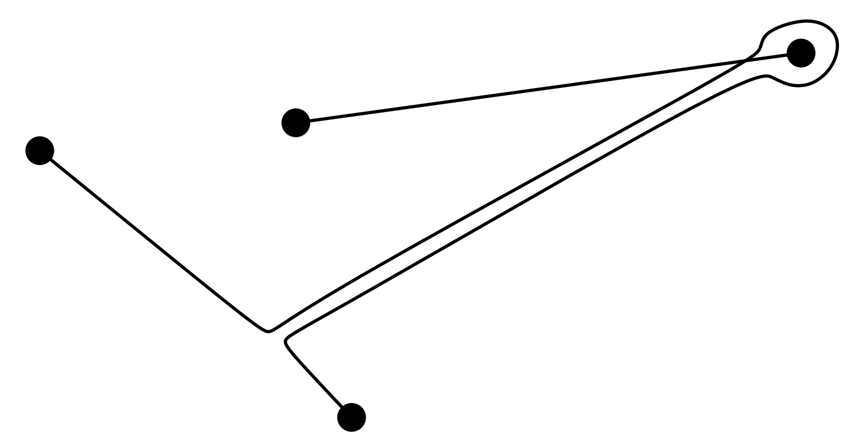
Generalization



Remarks

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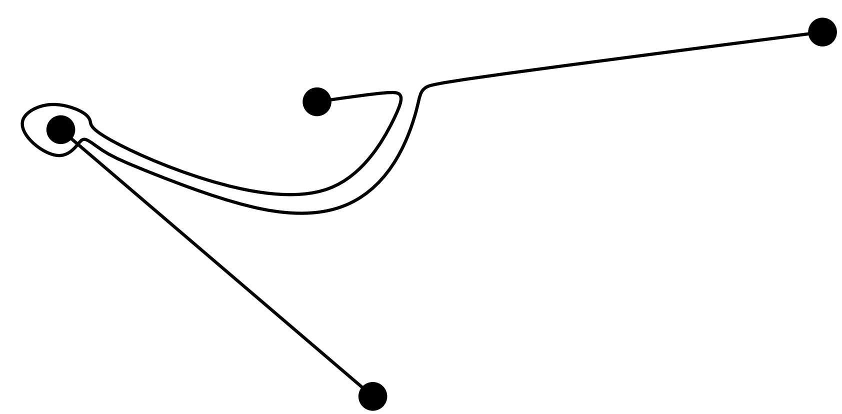
Generalization

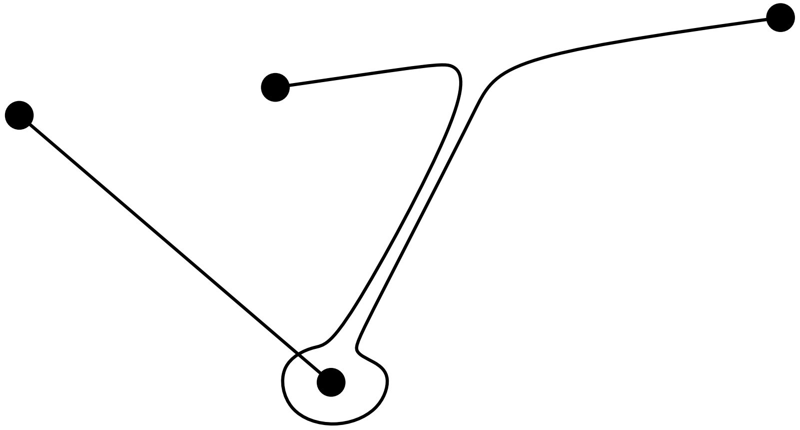


Remarks

Complexity

- $O(|V|^2)$
- iteration over pairs of edges
- for planar graphs $|E| \leq 3|V|$
- 2-SAT $\in O(n)$

Generalization



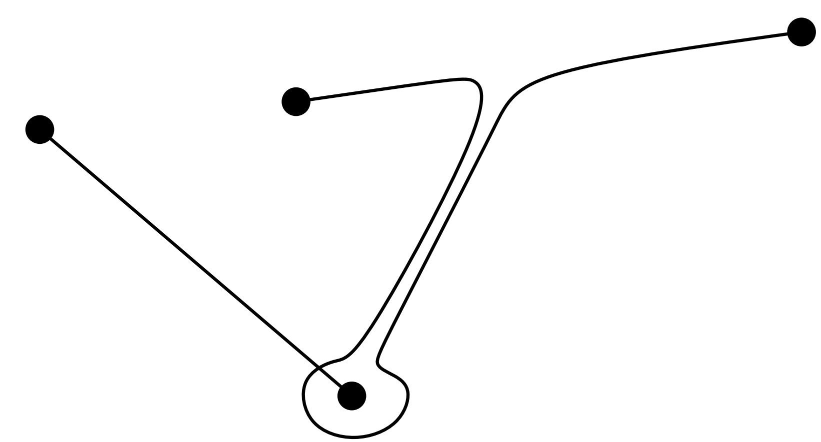
Remarks

Complexity

- $O(|V|^2)$
- iteration over pairs of edges
- for planar graphs $|E| \leq 3|V|$
- 2-SAT $\in O(n)$

Generalization

- 4 possible moves per edge pair



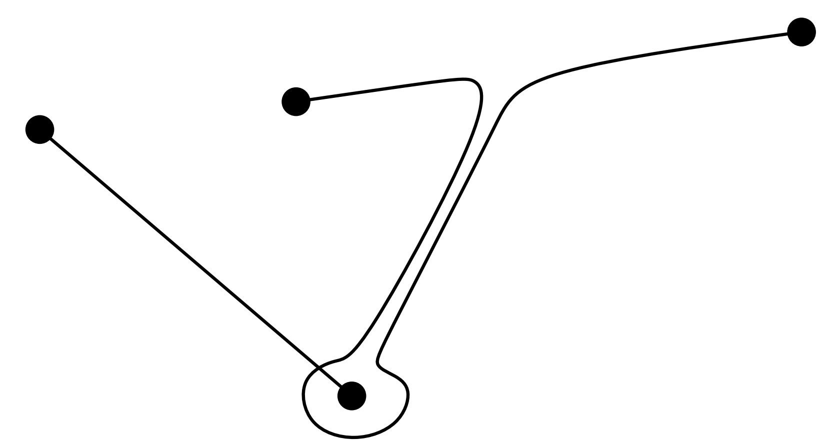
Remarks

Complexity

- $O(|V|^2)$
- iteration over pairs of edges
- for planar graphs $|E| \leq 3|V|$
- 2-SAT $\in O(n)$

Generalization

- 4 possible moves per edge pair
- must consider all edge pairs



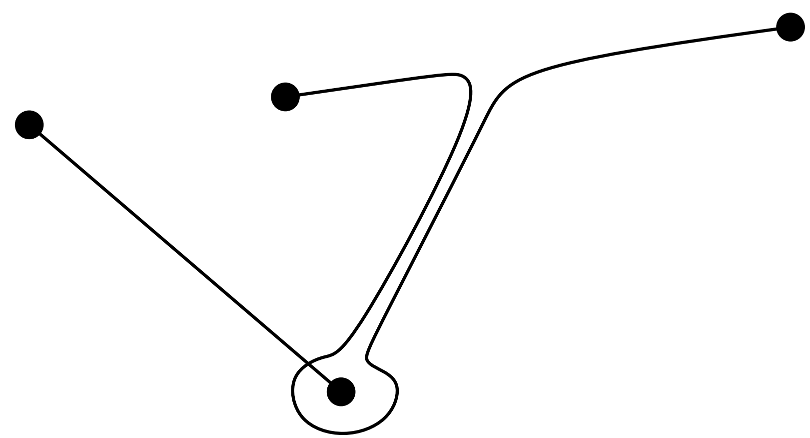
Remarks

Complexity

- $O(|V|^2)$
- iteration over pairs of edges
- for planar graphs $|E| \leq 3|V|$
- 2-SAT $\in O(n)$

Generalization

- 4 possible moves per edge pair
- must consider all edge pairs
- solve equation system over GF(2)



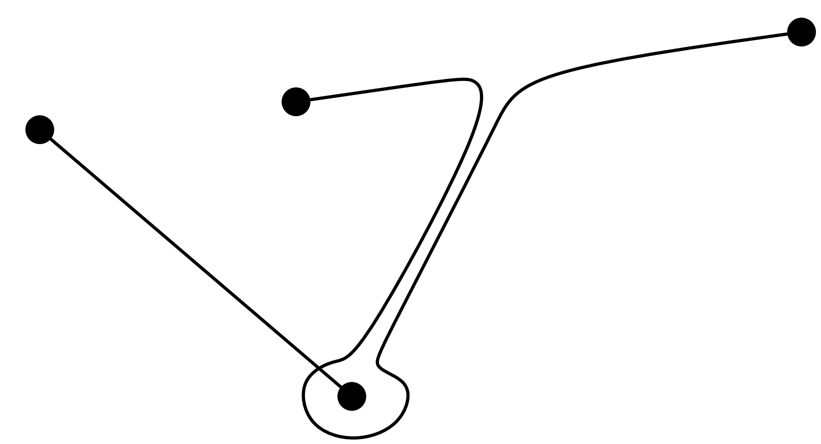
Remarks

Complexity

- $O(|V|^2)$
- iteration over pairs of edges
- for planar graphs $|E| \leq 3|V|$
- 2-SAT $\in O(n)$

Generalization

- 4 possible moves per edge pair
- must consider all edge pairs
- solve equation system over GF(2)
- Complexity: $O(n^6)$ with $n = (|E| + |V|)$



Questions